MAE

CONTROL SYSTEMS

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بسم الله الرحمن الرحيم، الحمد لله والصلاة والسلام على سيدنا محمد وعلى آله وصحبه اجمعين وبعد:

أضع بين يديكم دفتر مادة أنظمة التحكم الشامل لمادتنا في الجامعة، وضعت بطريقة مناسبة جدا للدراسة، وتحتوي على الكثير من الأسئلة بأجوبتها وبتفصيل للنقاط المهمة (المصادر في آخر صفحة).

ارجوا منكم نشر هذا الدفتر حتى تعم الفائدة لكل طلاب الهندسة الكهربائية (بشرط عدم وضع اي شعارات لاتحاداتٍ اخرى)، شاكرًا لكم، متمنيًا لكم التوفيق.

اخوكم محمد أبو عليان - فريق سبيكتروم.



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Table of Contents

1. Introduction 3
2. Laplace Transform 13
3. Mathematical Modelling 20
4. Electrical Modelling 26
5. Mechanical Modelling 38
6. Electromechanical Modelling 57
7. Block Diagrams 69
8. Signal Flow Graphs 89
2024-2025's MID Exam (2nd Semester) 107
9. Time Domain Analysis 108
10. Steady State Error 134
11. Routh-Hurwitz Criterion 145
12. Root Locus 153
13. PID Controller 168
2024-2025's Final Exam (2nd Semester) 182
References 183

1. Introduction

◈ Important Definitions:

- System: An interconnection of elements and devices for a desired purpose.
- Control System: An interconnection of components forming a system configuration that will provide a desired response.
- Process: The device, plant, or system under control. The input and output relationship represents the cause-and-effect relationship of the process.

input → Process → output

- Controlled Variable: The quantity or condition that is measured and controlled. It is normally the output of the control system.
- Manipulated Variable: The quantity of the condition that is varied by the controller to affect the value of controlled variable (controller variable).
- Control: The measuring the value of controlled variable of the system and applying the manipulated variable to the system to correct or limit the deviation of the measured value from a desired value.

- Set point: A target value that a system aims to maintain, it represents the reference input that the controller compares to the actual measured output.

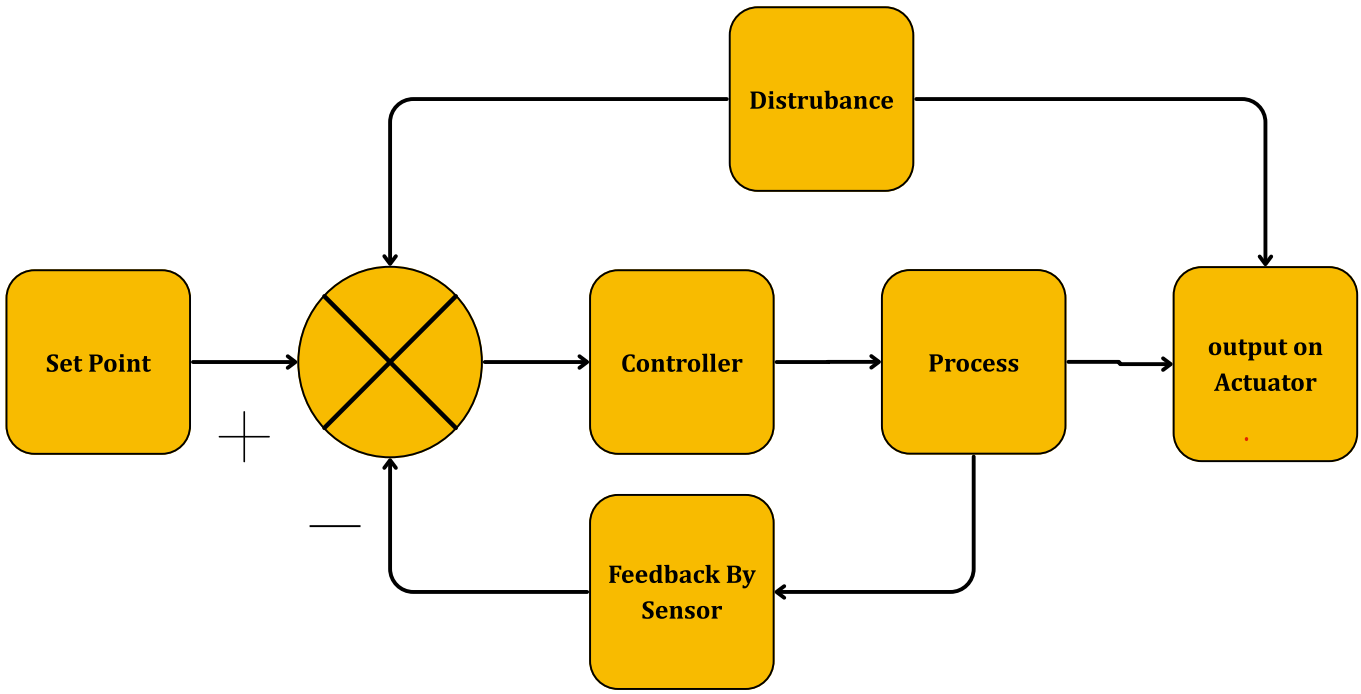
input or set point → controller → Process → output or controlled variable

- Disturbances: A signal that tends to adversely affect the value of the system. It is an unwanted input of the system.

There are 2 types of disturbance:

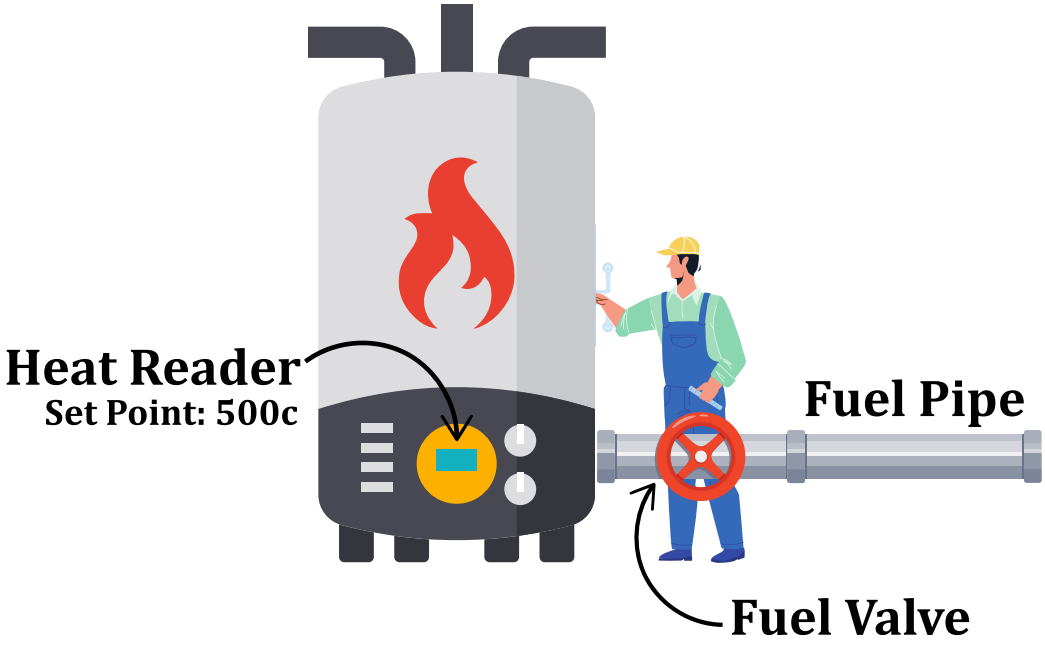
A. Internal disturbance: generated inside the system.

B. External disturbance: generated outside the system.



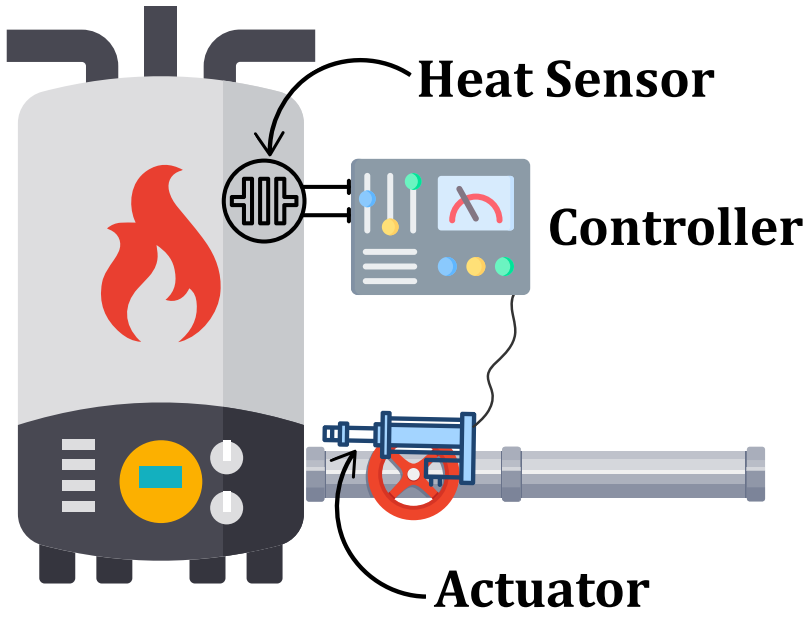
Example:

- Assume that you have a glass factory that has a heater, you want to make a glass by heating the sand at 500°c, so you brought a worker to monitor the temperature and adjust it. If it was less than 500°c, he would widen the fuel valve to increase the temperature to 500°c, and if it was higher than 500°c, he would narrow the valve to reduce the temperature.



- This system is not good because it is expensive and there is a possibility of error from the worker.

- So why don't we make an automatic system that senses the temperature by a sensor and from there controls the fuel valve temperature by an actuator?



- As shown up, the heat sensor sends to the controller heater's temperature and compares it with set point (500°c), also it programmed that if temperature is higher than 500°c it will control on fuel valve by an actuator, so it will reduce enough to allow less fuel to enter the heater to decrease temperature, and if it is less than 500°c it will wide enough to allow more fuel to enter the heater to increase temperature.

∴ How would we do and design this?

∴ What are the equations that do this?

- This is what we will learn in this course!

Classification of Control Systems:

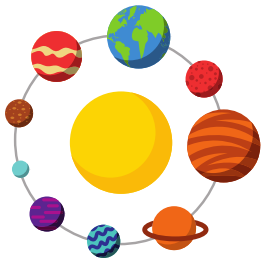
- To make it easy to understand, let's take them as parts:

1. **Neutral & Man-Made Systems:**

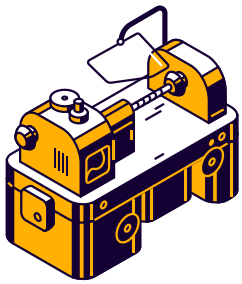
A. Neutral: A system without human creation.
Like universe and human bodies.

B. Man-Made: A system with human creation.
Like cars and machines.

Neutral



Man-Made

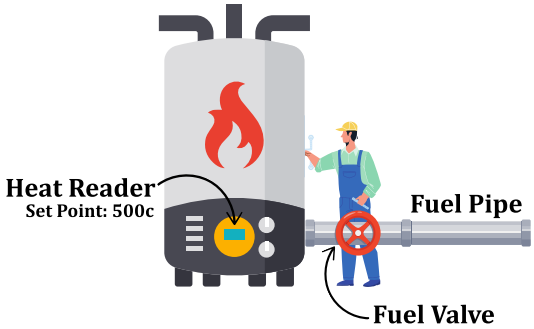


2. **Manual & Automatic Systems:**

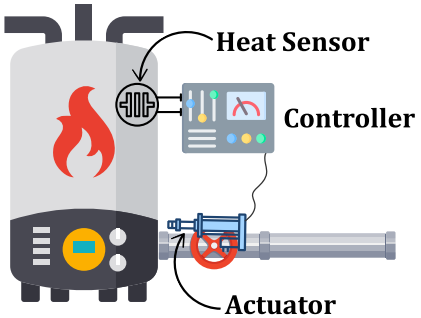
A. Manual: A system requires a human effort to control.
Like in 1st system in **Example** (open-loop system).

B. Automatic: A system doesn't require that to control.
Like in 2nd system in **Example** (closed-loop system).

Manual



Automatic



3. Feedback & Servo Systems:

A. Feedback: A system that maintains a prescribed relationship between the output and input by comparing them using the difference (error) to control,

B. Servo: A feedback system in which the output is a mechanical position, velocity or acceleration, like servo motors, radars and some of robotics components.

- The feedback in these systems can be negative or positive:

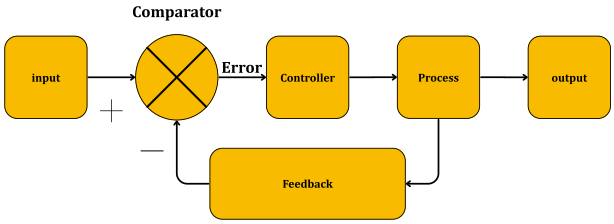
∴ If it was negative, it able to makes the system stable (The error can be zero):

$$error = Set\ Point - Feedback$$

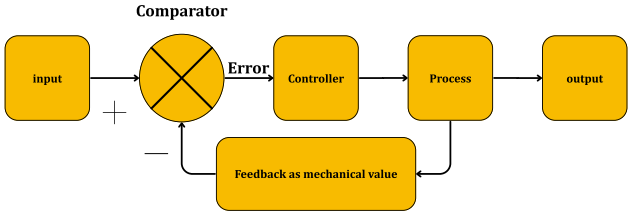
∴ If it was positive, it usually makes the system unstable (The error can't be zero):

$$error = Set\ Point + Feedback$$

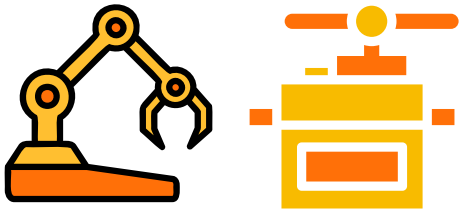
Feedback



Servo



Servo Applications

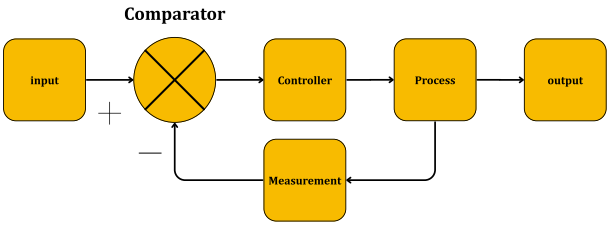


4. Single-Variable & Multi-Variable Systems:

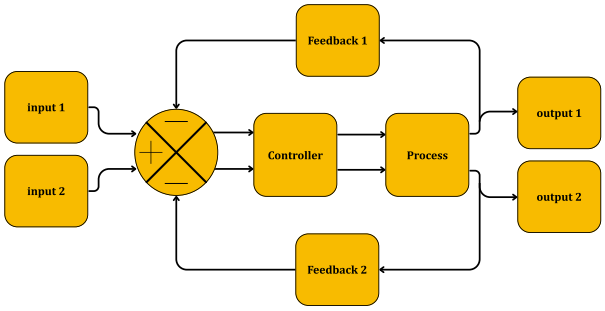
A. Single-Variable: A system that has only one input and one output, meaning it depends on a single variable. Like the automatic sytem in **Example** that can control temperature only.

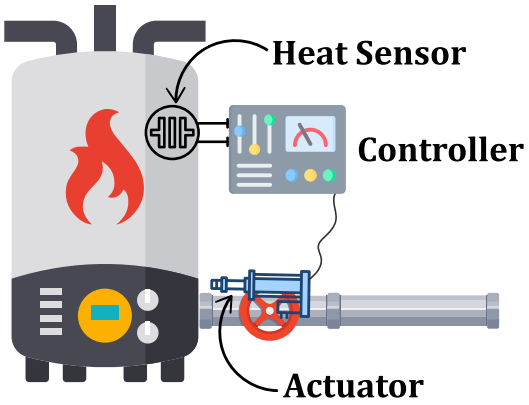
B. Multi-Variable: A system that has multiple inputs and/or multiple outputs, meaning it depends on more than one variable. Like that heater in **Example** but it can measure and control temperature and pressure at the same time.

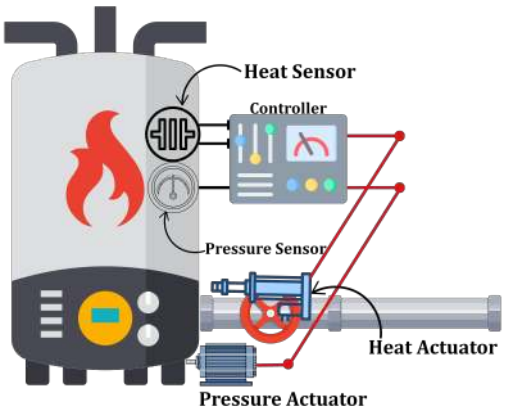
Single-Variable



Multi-Variable







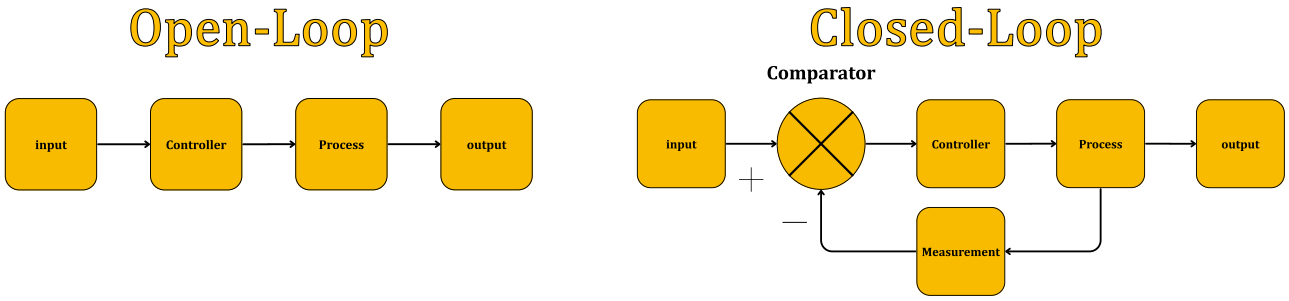
5. Open-Loop & Closed-Loop Systems:

A. Open-Loop: A system utilizes a controller or control actuator to obtain the desired response.

- The output has no effect on the control action, and it is neither measured nor fed back.
- It can't fix the disturbance of a system
- The manual system is an open-loop system.

B. Closed-Loop: A system utilizes feedback to compare the actual output to the desired output response.

- The output has effect on the control action, and it can be measured and fed back.
- It can fix the disturbance of a system.
- The feedback and auto systems are a closed-loop system.

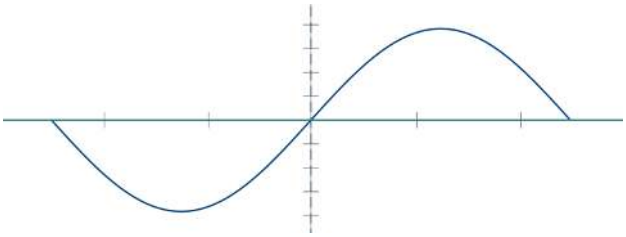


6. Continuous & Discrete Systems:

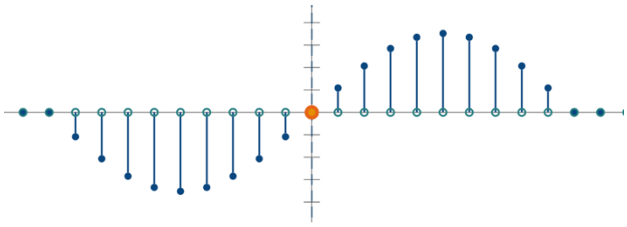
A. Continuous: A system involves one or more variables that are known at continuous time intervals.

B. Discrete: A system involves one or more variables that are known only at discrete time intervals.

Continuous



Discrete

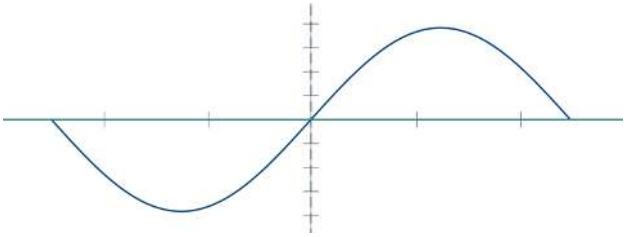


7. Deterministic & Stochastic Systems:

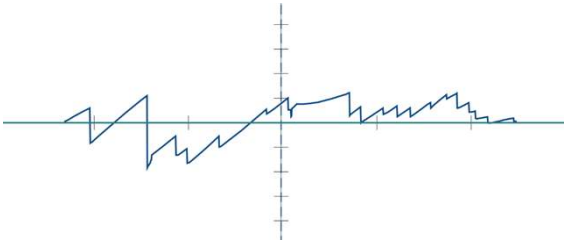
A. Deterministic: A system whose values are completely specified for any given time, response to input is predictable and repeatable.

B. Stochastic: A system whose values are completely random for any given time, response to input is unpredictable and unrepeatable.

Deterministic



Stochastic



8. Linear & Non-Linear Systems:

A. Linear: A system in which output varies linearly with the input.

B. Non-Linear: A system in which output varies non-linearly with the input.

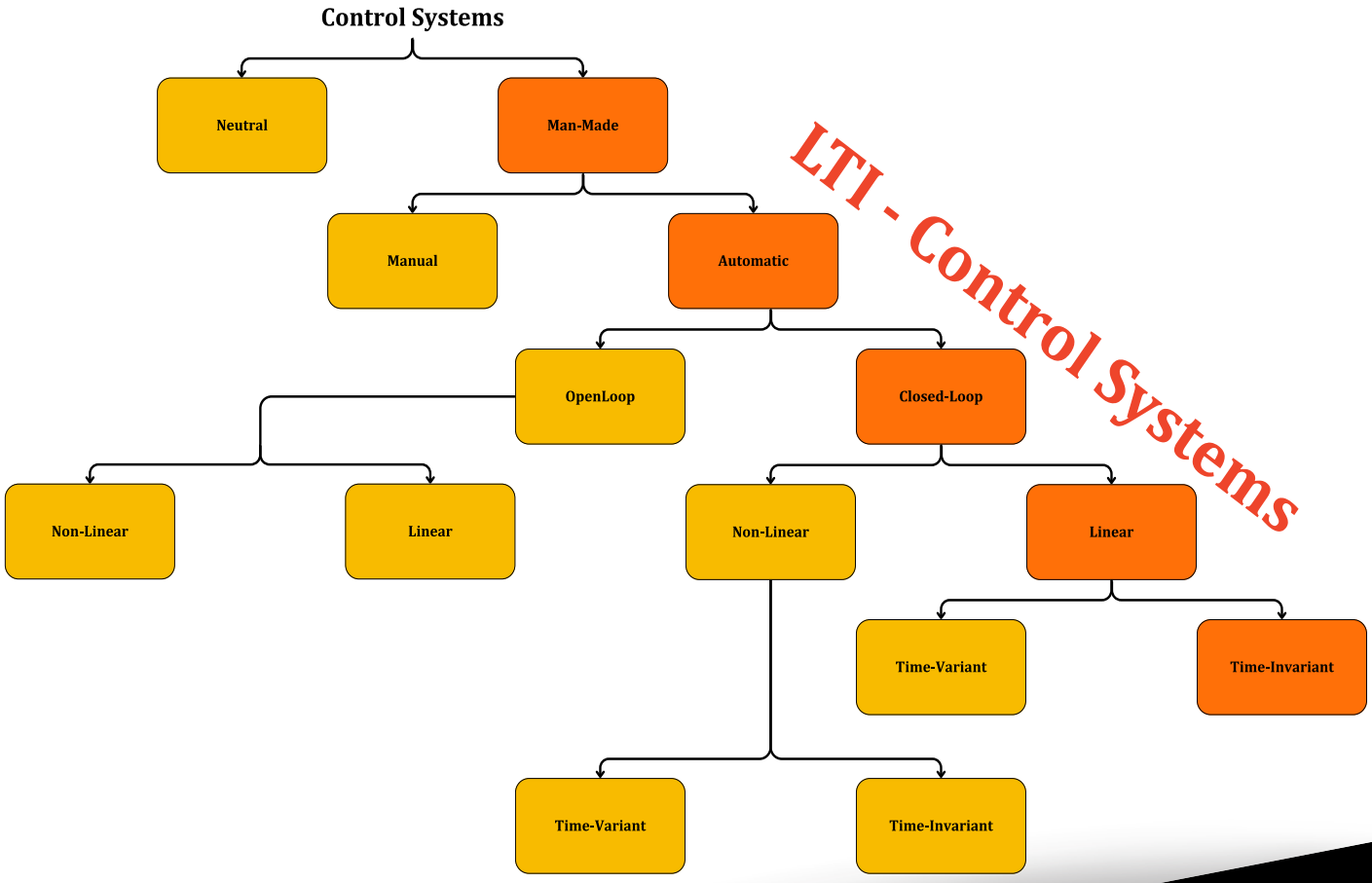
Linear	Non-Linear
$y(t) = 2x(t) + 3x(t-1)$	$y(t) = x^2(t) - 1$

9. Time-Invariant & Time-Variant Systems:

A. Time-Invariant: A system doesn't depend on time.

B. Time-Variant: A system depends on time.

Time-Invariant	Time-Variant
$y(t) = 2x(t) + 3$	$y(t) = t \cdot x(t)$



2. Laplace Transform

- We conclude Laplace Transforms in this table:

$x(t)$	$X(s)$	$x(t)$	$X(s)$
$\delta(t)$	1	$e^{-at}\,u(t)$	$\frac{1}{a+s}$
a	$\frac{a}{s}$	$e^{at}\,u(-t)$	$\frac{1}{a-s}$
$u(t)$	$\frac{1}{s}$	$e^{-a\lvert t\rvert}$	$\frac{2a}{a^2-s^2}$
$sgn(t)$	$\frac{2}{s}$	$rect\left(\frac{t}{\tau}\right)$	$\frac{\sinh\left(\frac{s\tau}{2}\right)}{\frac{s}{2}}$
$t^n u(t)$	$\frac{n!}{s^{n+1}}$	$tri\left(\frac{t}{\tau}\right)$	$\frac{\sinh^2\left(\frac{s\tau}{2}\right)}{\left(\frac{s}{2}\right)^2}$
$\sin(at)$	$\frac{a}{a^2+s^2}$	$\sinh(at)$	$\frac{a}{a^2-s^2}$
$\cos(at)$	$\frac{s}{a^2+s^2}$	$\cosh(at)$	$\frac{s}{a^2-s^2}$

- We conclude Laplace Transform properties in this table:

$x(t)$	$X(s)$
$x(at)$	$\frac{1}{\lvert a \rvert} X\left(\frac{s}{a}\right)$
$x(t - t_0)$	$e^{-st_0} X(s)$
$e^{s_0 t} x(t)$	$X(s - s_0)$
$x^{(n)}(t)$	$s^n X(s)$
$t^n \cdot x(t)$	$(-1)^n X^{(n)}(s)$
$\int_0^t x(\tau)\, d\tau$	$\frac{1}{s} \cdot X(s)$
$\frac{x(t)}{t}$	$\int_s^\infty X(s)\, ds$
$\lim_{t \to 0^+} x^{(n)}(t)$	$\lim_{s \to \infty} s^{n+1} \cdot X(s)$
$\lim_{t \to \infty} x^{(n)}(t)$	$\lim_{s \to 0} s^{n+1} \cdot X(s)$

▶ **Q. 2.1:** Find $\mathcal{L}\{x(t)\}$ for:

$\therefore\ x(t) = 3e^{-2t}u(t) + 4e^{t}u(-t)$

$$X(s) = 3\mathcal{L}\{e^{-2t}u(t)\} + 4\mathcal{L}\{e^{t}u(-t)\} = \frac{3}{s+2} + \frac{4}{s-1}$$

$\therefore\ x(t) = \sin\big(2(t-3)\big)$

$$\sin(at) \Leftrightarrow \frac{1}{a^2+s^2} \rightarrow a = 2, t_0 = 3$$

$$\sin\big(a(t-t_0)\big) \Leftrightarrow e^{-st_0}\frac{a}{a^2+s^2} = e^{-3s}\frac{2}{4+s^2}$$

$\therefore\ x(t) = t^2e^{-t}u(t)$

$$x(t) = e^{-t}u(t) \Leftrightarrow X(s) = \frac{1}{1+s}$$

$$t^2x(t) \Leftrightarrow X''(s) = \left(\frac{-1}{(1+s)^2}\right)' = \frac{2(1+s)}{(1+s)^4} = \frac{2}{(1+s)^3}$$

$\therefore\ x(t) = \int_0^t \sin 2\tau\, d\tau$

$$x(\tau) = \sin 2\tau \Leftrightarrow X(s) = \frac{2}{s^2+2^2}$$

$$\int_0^t x(\tau)\, d\tau \Leftrightarrow \frac{X(s)}{s} = \frac{2}{s(s^2+4)} = \frac{2}{s^3+4s}$$

▶ Q. 2.2: If $x(t) = t^2u(t)$, Find $\mathcal{L}\{x''(t)\}$

∴ *Solution*:

$$x(t) = t^2u(t) \iff X(s) = \frac{2}{s^3}$$

$$x''(t) \iff s^2X(s) = \frac{2}{s}$$

▶ Q. 2.3: Find $\lim_{t \to 0^+} x(t)$ if $X(s) = \frac{1}{s+2}$

∴ *Solution*:

$$\lim_{t \to 0^+} x(t) = \lim_{s \to \infty} s \cdot \left[\frac{1}{s + 2}\right] = 1$$

▶ Q. 2.4: Find $\lim_{t \to \infty} x'(t)$ if $X(s) = \frac{12}{s^2+6s+9}$

∴ *Solution*:

$$\lim_{t \to \infty} x'(t) = \lim_{s \to 0} s^2 \cdot \left[\frac{12}{s^2 + 6s + 9}\right] = 0$$

▶ **Q. 2.5:** Find $\mathcal{L}^{-1}\{X(s)\}$ for:

$\therefore X(s) = \frac{2}{3+s}$

$$\mathcal{L}^{-1}\left\{2/(3+s)\right\} = 2e^{-3t}u(t)$$

$\therefore X(s) = \frac{4}{s^2+4}$

$$\mathcal{L}^{-1}\left\{2 \cdot \frac{2}{s^2+2^2}\right\} = 2\sin 2t$$

$\therefore X(s) = \frac{6}{(s-1)^4}$

$$\mathcal{L}^{-1}\left\{\frac{3!}{(s-1)^4}\right\} = e^{t}t^{3}u(t)$$

$\therefore X(s) = e^{-4s}\frac{s}{s^2+9}$

$$\mathcal{L}^{-1}\left\{e^{-4s}\frac{s}{s^2+3^2}\right\} = \cos\big(3(t-4)\big)$$

$\therefore X(s) = \frac{s+2}{s^5}$

$$\mathcal{L}^{-1}\left\{\frac{s}{s^5}+\frac{2}{s^5}\right\} = \mathcal{L}^{-1}\left\{\frac{3!}{3!}\cdot\frac{1}{s^4}\right\} + \mathcal{L}^{-1}\left\{\frac{4!}{5!}\cdot\frac{2}{s^5}\right\} = \frac{t^3}{6}+\frac{t^4}{12}$$

$\therefore X(s) = \frac{s-1}{(s-3)^2+16}$

$$\mathcal{L}^{-1}\left\{\frac{s-1-3+3}{(s-3)^2+16}\right\} = \mathcal{L}^{-1}\left\{\frac{s-3+2}{(s-3)^2+16}\right\}$$

$$= \mathcal{L}^{-1}\left\{\frac{s-3}{(s-3)^2+4^2}\right\} + \mathcal{L}^{-1}\left\{\frac{4}{4}\frac{2}{(s-3)^2+4^2}\right\}$$

$$= e^{3t}\cos(4t) + \frac{1}{2}e^{3t}\sin(4t)$$

$\therefore X(s) = \frac{2(s+1)}{s(s+2)}$

$$\frac{2s+2}{s(s+2)} = \frac{A}{s} + \frac{B}{s+2} \rightarrow 2s+2 = A(s+2) + Bs$$

$s = 0 \rightarrow 2 = 2A \rightarrow A = 1$

$s = -2 \rightarrow -2 = -2B \rightarrow B = 1$

$$\frac{s+1}{s(s+2)} = \frac{1}{s} + \frac{1}{s+2}$$

$$\mathcal{L}^{-1}\left\{\frac{2s+2}{s(s+2)}\right\} = \mathcal{L}^{-1}\left\{\frac{1}{s}\right\} + \mathcal{L}^{-1}\left\{\frac{1}{s+2}\right\} = \left(1 + e^{-2t}\right)u(t)$$

$\therefore X(s) = \frac{s}{s^2+4s+3}$

$$\frac{s}{(s+1)(s+3)} = \frac{A}{s+1} + \frac{B}{s+3} \rightarrow s = A(s+3) + B(s+1)$$

$s = -1 \rightarrow -1 = 2A \rightarrow A = \frac{1}{2}$

$s = -3 \rightarrow -3 = -2B \rightarrow B = \frac{3}{2}$

$$\frac{s}{(s+1)(s+3)} = \frac{1}{2}\cdot\frac{1}{s+1} + \frac{3}{2}\cdot\frac{1}{s+3}$$

$$\mathcal{L}^{-1}\left\{\frac{s}{s^2+4s+3}\right\} = \frac{1}{2}\mathcal{L}^{-1}\left\{\frac{1}{1+s}\right\} + \frac{3}{2}\mathcal{L}^{-1}\left\{\frac{1}{3+s}\right\}$$

$$= \left(\frac{1}{2}e^{-t} + \frac{3}{2}e^{-3t}\right)u(t)$$

$\therefore X(s) = \frac{s+2}{s^2+2s+2}$

In this case, since the denominator can't be factored, so we use square completing $\left(\pm\frac{b^2}{2}\right)$:

$$\mathcal{L}^{-1}\left\{\frac{s+1}{s^2+2s+2}\right\} = \mathcal{L}^{-1}\left\{\frac{s+2}{s^2+2s+2-2+2}\right\} = \mathcal{L}^{-1}\left\{\frac{s+2}{(s^2+2s+4)-2}\right\}$$

$$= \mathcal{L}^{-1}\left\{\frac{s+2}{(s+2)^2-2}\right\} = e^{-2t}\cos\sqrt{2}\,t$$

▶ **Q. 2.6:** If:

$$i(t) = 2e^{-2t}\text{mA},$$
$$R = 5\text{k}\Omega, \qquad L = 2\text{mH}, \qquad C = 10\mu\text{F}$$

Find $V(s)$ when:

$\therefore\ v(t) = R \cdot i(t)$

$$I(s) = \mathcal{L}\{i(t)\} = \mathcal{L}\{2e^{-2t}\} = \frac{2}{2+s}\text{mA};$$

$$V(s) = \mathcal{L}\{R \cdot i(t)\} = R \cdot I(s) = 5\text{k} \cdot \frac{2}{2+s}\text{m} = \frac{10}{2+s}\text{V}$$

$\therefore\ v(t) = L\frac{di}{dt}$

$$V(s) = \mathcal{L}\left\{L\frac{di}{dt}\right\} = L \cdot s \cdot I(s) = 2\text{m} \cdot s \cdot \frac{2}{2+s}\text{m} = \frac{4s}{2+s}\mu\text{V}$$

$\therefore\ v(t) = \frac{1}{C}\int i(t)\,dt$

$$V(s) = \mathcal{L}\left\{\frac{1}{C}\int i(t)\,dt\right\} = \frac{1}{C} \cdot \frac{I(s)}{s} = \frac{1}{10\mu} \cdot \frac{2}{s(2+s)}\text{m}$$
$$= \frac{1}{5s^2 + 10s}\text{kV}$$

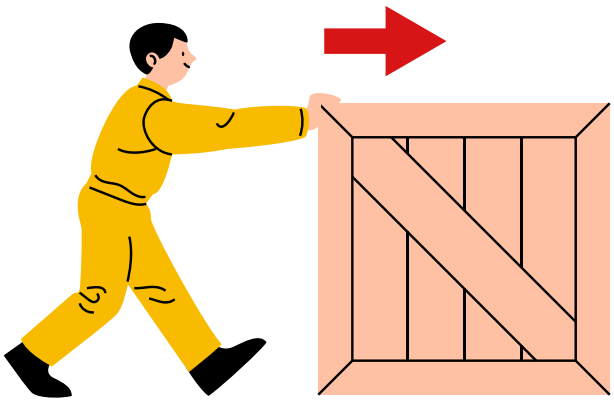
3. Mathematical Modelling

◈ Types of Systems:

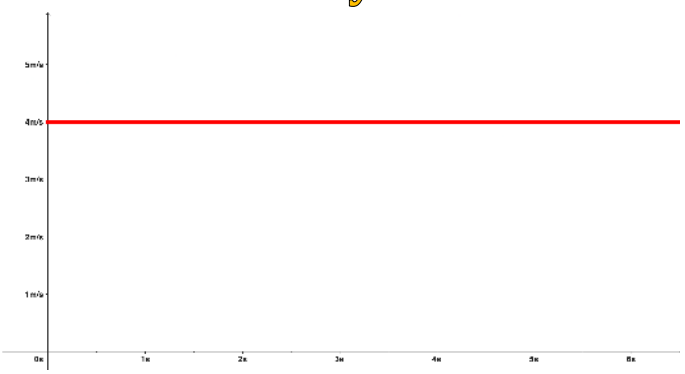
1. **Static System:** A system doesn't change with time.
2. **Dynamic System:** A system changes with time.

▶ Example:

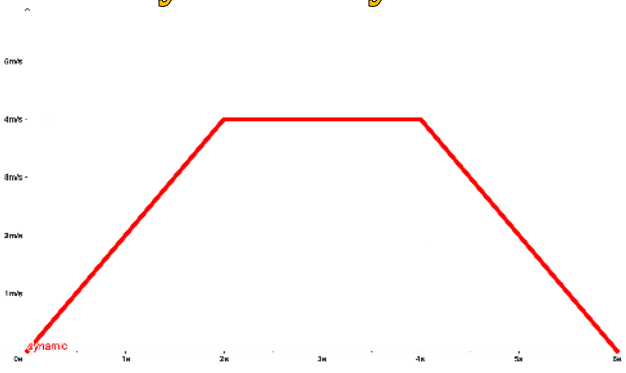
- "Static" word doesn't mean kinetic static. as when you move a box at a fixed speed, you see the box moving, but its speed doesn't change with time, this is what is meant. But when you move it at a different speed, it is a dynamic system.



Static System

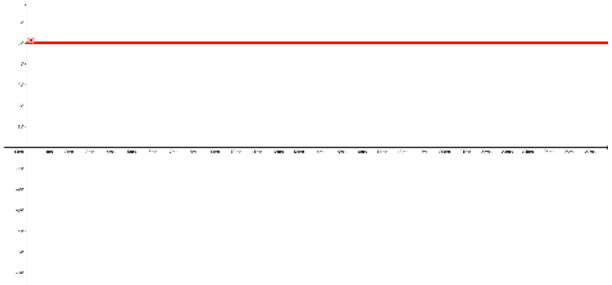


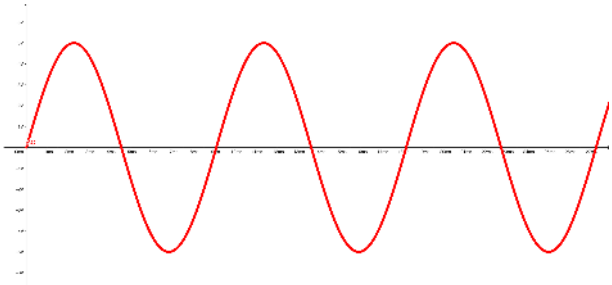
Dynamic System



▶ **Example:**

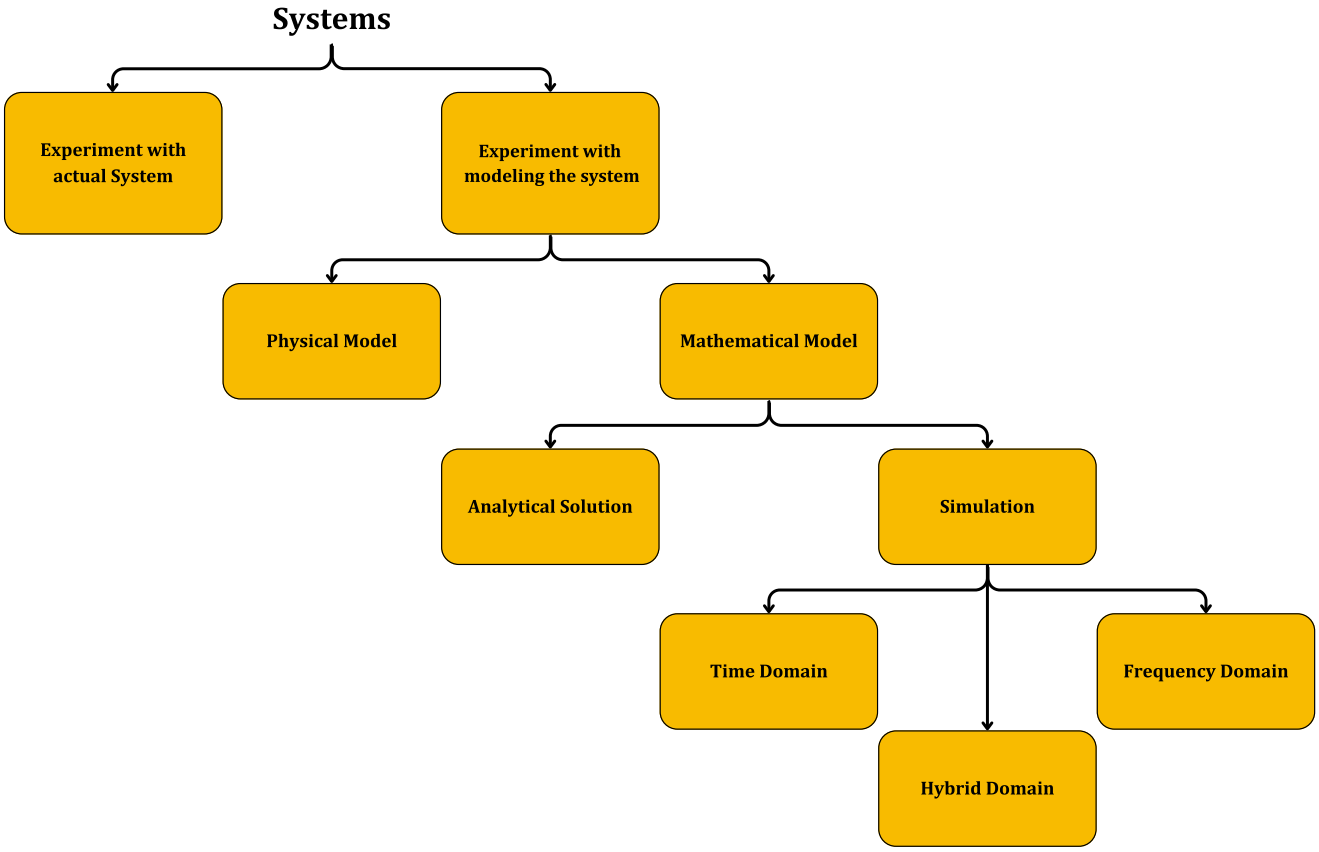
- It's similar with the idea of DC and AC; DC has an only one direction and one fixed value which is doesn't change over time **"static system"**, on the other side, AC has 2 directions (positive and negative) and its value changes over time with sine relation **"dynamic system"**.





- A system is said to be dynamic if its current output may depend on the past and the present values of the input variables.
- Input sometimes is represented by $u(t)$ or $r(t)$ instead of $x(t)$, don't get confused about $u(t)$ as input or as unit step function and $r(t)$ as input or as ramp function.
- Model: A simplified representation of reality.
- Reality is generally too complex to copy exactly and much of the complexity has no relation in problem solving.
- Mathematical Model: A set of differential equations that describes the input-output behavior of a system.

- To study a system, you would check it as following:



▶ Models are used for:

1. Simulation.
2. Control System Design.
3. Prediction/Forecasting.
4. Prognostics/Diagnostics.
5. Design/Performance Evaluation.

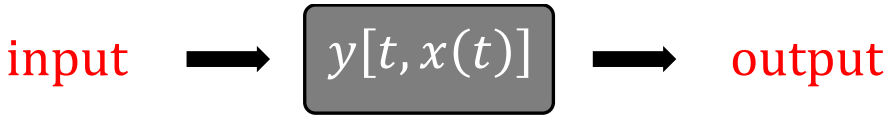
▶ Types of Models:

1. Black-Box Model:



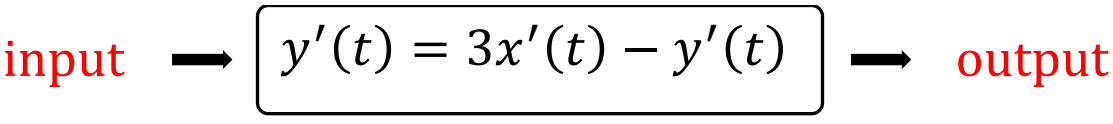
- Its properties:
  - A. The input and output are known.
  - B. The internal dynamics are unknown.
  - C. Easy to model.

2. Gray-Box Model:



- Its properties:
  - A. The input and output are known.
  - B. Some of the internal dynamics are known.
  - C. Harder than black box to model.

3. White-Box Model:



- Its properties:
  - A. The input and output are known.
  - B. The internal dynamics are known.
  - C. Hard to model.

Example:

- You have these processors:

1. CPU [Intel Core i7] **(Black Box)**:

- You know input and output, but you don't know how it processed since it closed source processor (too difficult).

input → CPU → output

2. SBC [Raspberry Pi 5] **(Gray Box)**:

- You know input, output and some of the internal dynamics (processing) by programing some of its component due to this SBC has a closed source component.

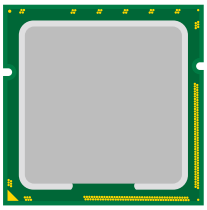
input → SBC → output

3. Microcontroller [Arduino UNO 3] **(White Box)**:

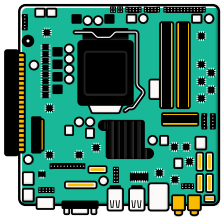
- You know input, output and the internal dynamics (processing) by programing it as you want.

input → UNO → output

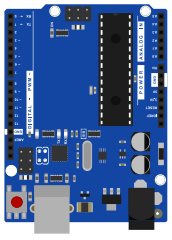
Intel Core i7



Raspberry Pi 5



Arduino UNO 3



Transfer Function:

- **Transfer Function:** A mathematical representation of the relationship between the input and output of LTI systems in the Laplace domain. Mathematically:

$$G(s) = \frac{\mathcal{L}\{\text{output function}\}}{\mathcal{L}\{\text{input function}\}}$$

- **Poles:** s values that makes the denominator of transfer function equal to 0.
- Poles can be real or complex and draw it in complex plane.

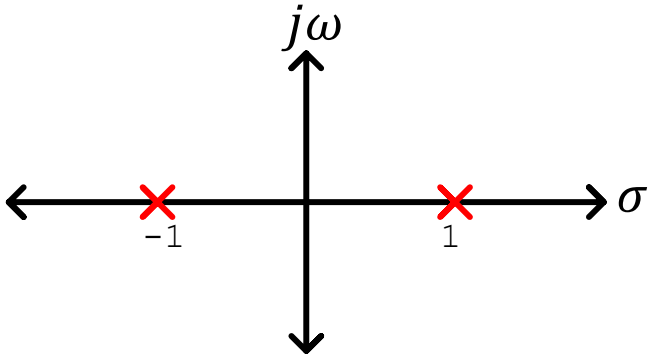
▶ **Q. 3.1:** If the input of a system is $R(s) = s^3 - s$ and the output is $C(s) = s$:

∴ Find the transfer function.

$$G(s) = \frac{C(s)}{R(s)} = \frac{s}{s^3 - s} = \frac{1}{s^2 - 1}$$

∴ Find and draw its poles.

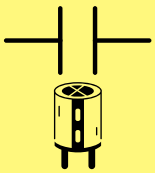
$$G(s) = \frac{1}{s^2-1} \rightarrow s^2 - 1 = 0,\ s = \pm 1$$



4. Electrical Modelling

Electrical Systems:

- Basic Elements of Electrical Systems:

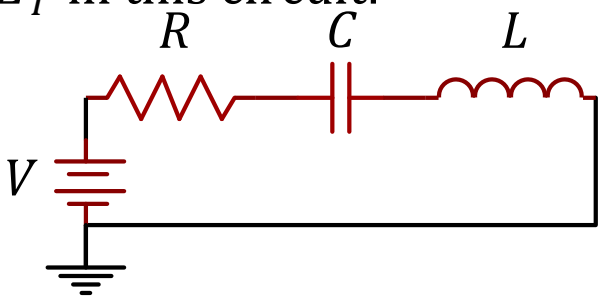
Element	Symbol	$v(t) - i(t)$ Relation	$V(s) - I(s)$ Relation	Z
Resistor R		$v_R(t) = R \cdot i_R(t)$	$V_R(s) = R \cdot I_R(s)$	R
Capacitor C		$v_C(t) = \frac{1}{C}\int i_C(t)\,dt$	$V_C(s) = \frac{I_C(s)}{C \cdot s}$	$\frac{1}{Cs}$
Inductor L		$v_L(t) = L\frac{di_L}{dt}$	$V_L(s) = L \cdot s \cdot I_L(s)$	Ls

- Transfer function for electrical systems is the output voltage divided by the input voltage, mathematically:

$$G(s) = \frac{V_o(s)}{V_i(s)}$$

- It's important to revision the impedance Z, that's due to find $V(s)$ if needed, and its name is <u>equivalent Laplace transform impedance</u> (Z_T).

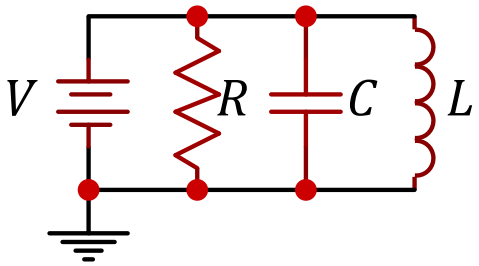
▶ **Q. 4.1:** Find Z_T in this circuit:



R, C and L are in series, so:

$$Z_T = Z_R + Z_C + Z_L = R + \frac{1}{Cs} + Ls$$

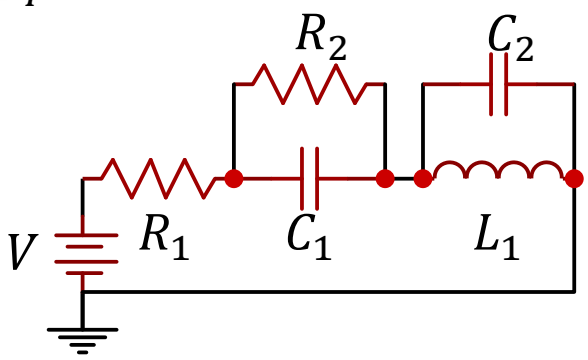
▶ **Q. 4.2:** Find Z_T in this circuit:



R, C and L are in parallel, so:

$$Z_T = \left(\frac{1}{Z_R} + \frac{1}{Z_C} + \frac{1}{Z_L}\right)^{-1} = \left(\frac{1}{R} + Cs + \frac{1}{Ls}\right)^{-1} = \frac{1}{\frac{1}{R} + Cs + \frac{1}{Ls}}$$

▶ Q. **4.3**: Find Z_T in this circuit:



As we see, Z_{C_1} is in parallel with Z_{R_2} [name it $Z_1 = Z_{C_1}||Z_{R_2}$], and Z_{L_1} is in parallel with Z_{C_2} [name it $Z_2 = Z_{L_1}||Z_{C_2}$], then $Z_{R_1} + Z_1 + Z_2$ are in series.

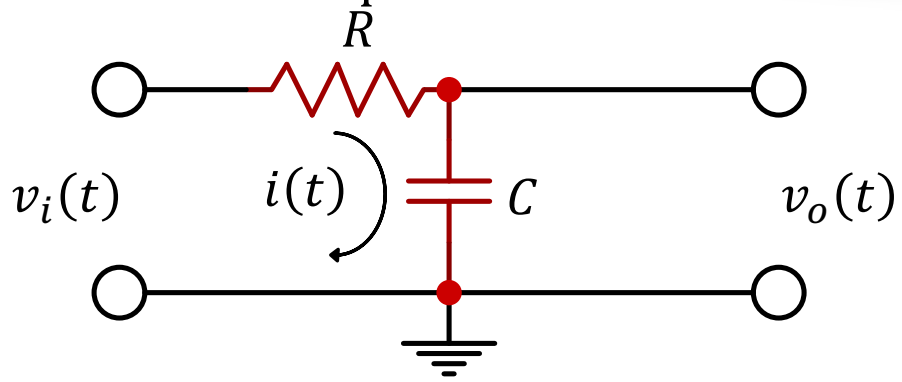
Thus, it would be $Z_{R_1} + Z_1 + Z_3$, so:

$$\begin{aligned} Z_T &= Z_{R_1} + (Z_{C_1}||Z_{R_2}) + (Z_{L_1}||Z_{C_2}) \\ &= R_1 + \left(\frac{1}{C_1 s}||R_2\right) + \left(L_1 s||\frac{1}{C_2 s}\right) \\ &= R_1 + \left(C_1 s + \frac{1}{R_2}\right)^{-1} + \left(\frac{1}{L_1 s} + C_2 s\right)^{-1} \end{aligned}$$

After simplifying, we conclude that:

$$Z_T = R_1 + \frac{R_2}{1 + C_1 R_2 s} + \frac{L_1 s}{1 + C_2 L_1 s^2}$$

▶ **Q. 4.4:** From this two-port network shown bellow:



Find:

∴ $v_i(t)$ and $v_o(t)$.

By KVL:

$$v_i(t) = v_R(t) + v_C(t) = Ri(t) + \frac{1}{C}\int i(t)\,dt$$

$$v_o(t) = v_C(t) = \frac{1}{C}\int i(t)\,dt$$

∴ $V_i(s)$ and $V_o(s)$.

$$V_i(s) = V_R(s) + V_C(s) = RI(s) + \frac{I(s)}{Cs}$$

$$V_o(s) = V_C(s) = \frac{I(s)}{Cs}$$

$\therefore$ Transfer function $G(s)$.

From previous info:

$$G(s) = \frac{V_o(s)}{V_i(s)} = \frac{\frac{I(s)}{Cs}}{RI(s) + \frac{I(s)}{Cs}} = \frac{\frac{1}{Cs}}{R + \frac{1}{Cs}} = \frac{1}{RCs + 1}$$

$\therefore$ Design this electrical system that would place a pole at -4.

Finding pole equation:

$$\frac{1}{RCs + 1} = \frac{1}{0} \rightarrow RCs + 1 = 0 \rightarrow s = -\frac{1}{RC}$$

We know that $s = -4$, so:

$$-4 = -\frac{1}{RC} \rightarrow RC = \frac{1}{4}$$

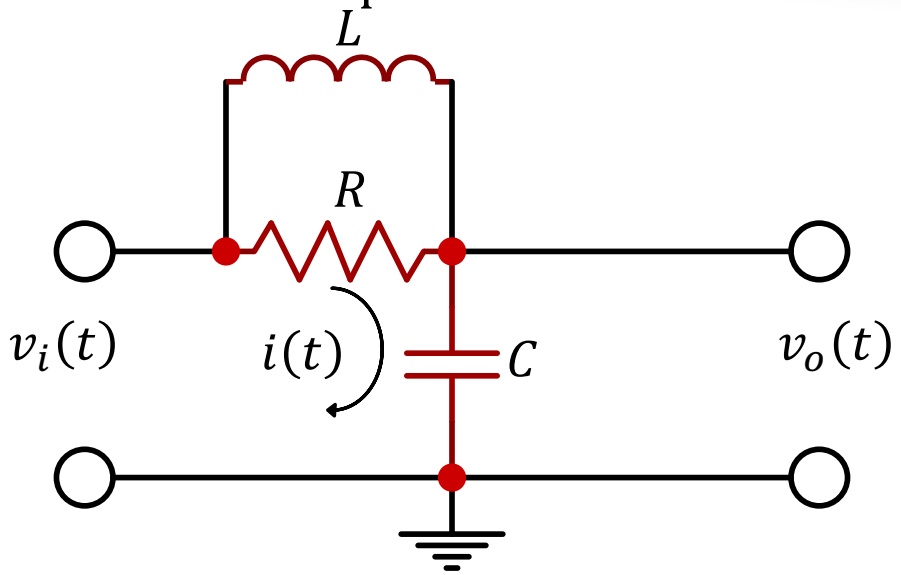
Let any value for R or C then find the other one, let assume that $R = 1\text{k}\Omega$, so:

$$1\text{k} \cdot C = \frac{1}{4} \rightarrow C = 250\mu\text{F}$$

So R and C values for design:

$$R = 1\text{k}\Omega, \qquad C = 250\mu\text{F}$$

▶ **Q. 4.5:** From this two-port network shown bellow:



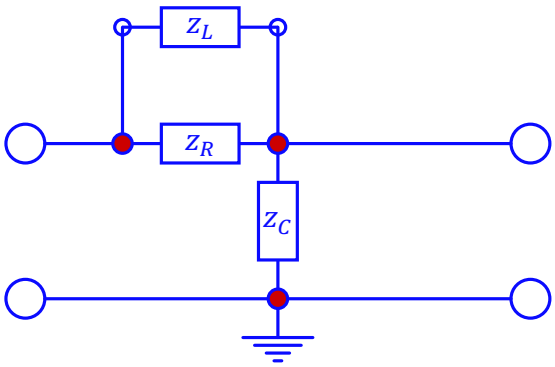
Find:

∴ Equivalent transfer impedance Z_T.

Simplify network by replacing components with their equivalent transform Z. First between R and L Then with C:

$$Z_T = (Z_R||Z_L) + Z_C = \left(\frac{1}{Z_R} + \frac{1}{Z_L}\right)^{-1} + Z_C = \left(\frac{1}{R} + \frac{1}{Ls}\right)^{-1} + \frac{1}{Cs}$$

$$= \frac{RLs}{R + Ls} + \frac{1}{Cs}$$



$\therefore$ $V_i(s)$ and $V_o(s)$.

Now we can find $V_i(s)$ using $V_i(s) = Z_{RLC} \cdot I_i(s)$, so:

$$V_i(s) = \left(\frac{RLs}{R + Ls} + \frac{1}{Cs}\right) I_i(s)$$

$V_i(s)$ is the voltage on the capacitor $V_C(s)$, so:

$$V_o(s) = \frac{I_i(s)}{Cs}$$

$\therefore$ $G(s)$.

From previous info:

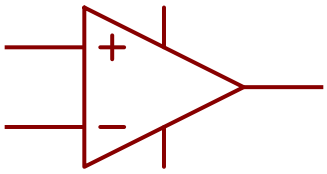
$$\frac{V_o(s)}{V_i(s)} = \frac{\frac{I_i(s)}{Cs}}{\left(\frac{RLs}{R + Ls} + \frac{1}{Cs}\right) I_i(s)} = \frac{1}{Cs}\left(\frac{1}{\frac{RLs}{R + Ls} + \frac{1}{Cs}}\right)$$

After simplifying:

$$G(s) = \frac{R + Ls}{RLCs^2 + R + Ls}$$

◈ Operational Amplifiers:

• **Operational Amplifier (Op-Amp):** An electronic circuit used to amplify signals. It is widely used in filters, comparators, and signal processing. It has three main terminals:



∴ V^+: Non-inverting input.

∴ V^-: Inverting input.

• When powered correctly, an Op-Amp follows the principle of negative feedback, meaning it tries to make the voltage difference between V^+ and V^- close to zero.

▶ We will study about these 2 amplifires:

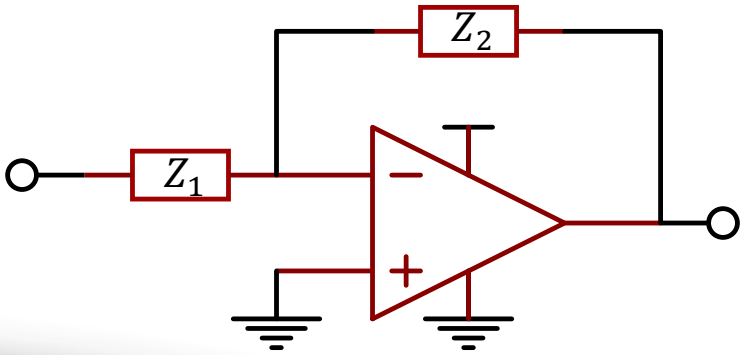
1. Inverting Amplifier:

• The input is connected to the inverting terminal V^-,which causes the output signal to be inverted and amplified.

• Its transfer function:

$$\frac{V_o(s)}{V_i(s)} = -\frac{Z_2}{Z_1}$$

• The negative sign means the output is inverted.

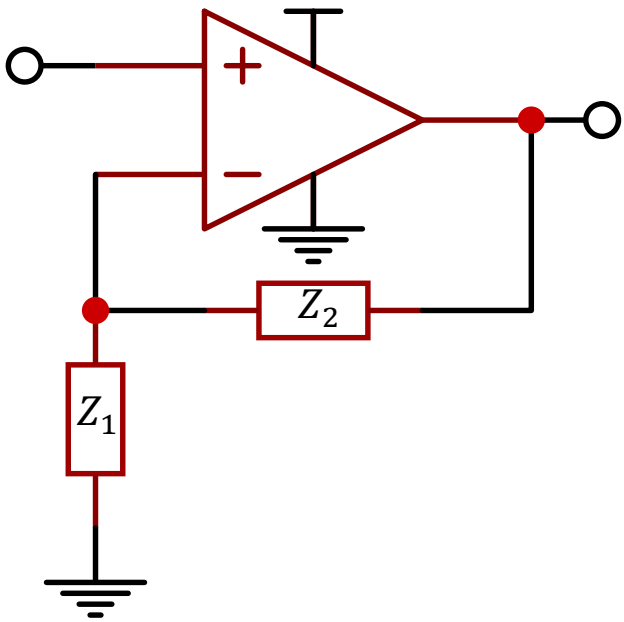


2. Non-Inverting Amplifier

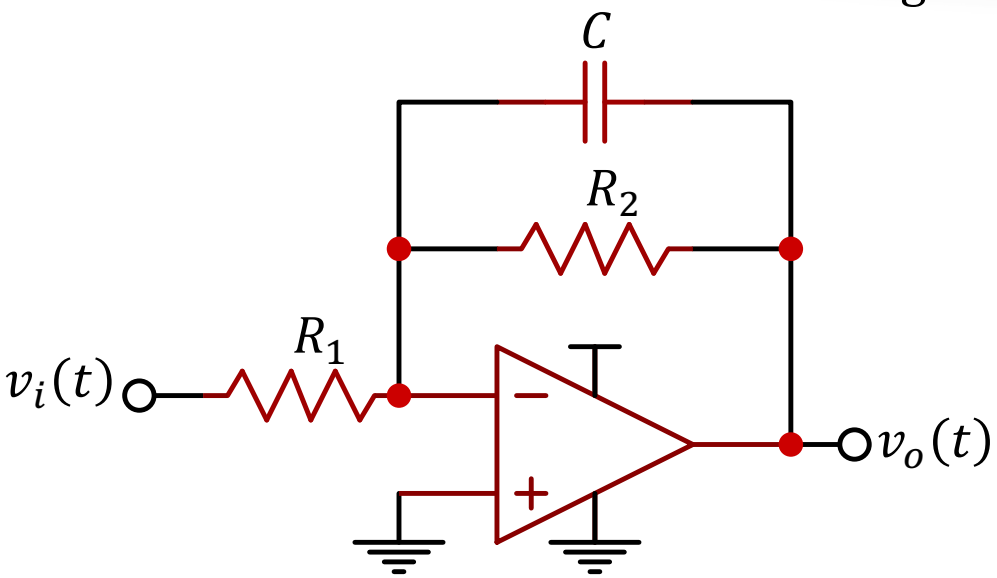
- The input is connected to the non-inverting terminal V^+, meaning the output is amplified without inversion.
- Its transfer function:

$$\frac{V_o(s)}{V_i(s)} = 1 + \frac{Z_2}{Z_1}$$

- There is no negative sign, so the output keeps the same polarity as the input.



▶ **Q. 4.6: Find transfer function of the following circuit.**

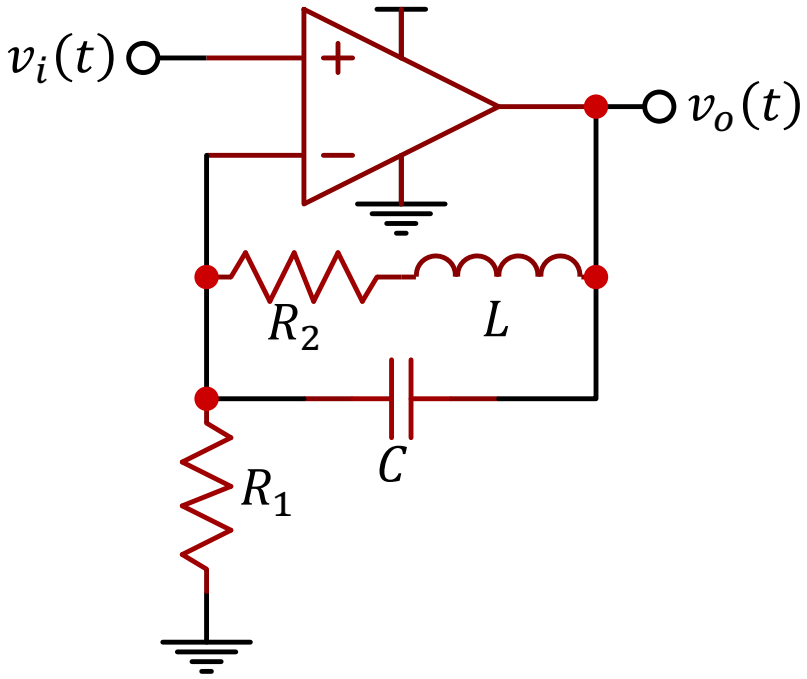


∴ *solution*:

$$\frac{V_o(s)}{V_i(s)} = -\frac{Z_2}{Z_1} = -\frac{(Z_{R_2} \| Z_C)}{Z_{R_1}} = -\frac{\left(\frac{1}{R_2} + Cs\right)^{-1}}{R_1}$$

$$= \frac{-R_2}{R_1(1 + R_2 Cs)}$$

▶ Q. 4.7: Find transfer function of the following circuit.

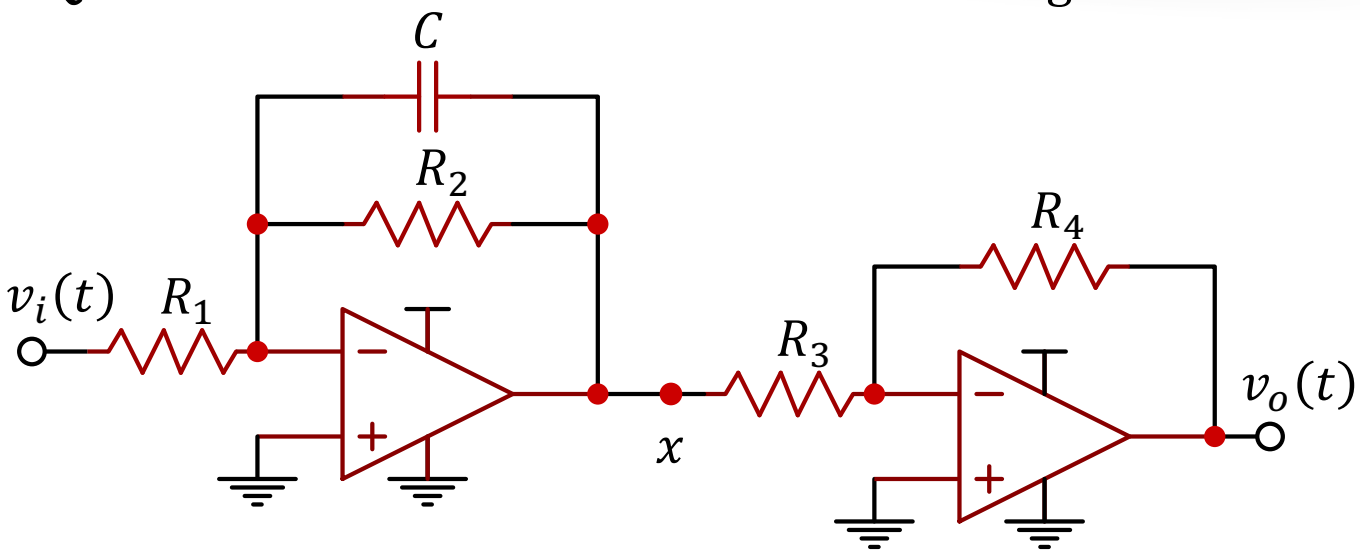


$\therefore$ *solution*:

$$\frac{V_o(s)}{V_i(s)} = 1 + \frac{Z_2}{Z_1} = 1 + \frac{\left((Z_{R_2} + Z_L)||Z_C\right)}{Z_{R_1}}$$

$$= 1 + \frac{\left(\frac{1}{R_2 + Ls} + Cs\right)^{-1}}{R_1} = 1 + \frac{R_2 + Ls}{R_1(1 + (R_2 + Ls)Cs)}$$

▶ **Q. 4.8:** Find transfer function of the following circuit.



∴ *solution*:

As we see, we have 2 Op-Amps, so we do this operation 2 times, the first one between V_i with x point as an output, then between x as an input with V_o:

$$\frac{x}{V_i} = -\frac{Z_2}{Z_1} = -\frac{(Z_{R_2} || Z_C)}{Z_{R_1}} = -\frac{R_2}{R_1(1+R_2Cs)}$$

$$\frac{V_o}{x} = -\frac{Z_4}{Z_3} = -\frac{R_4}{R_3}$$

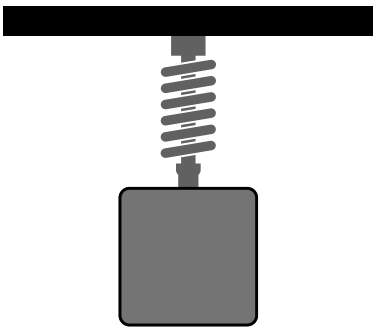
$$\frac{V_o}{V_i} = \frac{V_o}{x} \cdot \frac{x}{V_i} = \frac{-R_4}{R_3} \cdot \frac{-R_2}{R_1(1+R_2Cs)} = \frac{R_4R_2}{R_3R_1(1+R_2Cs)}$$

5. Mechanical Modelling

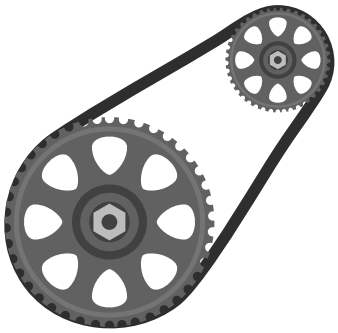
▶ Types of Mechanical Systems:

1. **Translational System:** A system with linear motion (occurs in a straight line).
2. **Rotational System:** A system with rotational motion (occurs around a fixed center).
3. **Linkage System:** A system consisting of interconnected rigid bodies that transfer motion and force through joints using gears.

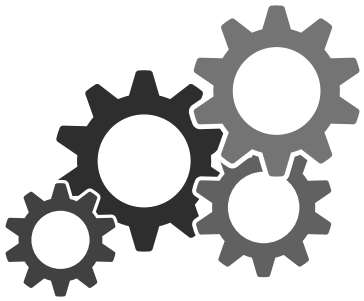
Translational



Rotational

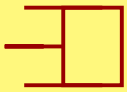
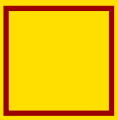


Linkage



Translational Mechanical System:

- Basic Elements of Translational Mechanical Systems:

Element	Symbol	Definition	Force
Translational Spring		A mechanical element that can be deformed by an external force such that the deformation is directly proportional to the force applied to it.	$f(t) = K \cdot x(t)$ $K \equiv \frac{\mathrm{N}}{\mathrm{m}}$
Translational Damper		A mechanical element used to dissipate energy and reduce oscillations, vibrations, or shocks.	$f(t) = B \cdot x'(t)$ $B \equiv \frac{\mathrm{N}}{\mathrm{ms}^{-1}}$
Translational Mass		A mechanical element and without it the system does not exist.	$f(t) = M \cdot x''(t)$ $M \equiv \frac{\mathrm{N}}{\mathrm{ms}^{-2}}$

- The external force $f(t)$ is the input of a system and we get the distance $x(t)$ as output.
- K, B and M are constants for each $f(t)$ law above.
- Damper (or dashpot in other words) and friction forces have almost same laws but the constant B and C are different ($B \rightarrow$ Damper constant, $C \rightarrow$ Friction constant):

$$f_{Damper} = B \cdot x'(t), \qquad f_{Friction} = C \cdot x'(t)$$

- Since the force $f(t)$ is the input and the distance $x(t)$ is the output, the transfer function for mechanical systems is:

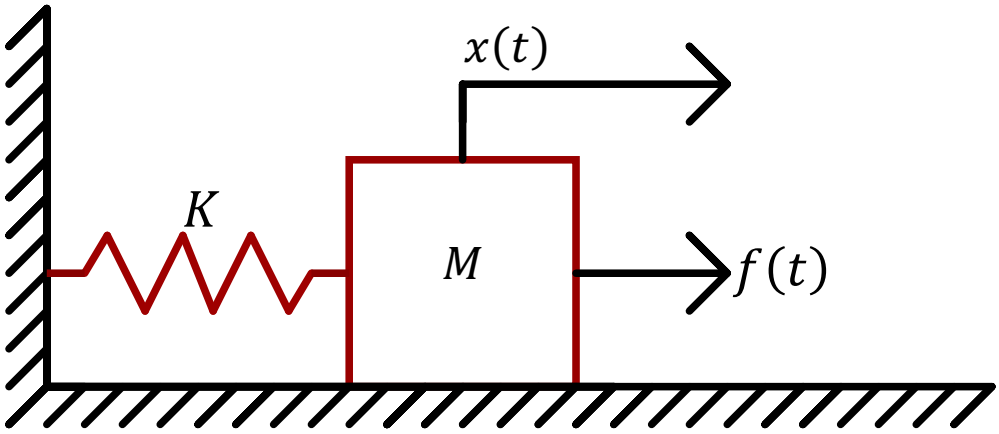
$$G(s) = \frac{X(s)}{F(s)}$$

- It's important to remember that the acceleration a is the second derivative of the distance $x(t)$.
- So, we can rewrite newton's second law as:

$$\sum f_{\text{on } M} = f_M(t) = M \cdot a = M \cdot x''(t)$$

- To know $\sum f_m$ easily, draw a free body diagram for the system and draw arrows for each force.

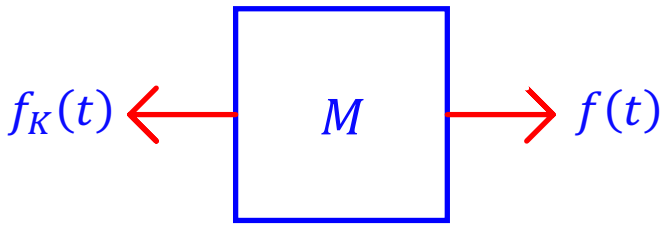
▶ **Q. 5.1:** Consider a simple horizontal spring-mass system on a frictionless surface, as shown below:



Find:

∴ Transfer function $\frac{X(s)}{F(s)}$.

As we see, we have a body with mass M connected with a spring that have a constant K then we affect to it with force $f(t)$, it will move a distance $x(t)$, so to get transfer function, lets draw a free body diagram to understand it easily:



From newton's second law:

$$\sum f_m(t) = Mx''(t) \rightarrow f(t) - f_K(t) = Mx''(t)$$

$$f(t) - Kx(t) = Mx''(t) \rightarrow f(t) = Mx''(t) + Kx(t)$$

Then, take the Laplace transform:

$$F(s) = Ms^2X(s) + KX(s)$$

Then transfer function is:

$$\frac{X(s)}{F(s)} = \frac{X(s)}{Ms^2X(s) + KX(s)} = \frac{1}{Ms^2 + K}$$

$\therefore$ If $M = 50$ kg and $K = 200\,\frac{\text{N}}{\text{m}}$, find the system's poles.

$$50s^2 + 200 = 0 \rightarrow s = \sqrt{-\frac{200}{50}}$$

$$s = \pm 2j$$

$\therefore$ Design this system that would place a pole at $\pm 4j$.

$$\sqrt{-\frac{K}{M}} = \pm 4j \rightarrow \sqrt{\frac{K}{M}} = 4 \rightarrow \frac{K}{M} = 16 \rightarrow K = 16M$$

Let $M = 1\text{kg}$, so:

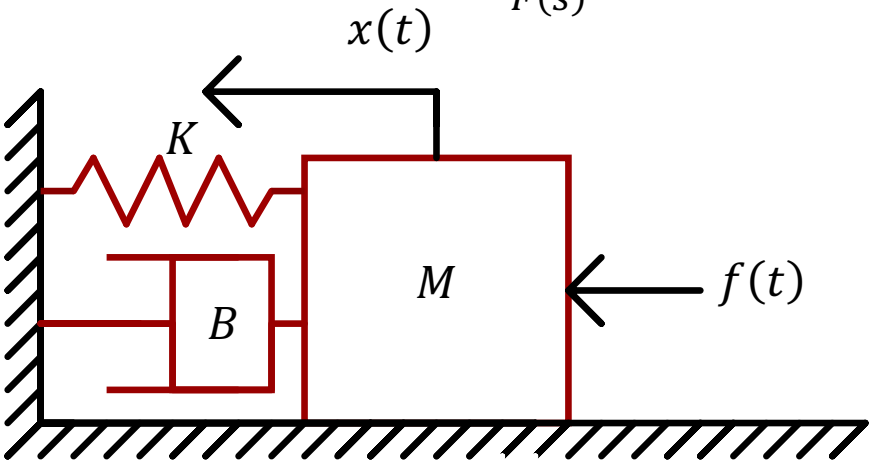
$$M = 1\text{kg}, \qquad K = 16\,\frac{\text{N}}{\text{m}}$$

- We will shortcut functions of t and write it without "(t)":

$$x(t) \rightarrow x, \qquad f(t) \rightarrow f$$

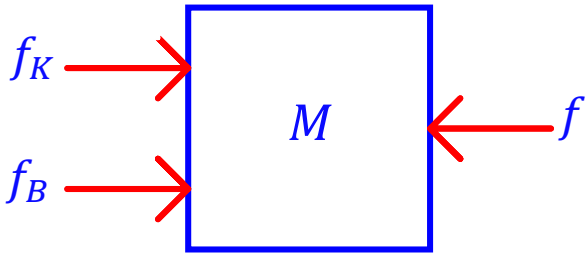
- We will shortcut Newton's second law by $\sum f = f_M = Mx''$

▶ **Q. 5.2:** Find transfer function $\frac{X(s)}{F(s)}$ for this system:



$\therefore$ *solution*:

Free body diagram is:



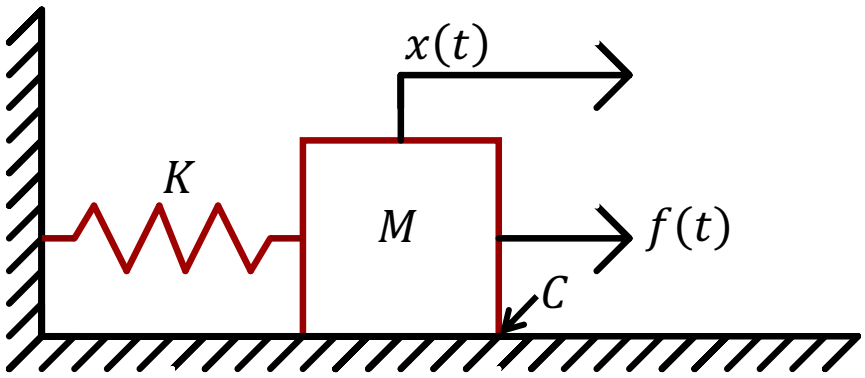
$$\sum f = f_M \rightarrow f - f_K - f_B = f_M \rightarrow f - Kx - Bx' = Mx''$$

$$\rightarrow f = Mx'' + Bx' + Kx$$

$$\rightarrow_{\mathcal{L}} F(s) = Ms^2X(s) + BsX(s) + KX(s)$$

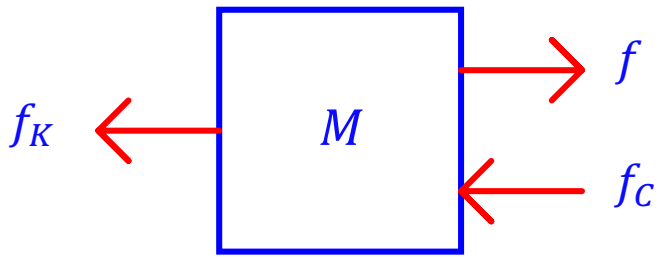
$$\frac{X(s)}{F(s)} = \frac{X(s)}{Ms^2X(s) + BsX(s) + KX(s)} = \frac{1}{Ms^2 + Bs + K}$$

▶ Q. 5.3: Find transfer function $\frac{X(s)}{F(s)}$ for this system:



$\therefore$ *solution*:

Friction force with constant C is acting on the opposite direction of $f(t)$, so free body diagram is:



$$\sum f = f_M \rightarrow f - f_K - f_C = f_M$$

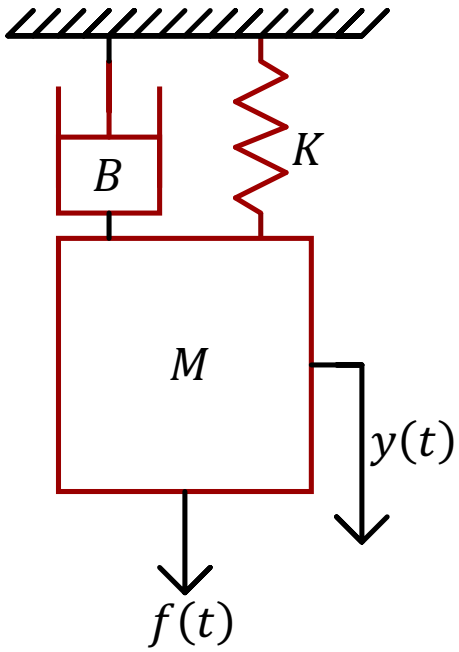
$$\rightarrow f - Kx - Cx' = Mx''$$

$$\rightarrow f = Mx'' + Cx' + Kx$$

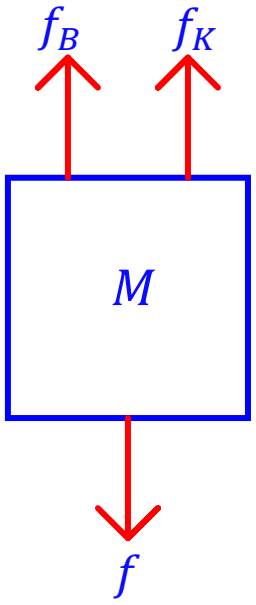
$$\mathcal{L} \rightarrow F(s) = Ms^2X(s) + CsX(s) + KX(s)$$

$$\frac{X(s)}{F(s)} = \frac{X(s)}{Ms^2X(s) + CsX(s) + KX(s)} = \frac{1}{Ms^2 + Cs + K}$$

▶ **Q. 5.4:** Find transfer function $\frac{Y(s)}{F(s)}$ for this system:



$\therefore$ *solution*:



As we solve the previous questions on $x-axis$, there are no differences if you want to find transfer function on $y-axis$

$$\sum f = f_M \rightarrow f - f_K - f_B = f_M$$

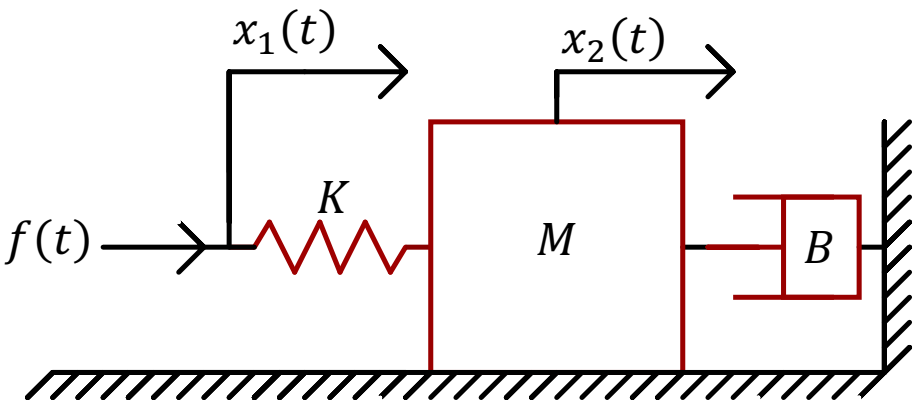
$$f - Ky - By' = My''$$

$$\rightarrow f = My'' + Cy' + Ky$$

$$\mathcal{L} \rightarrow F(s) = Ms^2Y(s) + BsY(s) + KY(s)$$

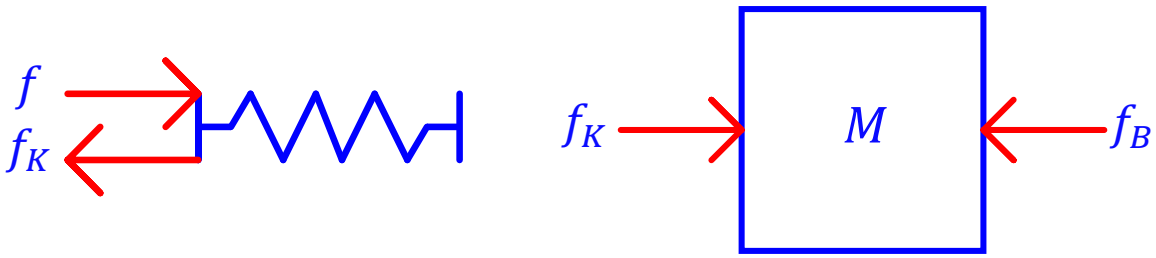
$$\frac{Y(s)}{F(s)} = \frac{Y(s)}{Ms^2Y(s) + BsY(s) + KY(s)} = \frac{1}{Ms^2 + Bs + K}$$

▶ Q. 5.5: Model this system using differential equation:



$\therefore$ *solution*:

Depending on newton's third law we say the forces f and f_K are equal in values but opposite in direction.



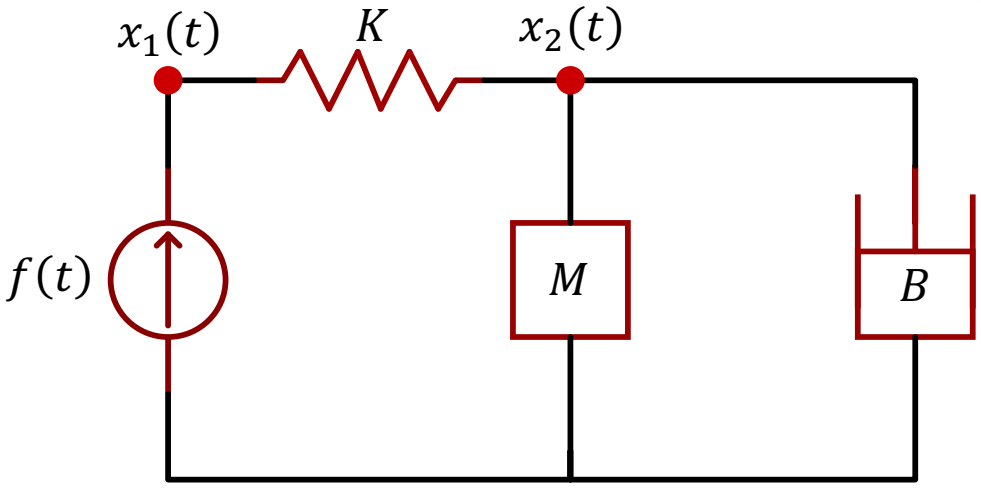
The force f is acting on the spring directly, it will be compressed on x_1 node, then it will expand at node x_2,when we say f_K it is the difference that spring is compressed or get expanded $(x_1 - x_2)$, as we see:

$$f = f_K \rightarrow f = K(x_1 - x_2), \qquad \mathcal{L} \rightarrow F(s) = K\big(X_1(s) - X_2(s)\big)$$

$$f_K - f_B = f_M \rightarrow Mx'' + Bx' + K(x_1 - x_2) = 0$$

$$\mathcal{L} \rightarrow Ms^2X_2(s) + BsX_2(s) - K\big(X_1(s) - X_2(s)\big) = 0$$

- We can describe that system with another way:

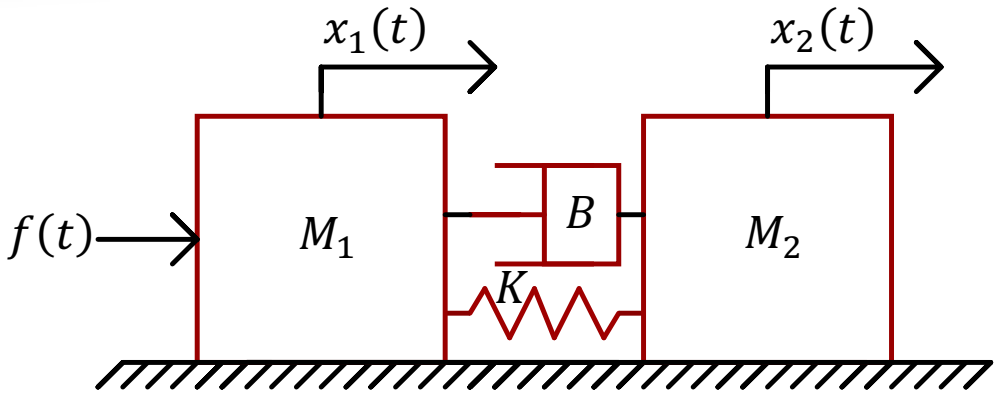


- This is called mechanical network. It's similar with electrical network, as we see spring K between node $x_1(t)$ and $x_2(t)$, the force is directly acting on $x_1(t)$ so it will be compressed at this node and will expand on $x_2(t)$ causing pushing force on the mass that compress the dashpot causing opposite forces f_M and f_B.

▶ **Why it $x_1 - x_2$ not $x_2 - x_1$?**

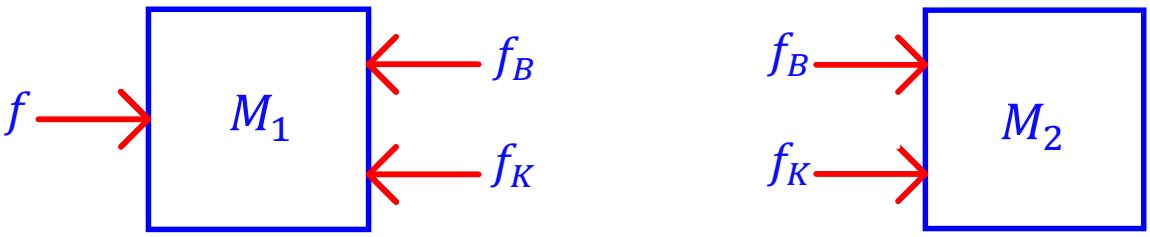
- This is because x_1 is always greater than x_2, to understand it easily, assume that the value of mass M is very high, if you acting on the spring on x_1 it will move depending on the force you act and assume it moved, but on x_2 the moving distance will be very short and approaching to zero.
- So, you subtract the longer distance from the shorter one to keep it positive.

▶ **Q. 5.6:** Model this system using the differential equation:



$\therefore$ *solution*:

The spring K and dashpot B will be compressed and expand by $x_1 - x_2$, so in free body diagram:



on M_1 : $f - f_K - f_B = f_{M_1}$

$$\rightarrow f = M_1 x_1'' + B(x_1' - x_2') + K(x_1 - x_2)$$

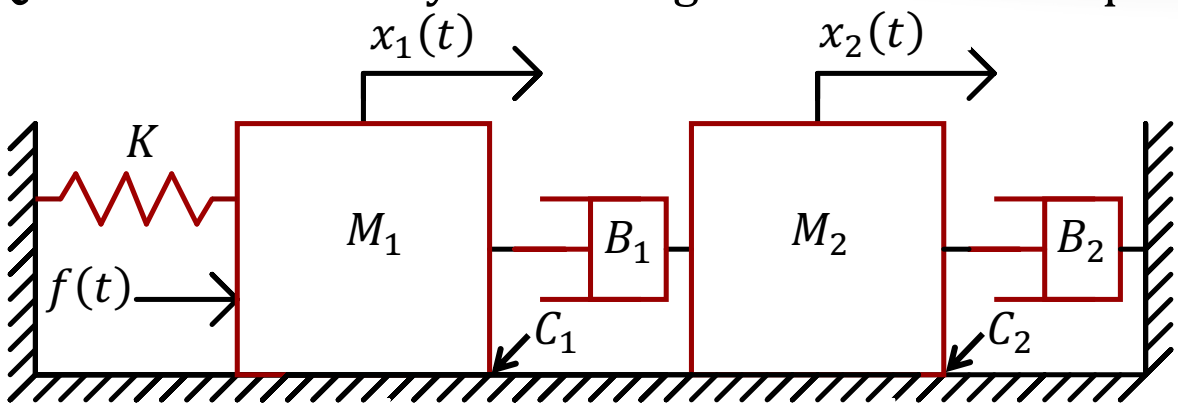
$$\mathcal{L} \rightarrow F(s) = M_1 s^2 X_1(s) + Bs\big(X_1(s) - X_2(s)\big) + K\big(X_1(s) - X_2(s)\big)$$

on M_2 : $f_K + f_B = f_{M_2}$

$$\rightarrow B(x_1' - x_2') + K(x_1 - x_2) = M_2 x_2''$$

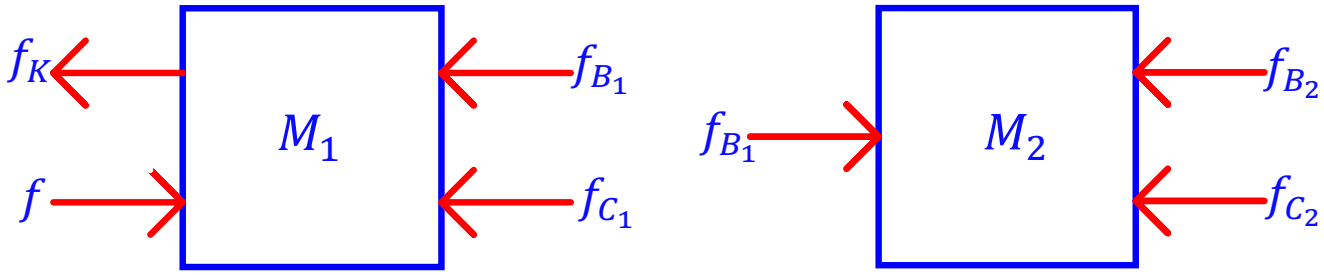
$$\mathcal{L} \rightarrow Bs\big(X_1(s) - X_2(s)\big) + K\big(X_1(s) - X_2(s)\big) = M_2 s^2 X_2(s)$$

▶ **Q. 5.7:** Model this system using the differential equation:



$\therefore$ *solution*:

Free body diagrams:



on M_1 : $f - f_K - f_{B_1} - f_{C_1} = f_{M_1}$

$$\rightarrow f = M_1 x_1'' + B_1(x_1' - x_2') + C_1(x_1' - x_2') + K(x_1 - x_2)$$

$$\mathcal{L} \rightarrow F(s) = M_1 s^2 X_1(s) + (B_1 + C_1)s\big(X_1(s) - X_2(s)\big) + K\big(X_1(s) - X_2(s)\big)$$

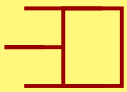
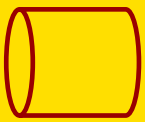
on M_2 : $f_{B_1} - f_{B_2} - f_{C_2} = f_{M_2}$

$$\rightarrow B_1(x_1' - x_2') - B_2(x_1' - x_2') - C_2(x_1' - x_2') = M_2 x_2''$$

$$\mathcal{L} \rightarrow (B_1 - B_2 - C_2)s\big(X_1(s) - X_2(s)\big) = M_2 s^2 X_2(s)$$

Rotational Mechanical System:

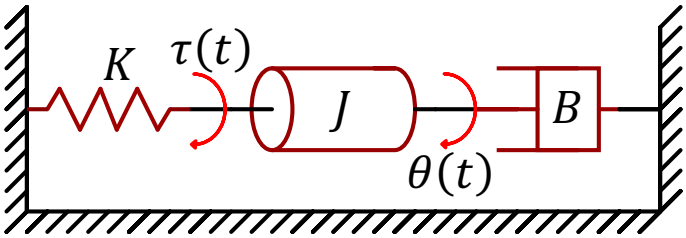
- Basic Elements of Rotational Mechanical Systems:

Element	Symbol	Torque
Rotational Spring		$\tau(t) = K \cdot \theta(t)$ $K \equiv \frac{\mathrm{N}}{\mathrm{m}}$
Rotational Damper		$\tau(t) = B \cdot \theta'(t)$ $B \equiv \frac{\mathrm{N}}{\mathrm{ms}^{-1}}$
Rotational Inertia		$\tau(t) = J \cdot \theta''(t)$ $J \equiv \frac{\mathrm{N}}{\mathrm{ms}^{-2}}$

- As we learn before how to model the translation mechanical system, the rotational mechanical system is not much different from translation one.
- Torque is the ability of rotate in rotational system as we know the ability to move a body is force.
- K, B and J are constants.
- There are no force, distance and mass in these systems, because these definitions aren't enough to describe rotational motion.

Translational System		Rotational System	
Force	$f(t), F(s)$	Torque	$\tau(t), T(s)$
Distance	$x(t), X(s)$	Degree	$\theta(t), \phi(s)$
Mass	M	Inertia	J

▶ **Q. 5.8:** Find transfer function $\frac{\phi(s)}{T(s)}$ for this system:

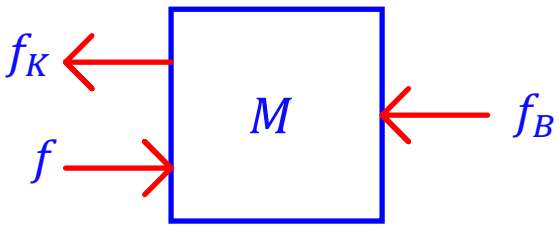


$\therefore$ *solution*:

We can solve it by assuming it as a translational system and solving it as we do before then replace $f \to \tau,\ x \to \theta,\ M \to J$ or drawing free body diagram for this system in rotational moves and continue as we do before.

1. As a translational system:

Free body diagram:



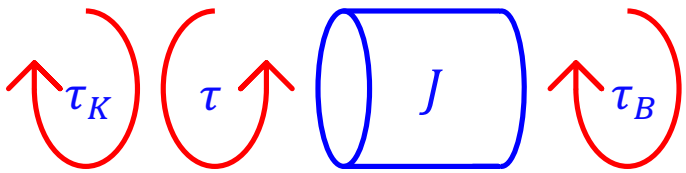
$$f_M = f - f_K - f_B$$

$$\to f = Mx'' + Bx' + Kx \ \to\ \tau = J\theta'' + B\theta' + K\theta$$

$$\mathcal{L} \to T(s) = Js^2\phi(s) + Bs\phi(s) + K\phi(s)$$

2. As a rotational system:

Free body diagram:

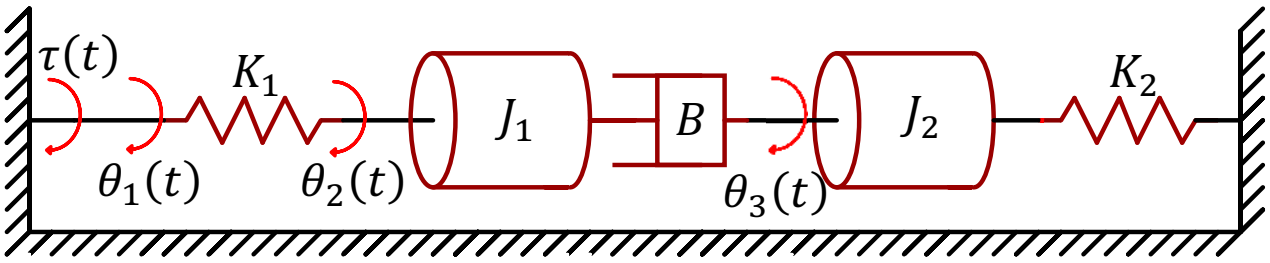


$$\tau_J = \tau - \tau_K - \tau_B$$

$$\to\ \tau = J\theta'' + B\theta' + K\theta$$

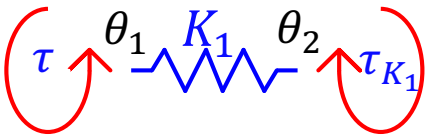
$$\mathcal{L} \to T(s) = Js^2\phi(s) + Bs\phi(s) + K\phi(s)$$

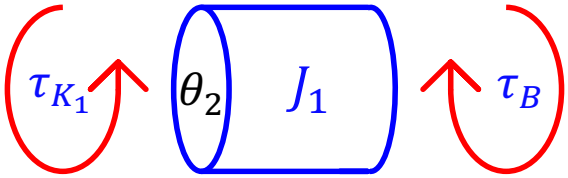
▶ Q. 5.9: Model this system using differential equations:

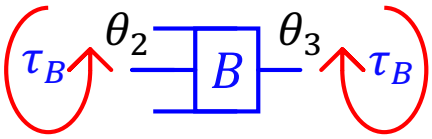


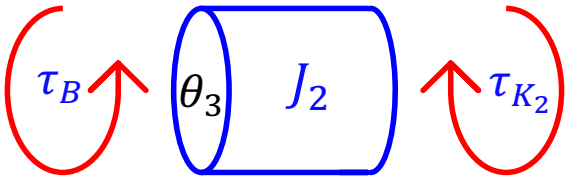
∴ *solution*:

Free body diagrams:









$$\tau = \tau_{K_1} = K(\theta_1 - \theta_2), \qquad \mathcal{L} \rightarrow T(s) = K_1\left(\phi_1(s) - \phi_2(s)\right)$$

$$\text{on } J_1: \tau_{J_1} = \tau_{K_1} - \tau_B \rightarrow \tau_{K_1} = \tau = \tau_{J_1} + \tau_B$$

$$\rightarrow \tau = J_1\theta_2'' + B(\theta_2' - \theta_3')$$

$$\mathcal{L} \rightarrow T(s) = J_1 s^2\phi_2(s) Bs\left(\phi_2(s) - \phi_3(s)\right)$$

$$\text{on } J_2: \tau_{J_2} = \tau_B - \tau_{K_2} \rightarrow \tau_{J_2} - \tau_B + \tau_{K_2} = 0$$

$$\rightarrow J_2\theta_3'' - B(\theta_2' - \theta_3') + K_2\theta_3 = 0$$

$$\mathcal{L} \rightarrow T(s) = Js^2\phi_2(s) - Bs(\phi_2 - \phi_3) + K_2(s) = 0$$

◈ Linkage Mechanical System:

• **Gear:** a toothed machine parts that mesh to transmit motion, change speed, or alter direction.

• When two gears interact, they rotate in opposite directions, and if they have different numbers of teeth, they revolve at different speeds.

▶ Key Points:

• The relationship between angular velocity ω and torque τ is inverse, meaning that as one increase, the other decreases when power is constant:

$$\omega \propto \frac{1}{\tau}$$

• Rotational power P is calculated as the product of ω and τ:

$$\omega = \frac{\theta}{t}, \qquad \tau = fd, \qquad P = \omega \cdot \tau$$

• When power is transferred between two gears, energy is conserved, so the product of angular velocity and torque in the first gear equals that in the second gear.

$$P_1 = P_2 \rightarrow \omega_1 \tau_1 = \omega_2 \tau_2$$

- Gear ratio help us to find the relation between input and output to find transfer function:

$$\text{Gear Ratio} \rightarrow \frac{N_1}{N_2} = \frac{\tau_1}{\tau_2} = \frac{\omega_2}{\omega_1} = \frac{\theta_2}{\theta_1}$$

- We note the driver gear by 1 and the follower by 2.

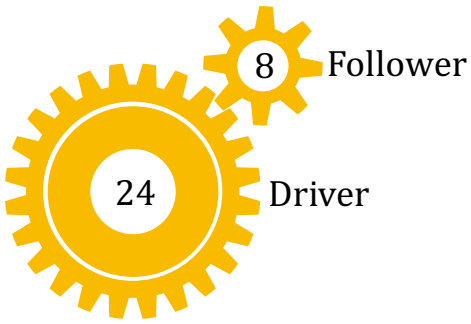
- Gearing up :increase velocity and decrease torque.

$$N_{Driver} > N_{Follower}$$

$$N_1 > N_2 \rightarrow \tau_1 > \tau_2 \rightarrow \omega_2 > \omega_1$$

Like in this system the ratio is:

$$24 : 8 \rightarrow 3 : 1$$



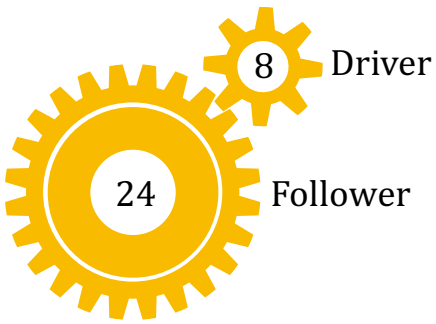
- Gearing down: increase torque and decrease velocity.

$$N_{Driver} < N_{Follower}$$

$$N_1 < N_2 \rightarrow \tau_1 < \tau_2 \rightarrow \omega_2 < \omega_1$$

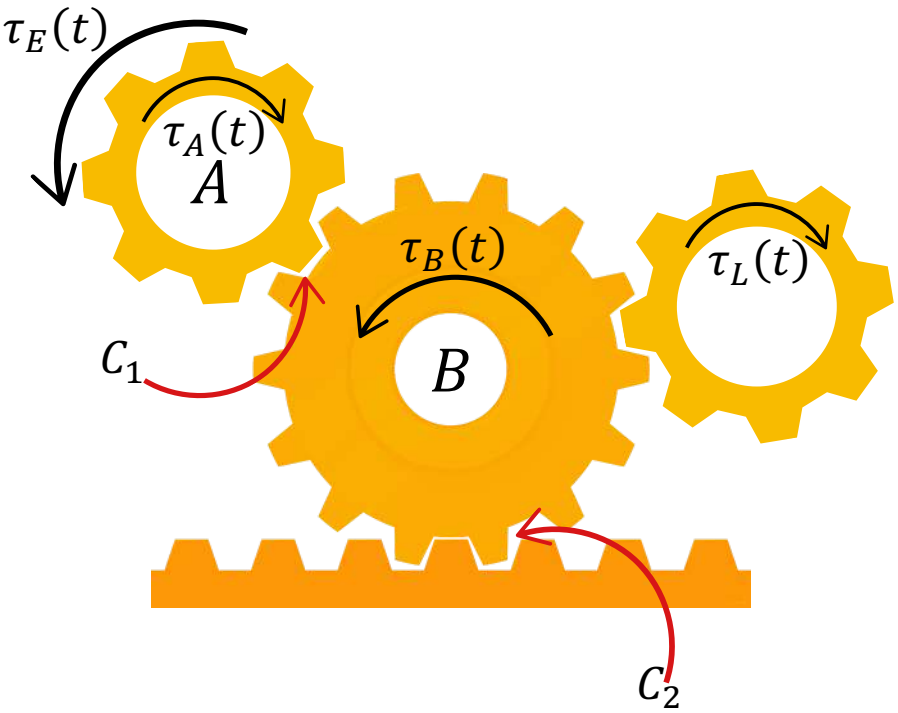
Like in this system the ratio is:

$$8 : 24 \rightarrow 1 : 3$$



- DC motors operate at high speeds and low torques, making gear reduction essential.

▶ **Q. 5.10:** Model this system that defined between driver and follower gears (A, B in series) using differential equations:



As we see, there is an external torque acting on the gear A, this torque will generate a torque has the opposite direction of the external one, and there is a friction C_1 between the gears A and B:

$$\text{sum of } \tau = \text{External } \tau - \text{Reverse } \tau - \text{Frictional } \tau$$

So, from Newton's second law:

$$J_1\theta_1'' = \tau_E - \tau_A - \tau_{C_1}$$

$$\rightarrow J_1\theta_1'' = \tau_E - \tau_A - C_1\theta'_1 \rightarrow \tau_E = J_1\theta_1'' + C_1\theta'_1 + \tau_A$$

$$\mathcal{L} \rightarrow T_E(s) = J_1 s^2\phi_1(s) + C_1 s\phi_1(s) + T_A(s)$$

Now looking on gear B, the total torque has the same direction of τ_B, in opposite direction we see the friction on C_2 and the load torque τ_L:

$$J_2\theta_2'' = \tau_B - \tau_{C_2} - \tau_L$$

$$J_2\theta_2'' = \tau_B - C_2\theta_2' - \tau_L \rightarrow \tau_B = J_2\theta_2'' + C_2\theta_2' + \tau_L$$

$$\mathcal{L} \rightarrow T_B(s) = J_2 s^2\phi_2(s) + C_2 s\phi_2(s) + T_L(s)$$

Finally, we can find a relation between $\theta_1 and \theta_2$ by this rule:

$$\tau_1\omega_1 = \tau_2\omega_2$$

$$N_1\theta_1 = N_2\theta_2 \rightarrow \theta_1 = \left(\frac{N_1}{N_2}\right)\theta_2$$

$$\mathcal{L} \rightarrow \phi_1(s) = \left(\frac{N_1}{N_2}\right)\phi_2(s)$$

So, modeling differential equations are:

$$T_E(s) = J_1 s^2\phi_1(s) + C_1 s\phi_1(s) + T_A(s)$$

$$T_B(s) = J_2 s^2\phi_2(s) + C_2 s\phi_2(s) + T_L(s)$$

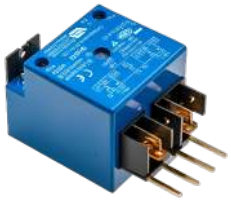
$$\phi_1(s) = \left(\frac{N_1}{N_2}\right)\phi_2(s)$$

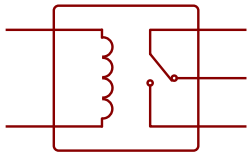
6. Electromechanical Modelling

- Electromechanical system: A system combines electrical and mechanical processes which carry out electrical operations by using moving parts.

▶ **Examples for Electromechanics:**

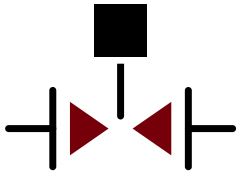
- Relay





- Solenoid



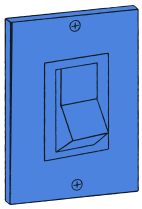


- DC Motor



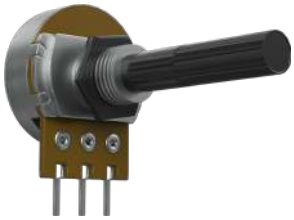


- Switch





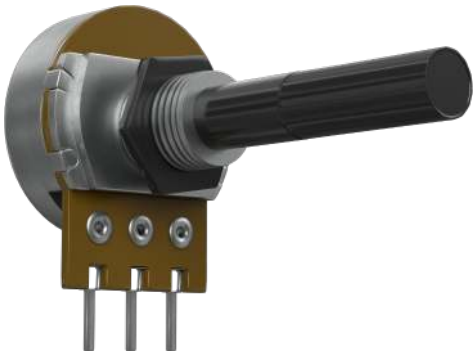
- Potentiometer





Potentiometer:

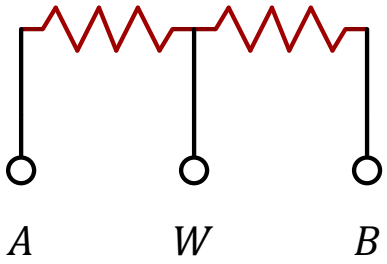
- Potentiometer: An adjustable resistor that control V and I by varying its value by sliding the wiper.
- It has three terminals: two connected to the ends of a resistive element (A and B) and one connected to a wiper that slides along the resistive path, allowing variable resistance (W).

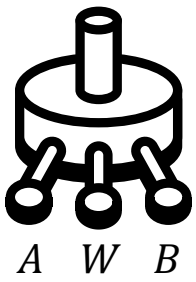


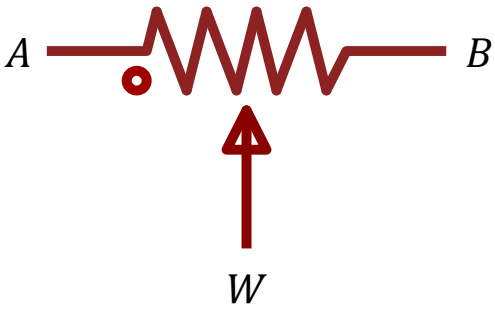
- The resistance between the wiper W and A is labeled R_1, the resistance between the wiper W and B is labeled R_2.
- The total resistance between A and B is constant:

$$R_T = R_1 + R_2$$

- You can assume it as a voltage divider like this:

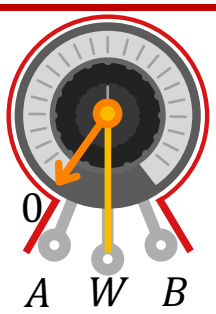






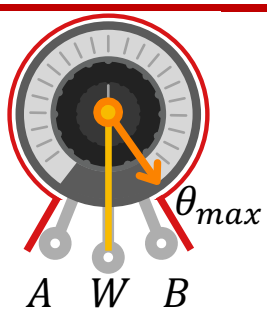
- If the wiper is turned to the extreme counterclockwise position such that W is touching A, in this position:

$$\theta = 0, \qquad R_1 = 0, \qquad R_2 = R_T$$



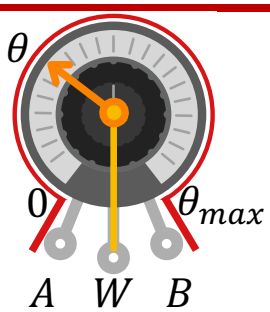
- If the wiper is turned to the extreme clockwise position such that W is touching B, in this position:

$$\theta = \theta_{max}, \qquad R_1 = R_T, \qquad R_2 = 0$$

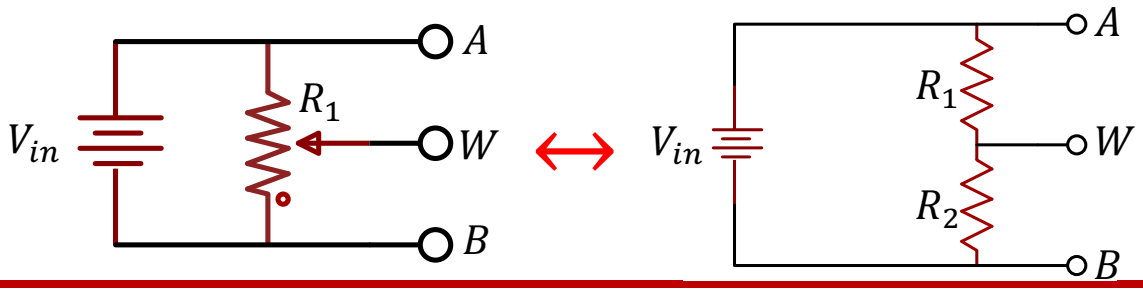


- R_1 and R_2 vary linearly with θ between the two extremes:

$$R_1 = \frac{\theta}{\theta_{max}} \cdot R_T, \qquad R_2 = \frac{\theta_{max} - \theta}{\theta_{max}} \cdot R_T$$



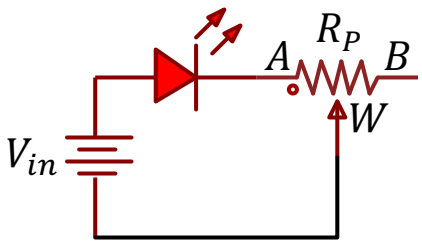
- Voltages between each terminal as VDR:



$$V_{AW} = V_{R_1} = V_{in} \cdot \frac{R_1}{R_T} \rightarrow V_1 = V_{in} \cdot \frac{\theta}{\theta_{max}}$$

$$V_{WB} = V_{R_2} = V_{in} \cdot \frac{R_2}{R_T} \rightarrow V_2 = V_{in} \cdot \frac{\theta_{max} - \theta}{\theta_{max}}$$

▶ **Q. 6.1:** In this circuit, this 1kΩ potentiometer has maximum degree that is 300° wiped from 120° to 200°:



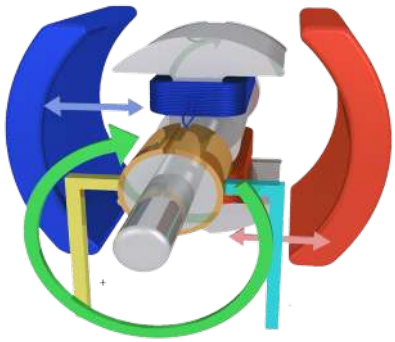
∴ Find $R_P|_{\theta=120°}$

$$R_P|_{\theta=120°} = 1\text{k} \cdot \frac{120°}{300°} = 1\text{k} \cdot \frac{2}{5} = 400\Omega$$

∴ Find $R_P|_{\theta=200°}$

$$R_P|_{\theta=200°} = 1\text{k} \cdot \frac{200°}{300°} = 1\text{k} \cdot \frac{2}{3} = 666\Omega$$

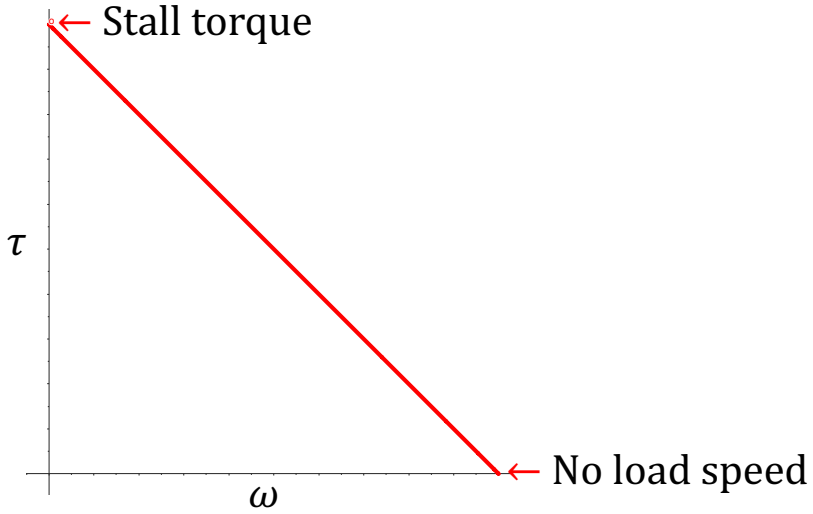
◈ DC Drives:



- Speed control can be achieved using DC drives in several ways:
- Armature control: A variable voltage can be applied to the armature terminals of the DC motor that involves adjusting the motor's armature.
- Field Control: to vary the flux per pole of the motor that involves adjusting the motor field.

▶ Motors characteristics:

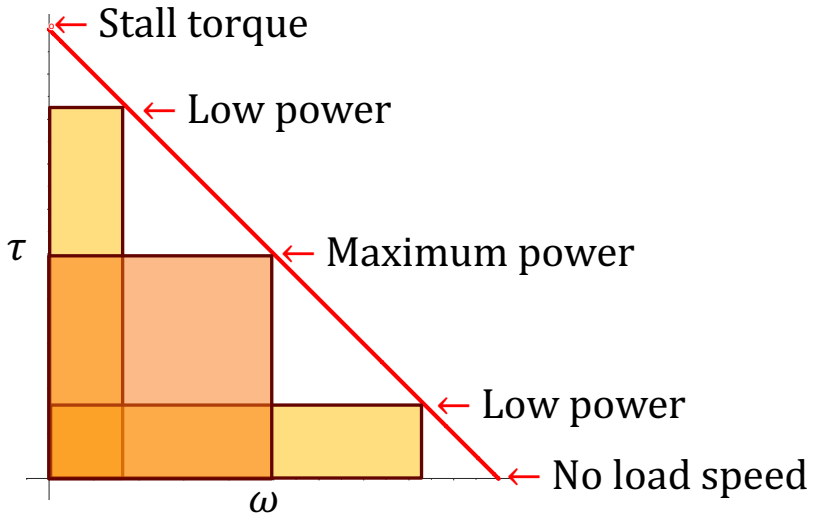
1. For every motor, there is a specific torque, speed and power curve that they are frequently given as two points on this graph:



2. Stall Torque: The point on the graph at which the torque is maximum, but the shaft is not rotating.
3. No Load Speed: The maximum speed of the motor.

- Because the power of this system is $P = \tau \cdot \omega$, so in this curve we see that the area is representing the power.
- As we see in this graph, we can obtain the maximum power P_{max} when we have the largest area by:

$$\tau = \frac{\tau_{stall}}{2}, \qquad \omega = \frac{\omega_{no\ load}}{2}$$



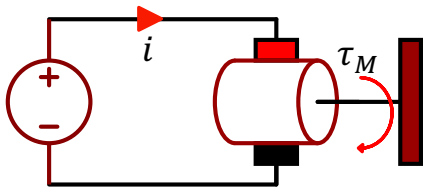
Armature-Controlled system:

- When a DC motor had an electrical power (V and I) it will move rotationally with conversation of power from electrical to mechanical causing a torque that given by:

$$\tau_M(t) = K_\tau \cdot i(t)$$

$\therefore$ $i(t)$ is the current that make the motor rotate.

$\therefore$ K_τ is a motor torque constant that $\equiv \text{N}\frac{\text{m}}{\text{A}}$.

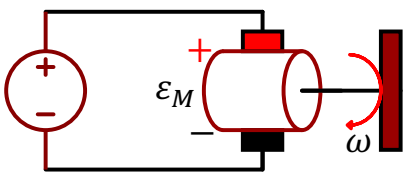


- When a DC motor had a mechanical power (τ and ω) it will make a charge over its terminals due to back electromotive force "EMF" with conversation of power from mechanical to electrical causing a back voltage that given by:

$$\varepsilon_M(t) = K_b \cdot \omega(t)$$

$\therefore$ $\omega(t)$ is the angular speed of the motor.

$\therefore$ K_b is a back EMF constant that $\equiv \text{V}\frac{\text{s}}{\text{rad}}$.

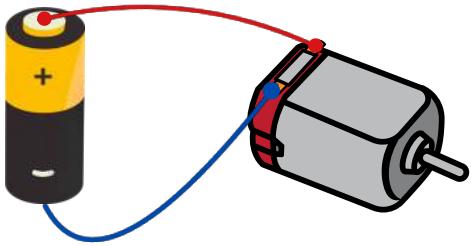


- In SI unit:

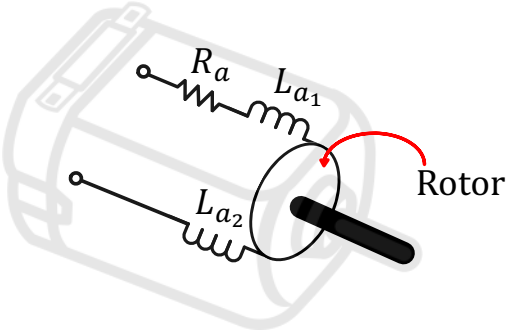
$$K_\tau \equiv K_b$$
$$\frac{\text{N}\cdot\text{m}}{\text{A}} \equiv \frac{\text{V}\cdot\text{s}}{\text{rad}}$$

Example:

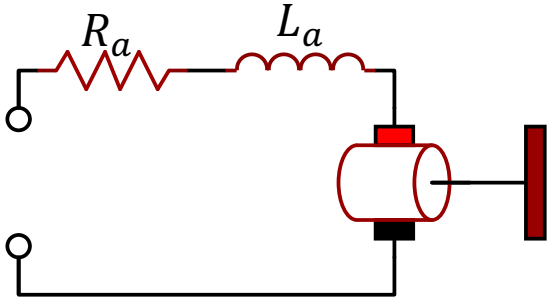
- Assume this simple motor system:



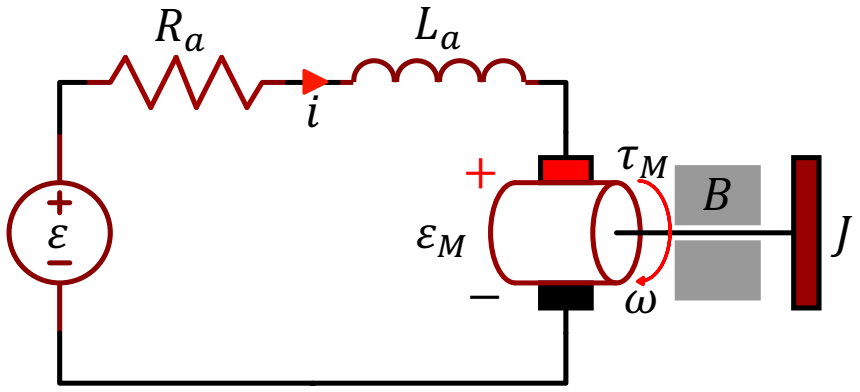
- while the motor is connecting to a voltage source, it will make a current that will enter to the motor.
- If we see the internal component of the motor, we will see that there are an internal resistor and inductors that make an electromagnetic induction to rotate the rotating part “rotor” of the motor as we see here:



- We can represent it like this:



▶ **Q. 6.2:** For this armature-controlled system:



Find:

$\therefore \frac{\Omega(s)}{E(s)}$

This system has an electrical and mechanical parts,
in electrical part, we can find its equation using KVL:

$$\varepsilon = V_{R_a} + V_{L_a} + \varepsilon_M \rightarrow \varepsilon(t) = R_a i(t) + L_a i'(t) + K_b \omega(t)$$

In mechanical part, we analyze it as we do before, but we make it with respect to ω instead of θ $[\omega = \theta']$ so:

$$J\theta'' = \tau_M - B\theta' \rightarrow J\theta'' = K_\tau i - B\theta'$$

$$\rightarrow i(t) = \frac{J\theta''(t) + B\theta'(t)}{K_\tau} \rightarrow i(t) = \frac{J\omega'(t) + B\omega(t)}{K_\tau}$$

Laplace transform those equations "make all of them has the function $I(s)$ only to delete them when we divide $\Omega(s)$ over $E(s)$":

$$\mathcal{L} \rightarrow E(s) = RI(s) + LsI(s) + K_b\Omega(s)$$

$$\mathcal{L} \rightarrow I(s) = \frac{(Js + B)\Omega(s)}{K_\tau} \rightarrow \Omega(s) = \frac{K_\tau I(s)}{(Js + B)}$$

Now divide the output $\Omega(s)$ by the input $E(s)$

"the solution must be with constants and s only":

$$\frac{\Omega(s)}{E(s)} = \frac{\Omega(s)}{R_a I(s) + L_a s I(s) + K_b \Omega(s)}$$

$$= \frac{\frac{K_\tau I(s)}{(Js + B)}}{\left(R_a I(s) + L_a s I(s) + K_b \frac{K_\tau I(s)}{(Js + B)}\right)}$$

$$= \frac{K_\tau}{(Js + B)\left(R_a + L_a s + K_b \frac{K_\tau}{(Js + B)}\right)}$$

$$= \frac{K_\tau}{(Js + B)R_a + (Js + B)Ls + K_b K_\tau}$$

$$= \frac{K_\tau}{JLs^2 + (BL + JR_a)s + BR_a + K_b K_\tau}$$

$\therefore \frac{\phi(s)}{E(s)}$

$$\theta(t) = \int \omega(t)\, dt \rightarrow \phi(s) = \frac{\Omega(s)}{s}$$

$$\frac{\phi(s)}{E(s)} = \frac{\Omega(s)}{sE(s)} = \frac{K_\tau}{s(JL_a s^2 + (BL_a + JR_a)s + BR_a + K_b K_\tau)}$$

◈ Field-Controlled System:

- This system has 2 circuits:

∴ Field circuit f.

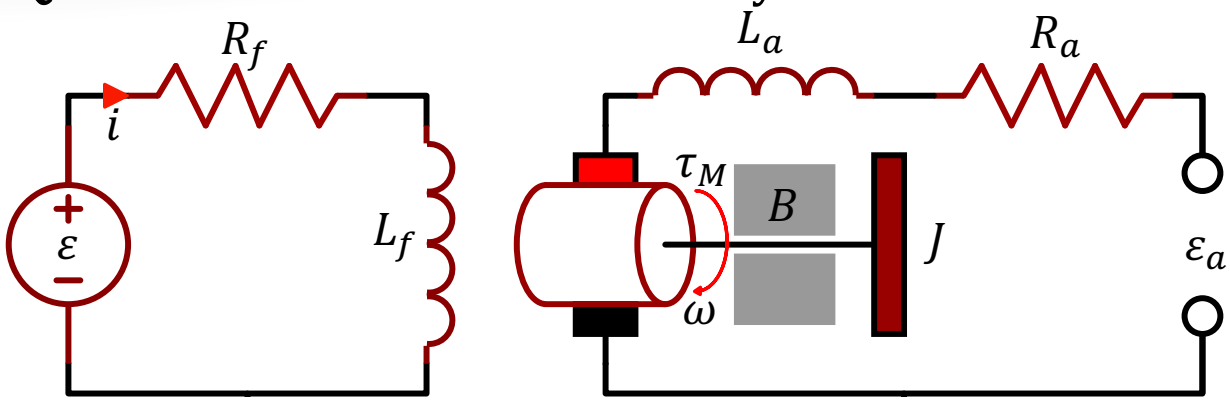
∴ Armature circuit a.

- Field Circuit has the input voltage ε that make a current through into the inductor causing a magnetic field. That magnetic field lines will cross with rotor that rotate it.
- The torque of the motor is created due to magnetic field that depending on the current of field circuit.

▶ Key Notes:

1. The internal inductance in the armature circuit is negated usually "$L_a \approx 0$".
2. It's not necessary that the current that rotate the motor be inside the motor, as we will see in in the follower question; the current i entering L_f makes magnetic field that make the rotor of motor rotate.
3. The transfer function must be with constants and s only, so make all of them has the function $I(s)$ only to delete them when we divide $\Omega(s)$ or $\phi(s)$ over $E(s)$.

▶ Q. 6.3: For this field-controlled system:



Find $\frac{\Omega(s)}{E(s)}$ and $\frac{\phi(s)}{E(s)}$.

Electrical part analysis:

$$\varepsilon = V_{R_f} + V_{L_f} \rightarrow \varepsilon(t) = R_f i(t) + L_f i'(t)$$

Mechanical part analysis:

$$J\theta'' = \tau_M - B\theta' \rightarrow J\theta'' = K_\tau i - B\theta'$$

$$\rightarrow i(t) = \frac{J\theta''(t) + B\theta'(t)}{K_\tau} \rightarrow i(t) = \frac{J\omega'(t) + B\omega(t)}{K_\tau}$$

Laplace transform:

$$\mathcal{L} \rightarrow E(s) = R_f I(s) + L_f s I(s)$$

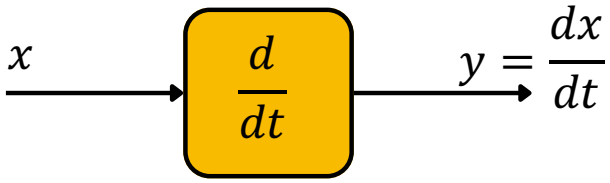
$$\mathcal{L} \rightarrow I(s) = \frac{(Js + B)\Omega(s)}{K_\tau} \rightarrow \Omega(s) = \frac{K_\tau I(s)}{(Js + B)}$$

$$\frac{\Omega(s)}{E(s)} = \frac{\frac{K_\tau I(s)}{(Js + B)}}{R_f I(s) + L_f s I(s)} = \frac{K_\tau}{(Js + B)(L_f s + R_f)}$$

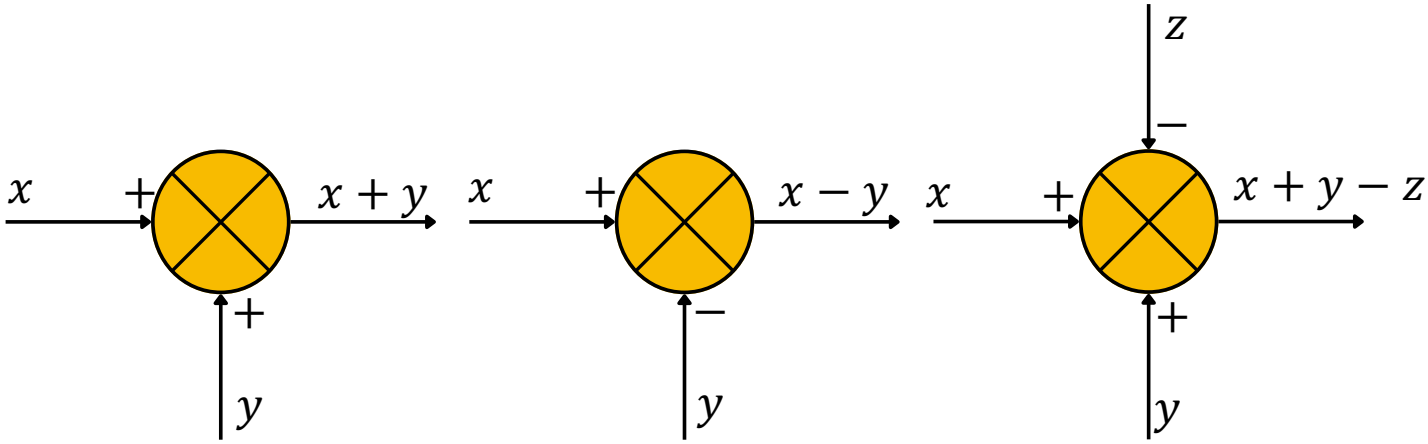
$$\frac{\phi(s)}{E(s)} = \frac{\Omega(s)}{sE(s)} = \frac{K_\tau}{s(Js + B)(L_f s + R_f)}$$

7. Block Diagrams

- **Block Diagram:** A shorthand pictorial representation of the cause-and-effect relation of a system.



- **The Rectangle** represents name or description of the element or symbol for the mathematical operation.
- **The Arrows** represent information direction or signal flow.
- **The Circle** (with cross or not) represents summing point and perform addition or subtraction and it known as **Adder.**
- The output is the algebraic sum of inputs.



- Takeoff points allow a signal to split and reach multiple destinations without alteration.



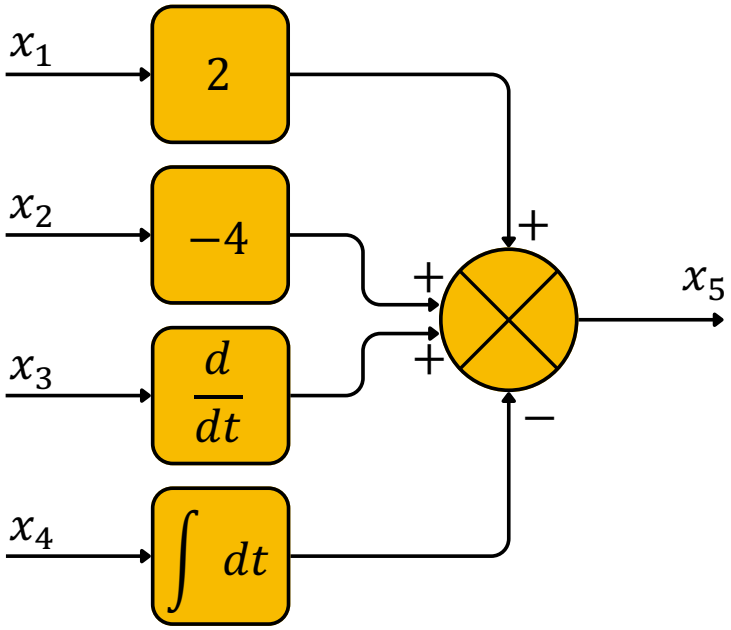
- To represent equations in this formula:

$$x_n = a_1x_1 \pm a_2x_2 \pm \cdots \pm a_{n-1}x_{n-1}$$

- We must know that:

x_n	→	the output.	→	Arrowhead
$x_1, x_2, \cdots, x_{n-1}$	→	the inputs.	→	Arrow tail
$a_1, a_2, \cdots, a_{n-1}$	→	the arithmetic operations	→	block

- Take this block diagram for example:



$$x_5 = 2x_1 - 4x_2 + \frac{dx_3}{dt} - \int x_4\, dt$$

▶ **Key Notes:**

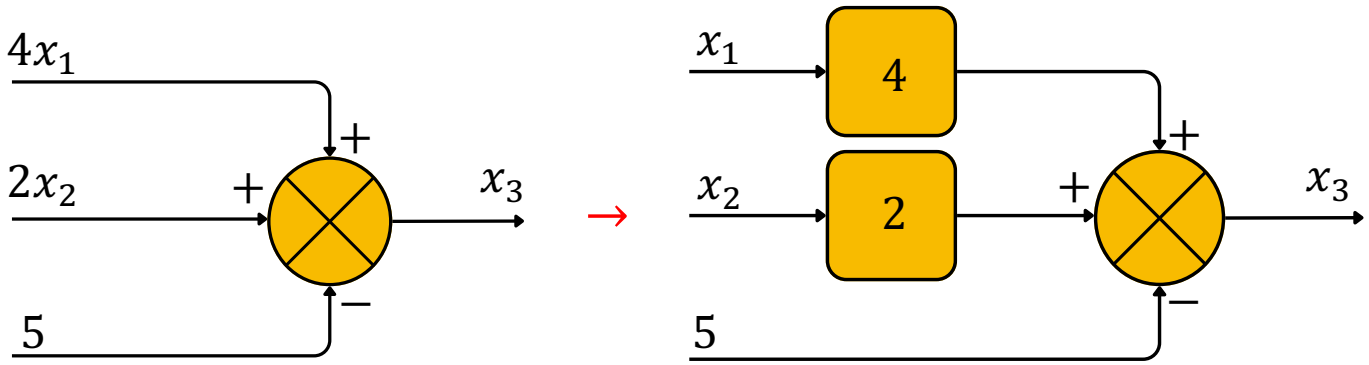
∴ Each arithmetic operation has 1 block only.

∴ The sign in the equation is depending on block's sign and comparator's sign $(+ * + = +,\ + * - = -,\ - * - = +)$.

▶ **Q. 7.1:** Represent this equation as a block diagram:

$$x_3 = 4x_1 + 2x_2 - 5$$

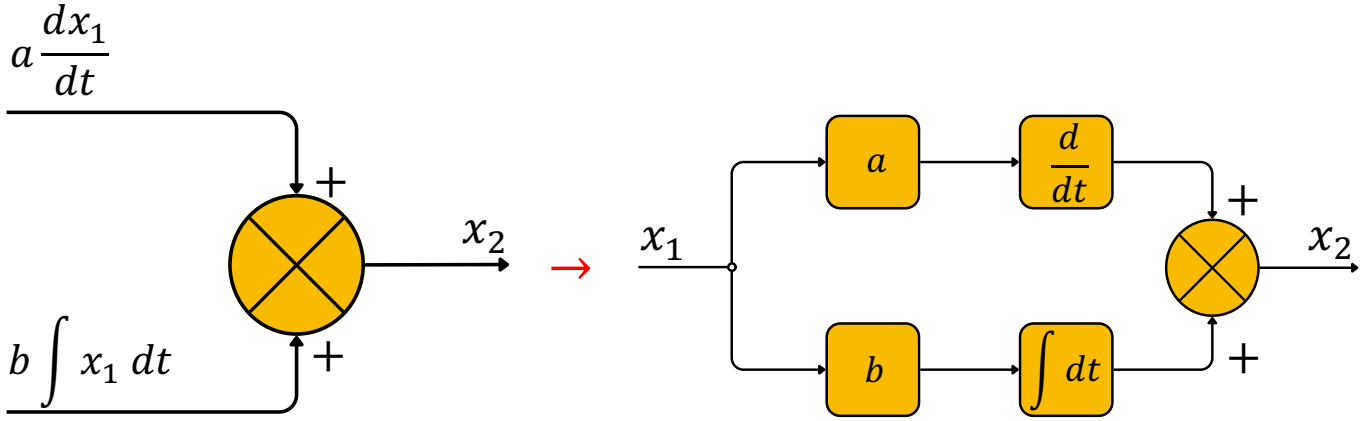
∴ *solution*:



▶ **Q. 7.2:** Represent this equation as a block diagram:

$$x_2 = a\frac{dx_1}{dt} + b\int x_1\,dt$$

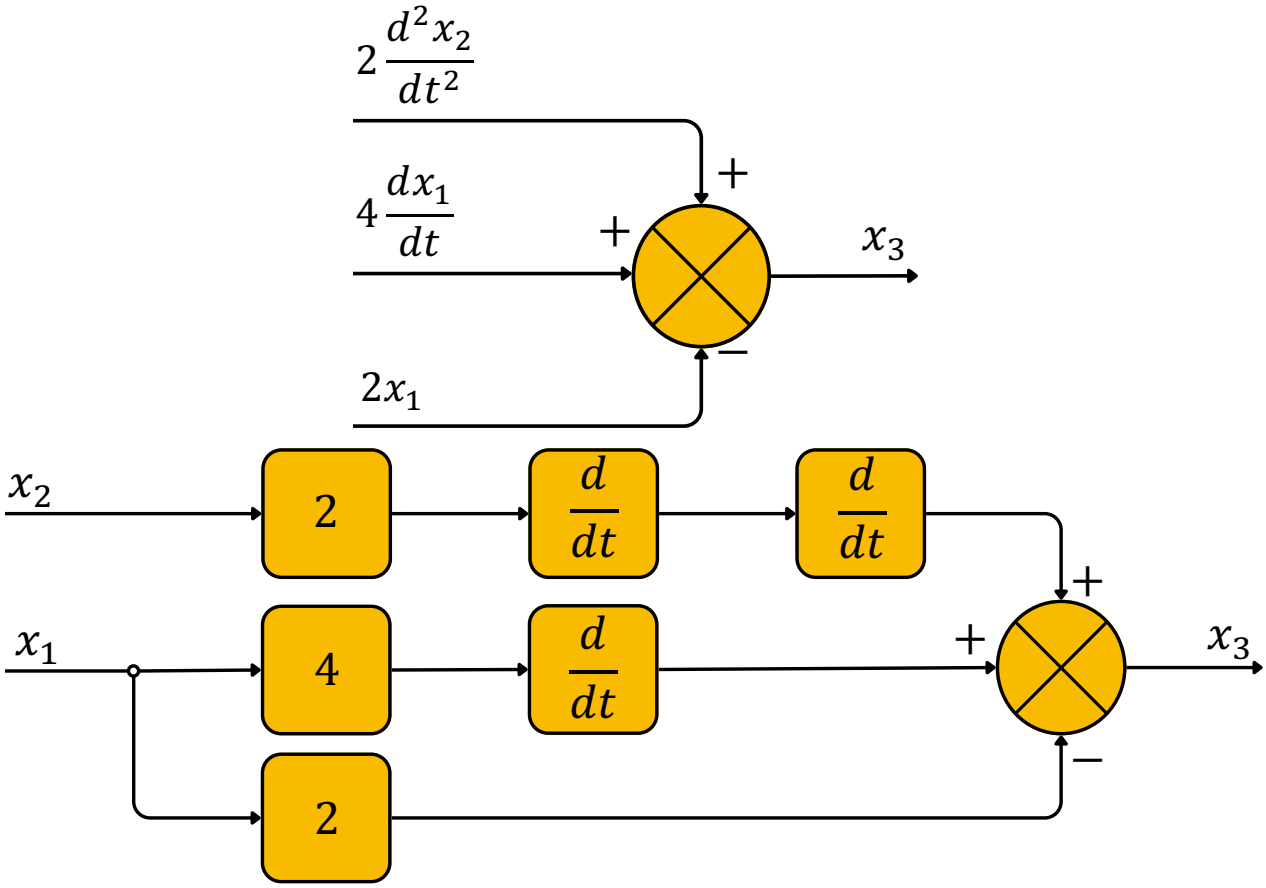
∴ *solution*:



▶ **Q. 7.3:** Represent this equation as a block diagram:

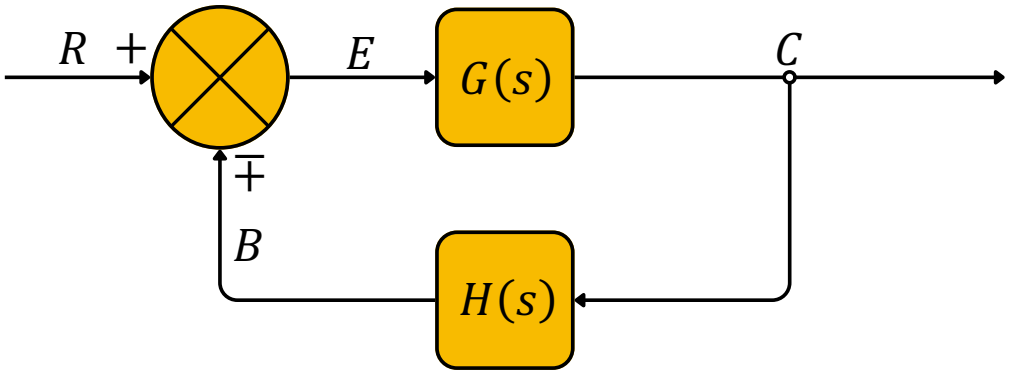
$$x_3 = 2\frac{d^2x_2}{dt^2} + 4\frac{dx_1}{dt} - 2x_1$$

∴ *solution*:



◈ Canonical Form:

- It given as this block diagram form:



- When:

∴ $G(s)$: Direct transfer function.

∴ $H(s)$: Feedback transfer function.

∴ R: Input of system "Reference".

∴ B: Output of feedback "Before Feedback Loop".

∴ E: Output of comparing R with B "Error".

∴ C: Output of system "Controlled Output".

▶ Key Notes:

- R, B, E and C are functions with respect to (s) "I ignore it in simplification and make it in $G(s)$ and $H(s)$ to sign that they are the main transfer functions of the system".
- The sign of B usually be $-$ and that known as negative feedback, I will assume that in all following rules that the main sign of B is $-$ in general.
- "DON'T FORGET THE SIGN!"

Important Rules:

- These relations help us to find any ratio.

$C = EG(s)$, $B = CH(s)$, $E = R \mp B \rightarrow R = E \pm B$

1. Control Ratio:

$$\frac{C(s)}{R(s)} = \frac{G(s)}{1 \pm G(s)H(s)}$$

- This ratio has a lot of names, like "Overall Transfer Function" or "Closed-Loop Transfer Function", and has a relation between the controlled output $C(s)$ divided by the reference $R(s)$, DON'T FORGET THE SIGN AGAIN!

$\pm \rightarrow$ Negative Feedback $\mp \rightarrow$ Positive Feedback

$\therefore proof:$

$$\frac{C(s)}{R(s)} = \frac{EG(s)}{E \pm B} = \frac{EG(s)}{E \pm CH(s)} = \frac{EG(s)}{E \pm EG(s)H(s)} = \frac{G(s)}{1 \pm G(s)H(s)}$$

2. Open-Loop Ratio:

$$\frac{B(s)}{E(s)} = G(s)H(s)$$

- This ratio has a relation between feedback loop $B(s)$ with the error's output $E(s)$.

3. Feedback Ratio:

$$\frac{B(s)}{R(s)} = \frac{G(s)H(s)}{1 + G(s)H(s)}$$

- This ratio has a relation between "before feedback loop" $B(s)$ with the reference $R(s)$.

4. Error Ratio:

$$\frac{E(s)}{R(s)} = \frac{1}{1 + G(s)H(s)}$$

- This ratio describes how the error is related to the input in a feedback system. It shows the error ratio when feedback is applied, and it depends on the open-loop gain "$G(s)H(s)$".

5. Characteristic Equation:

$$1 + G(s)H(s) = 0$$

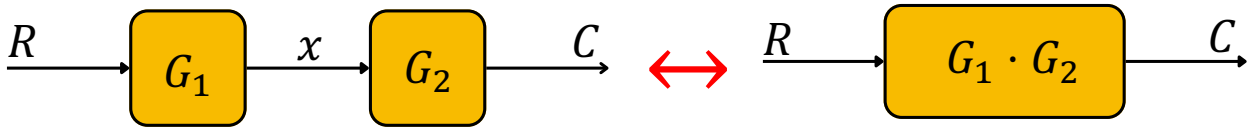
- It is the control function's denominator equaled zero, as we do with poles before, and we can find all characteristics of the system from this equation.

Reduction of Block Diagrams:

- A practical control system can be complex, involving multiple feedback and feedforward loops with several inputs.
- **Block diagram reduction** helps simplify such systems into a canonical form, making analysis and design easier
- Remember that G is transfer function which is the output over the input.

Reduction Rules:

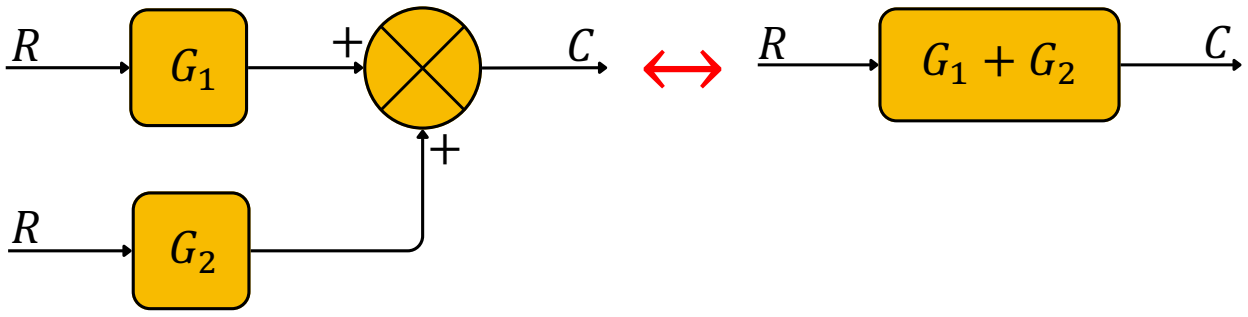
1. Series/Cascade Combination:



$\therefore proof:$

$$G_1 = \frac{x}{R}, \quad G_2 = \frac{C}{x} \quad \rightarrow \quad \frac{C}{R} = \frac{x}{R} \cdot \frac{C}{x} = G_1 \cdot G_2$$

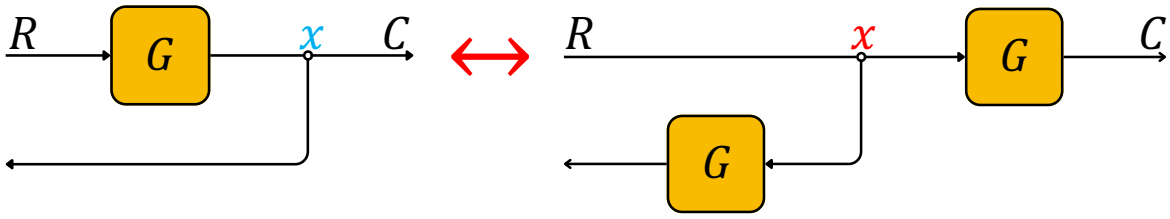
2. Parallel Combination:



$\therefore proof:$

$$C = RG_1 + RC_2 \quad \rightarrow \quad \frac{C}{R} = \frac{RG_1 + RC_2}{R} = G_1 + G_2$$

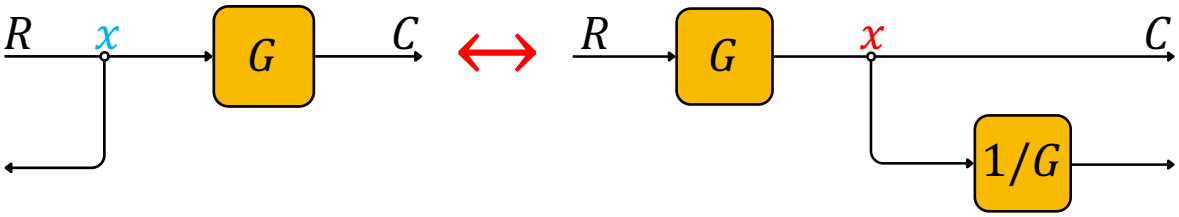
3. Shifting of Takeoff Point Ahead a Block:



$\therefore\ proof:$

- Before shifting the takeoff point x it was equal to $R \cdot G$, after shifting, x becomes equal to R and this isn't correct, so to make it equivalent to its value before shifting "which is RG" you must set a block with G value to make it $x = R \cdot G$.
- Takeoff point's output must be same before or after shifting.

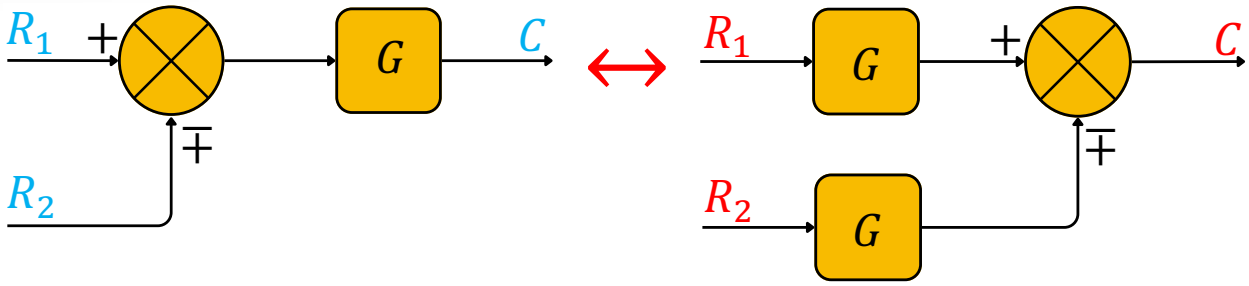
4. Shifting of Takeoff Point Behind a Block:



$\therefore\ proof:$

- Before shifting the takeoff point x it was equal to R, after shifting, x becomes equal to $R \cdot G$ and this isn't correct, so to make it equivalent to its value before shifting "which is R" you must set a block with $\frac{1}{G}$ value to make it $x = R$.

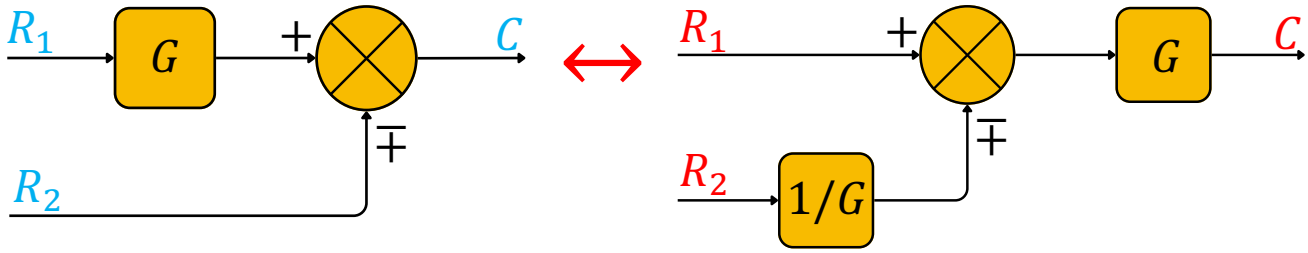
5. Shifting of Adder behind a Block:



$\therefore proof:$

- Before shifting adder, C was equal to $R_1G + R_2G$, after shifting, C becomes equal to $R_1G + R_2$ and this isn't correct, so to make C equivalent to its value before shifting "which is $R_1G + R_2G$" you must set a block with G value on R_2 to make it $C = R_1G + R_2G$.
- The output must be same before or after shifting.

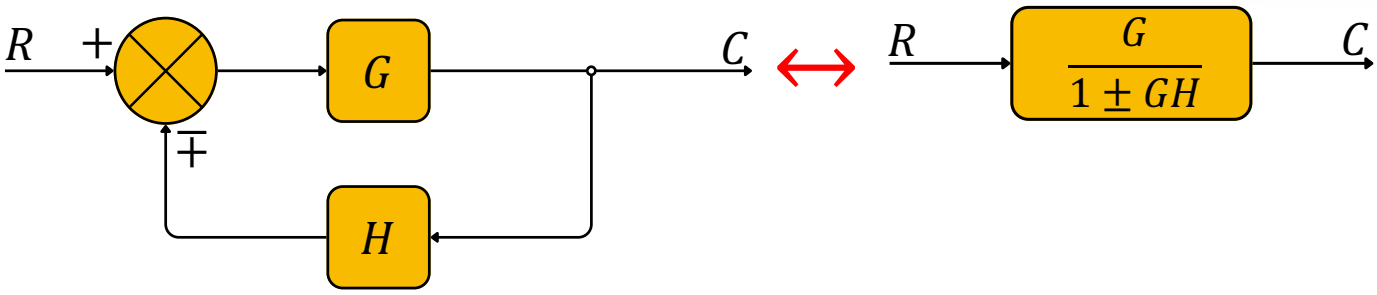
6. Shifting of Adder Point Ahead a Block:



$\therefore proof:$

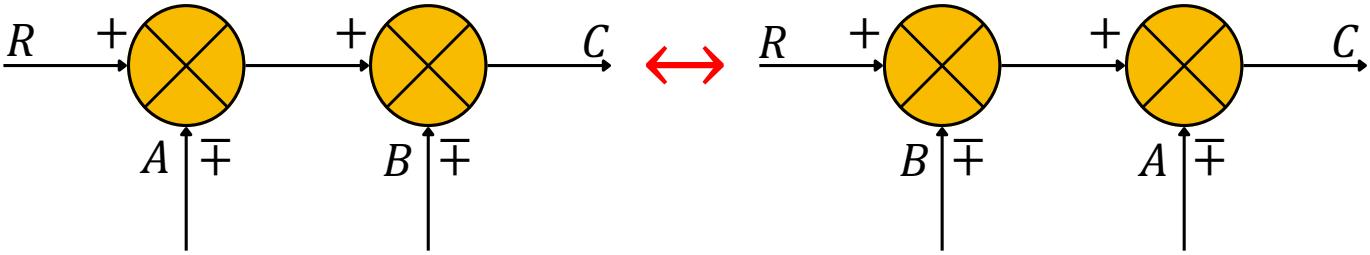
- Before shifting adder, C was equal to $R_1G + R_2$, after shifting, C becomes equal to $R_1G + R_2G$ and this isn't correct, so to make C equivalent to its value before shifting "which is $R_1G + R_2$" you must set a block with $\frac{1}{G}$ value on R_2 to make it $C = R_1G + R_2$.

7. Eliminating a Feedback Loop:



- As we learn before, you can replace this feedback system with its overall transfer function in one block.

8. Swap With 2 Summing Points:



$$(R \mp A) \mp B = (R \mp B) \mp A = R \mp A \mp B$$

▶ Key Notes:

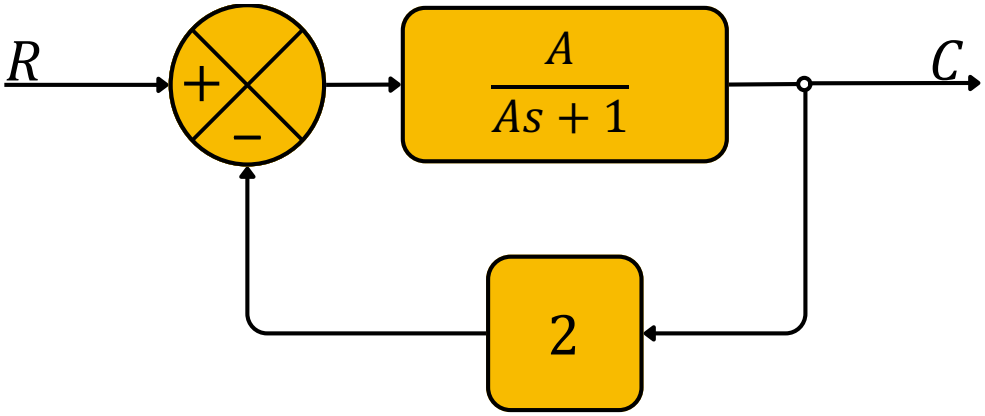
∴ Takeoff point's output must be same before or after shifting.

∴ The output must be same before or after shifting.

∴ Small trick: if $G = \frac{a}{b}$ and $H = c$:

$\frac{C}{R} = \frac{a}{ac + b}$	$\frac{B}{E} = \frac{ac}{b}$
$\frac{B}{R} = \frac{ac}{ac + b}$	$\frac{E}{R} = \frac{b}{b + ac}$

Q. 7.4: For the following block diagram:



Find:

∴ Closed loop transfer function.

$$\frac{C(s)}{R(s)} = \frac{G(s)}{1 + G(s)H(s)} = \frac{A}{As + 2A + 1}$$

∴ Open loop transfer function.

$$\frac{B(s)}{E(s)} = G(s)H(s) = \frac{2A}{As + 1}$$

∴ Feedback ratio.

$$\frac{B(s)}{R(s)} = \frac{G(s)H(s)}{1 + G(s)H(s)} = \frac{2A}{As + 1 + 2A}$$

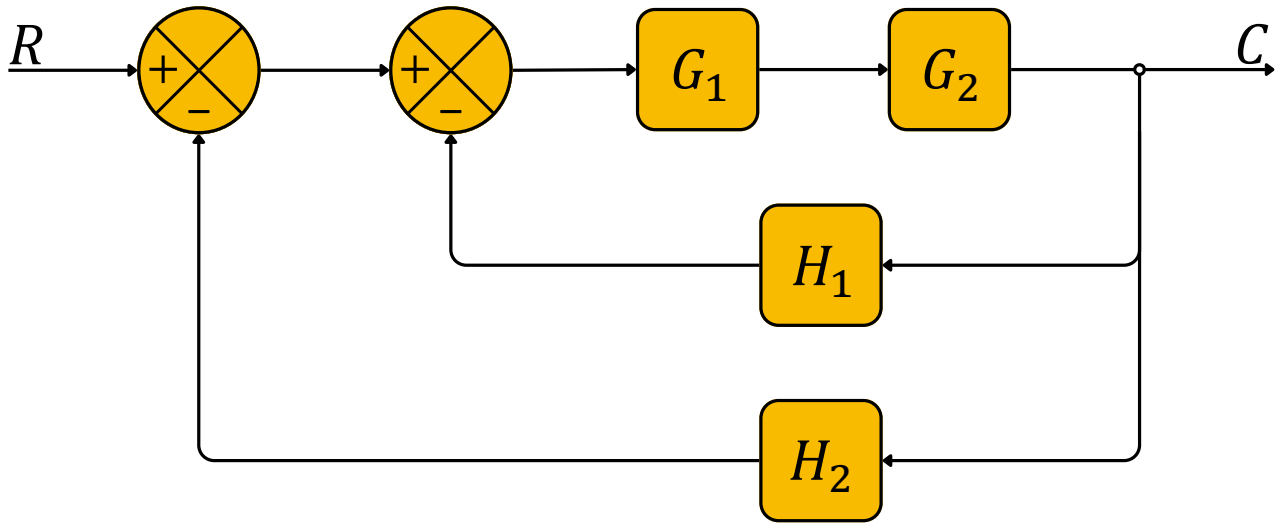
∴ Error ratio.

$$\frac{E(s)}{R(s)} = \frac{1}{1 + G(s)H(s)} = \frac{As + 1}{As + 1 + 2A}$$

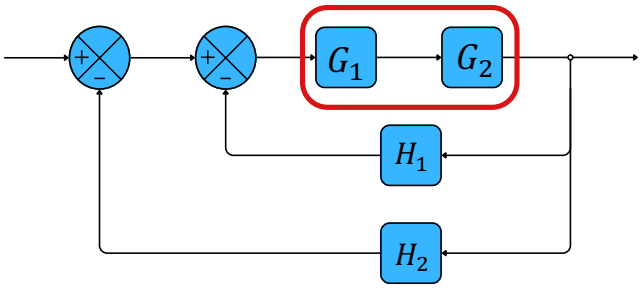
∴ Characteristic equation and pole when $A = 10$.

$$As + 2A + 1 = 0 \rightarrow 10s + 21 = 0 \rightarrow s = -\frac{21}{10} = -2.1$$

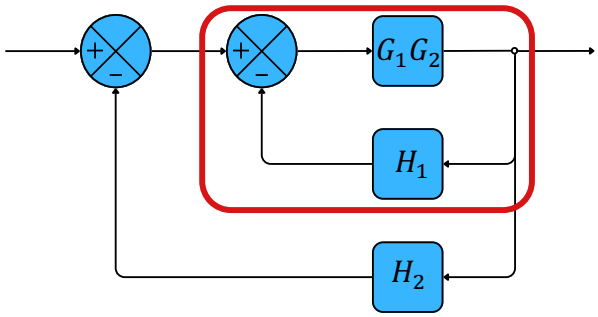
▶ **Q. 7.5:** Find T. F. $\frac{R(s)}{C(s)}$ of the following block diagram:



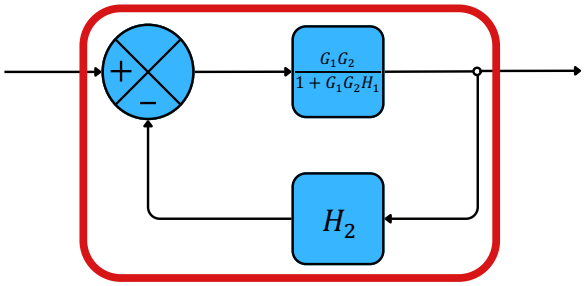
∴ *solution*:



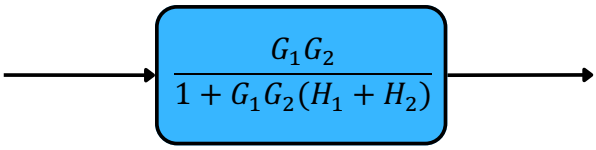
Rule 1



Rule 2

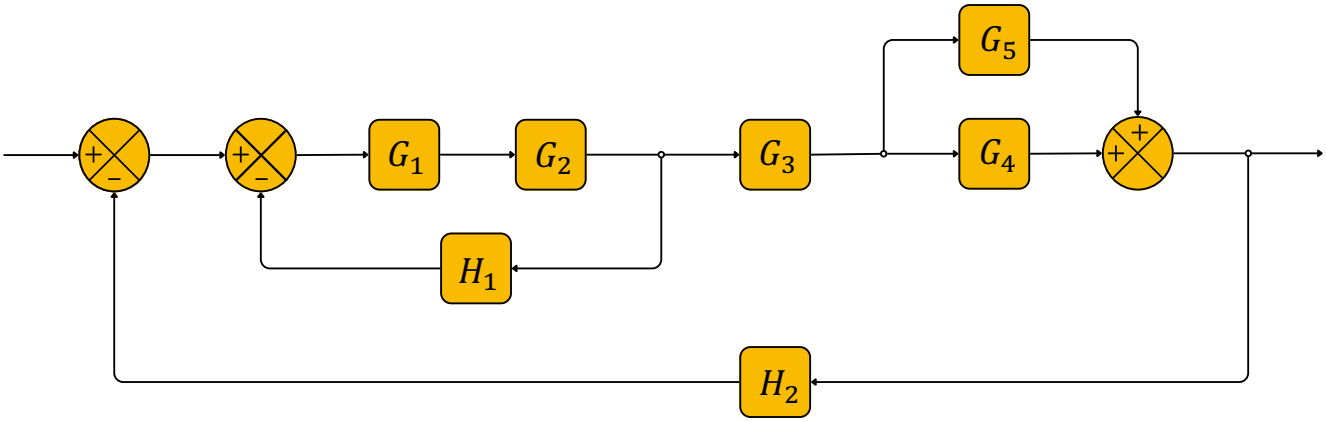


Rule 7

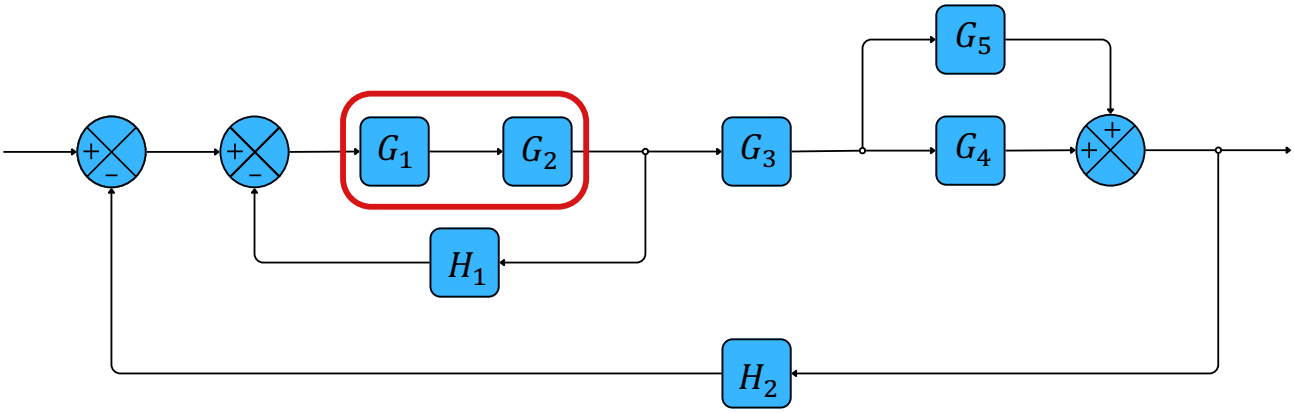


Final T.F.

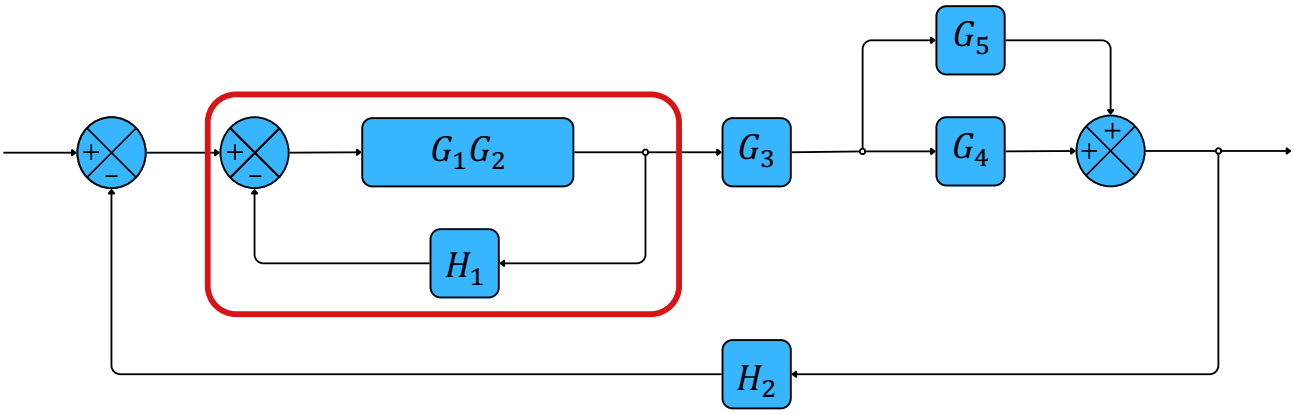
▶ Q. 7.6: Find overall T.F. of the following block diagram:



∴ *solution*:

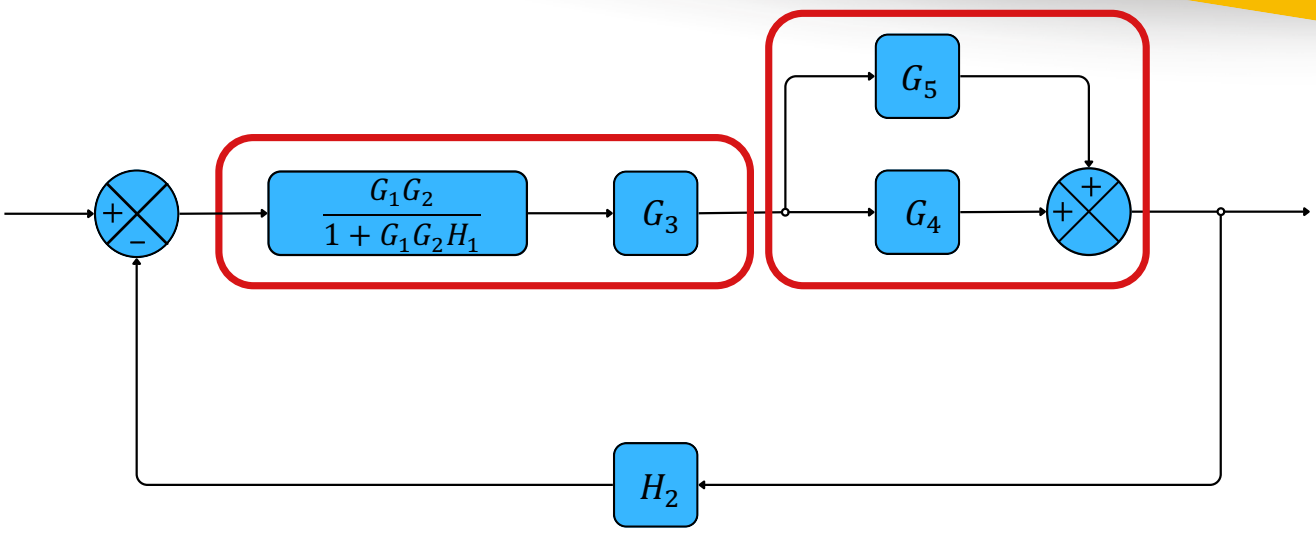


Rule 1 ↓

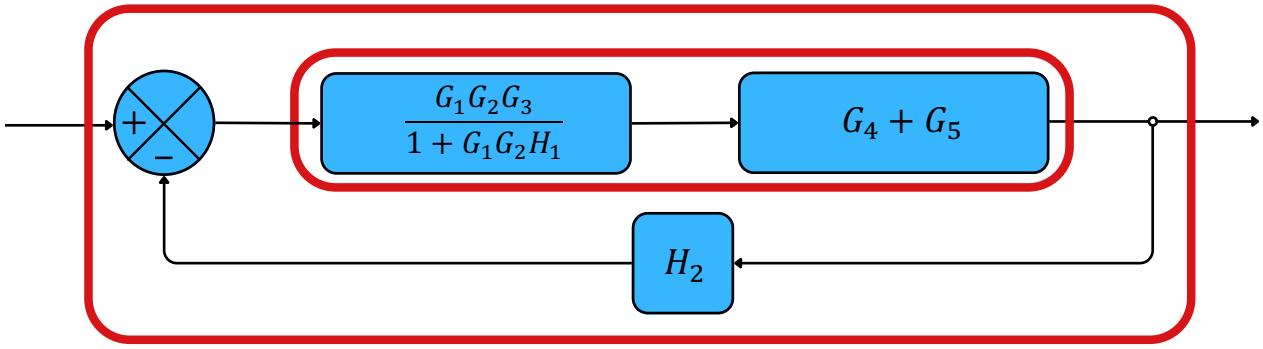


Rule 7 ↓

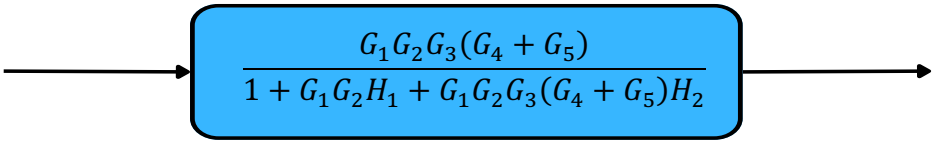
Next Page



Rule 1



Rule 1&7



Final T.F.

- For more questions, scan this QR code:

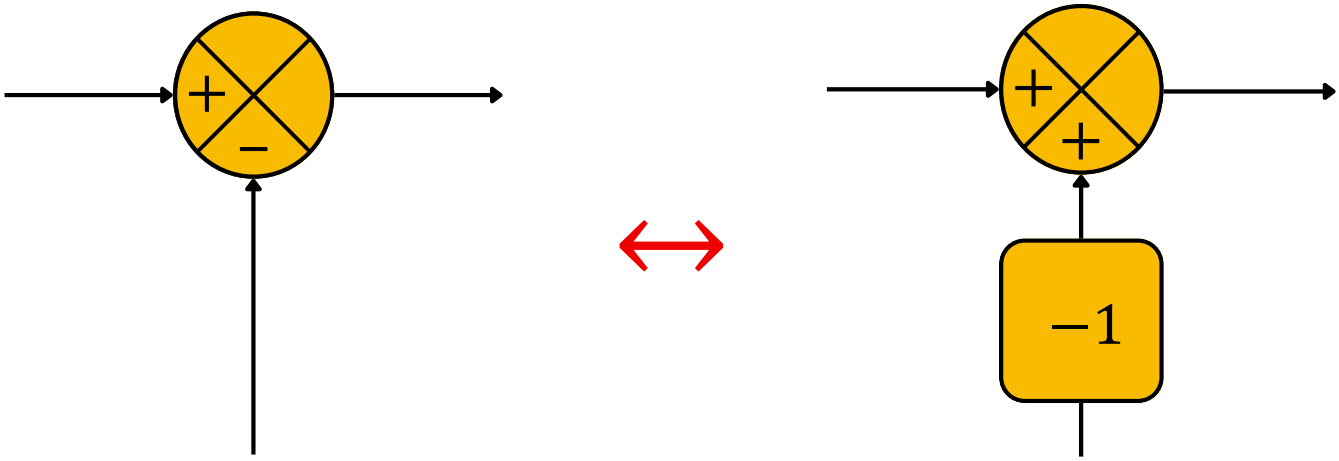
Multi Inputs-Outputs Block Diagram:

- Multi Inputs-Outputs System is a system has more than 1 input, output or them together.
- To obtain transfer function for a particular output due to a particular input, you use the superposition theorem. This theorem states that for linear systems, the total response is the sum of the responses caused by each input acting alone.

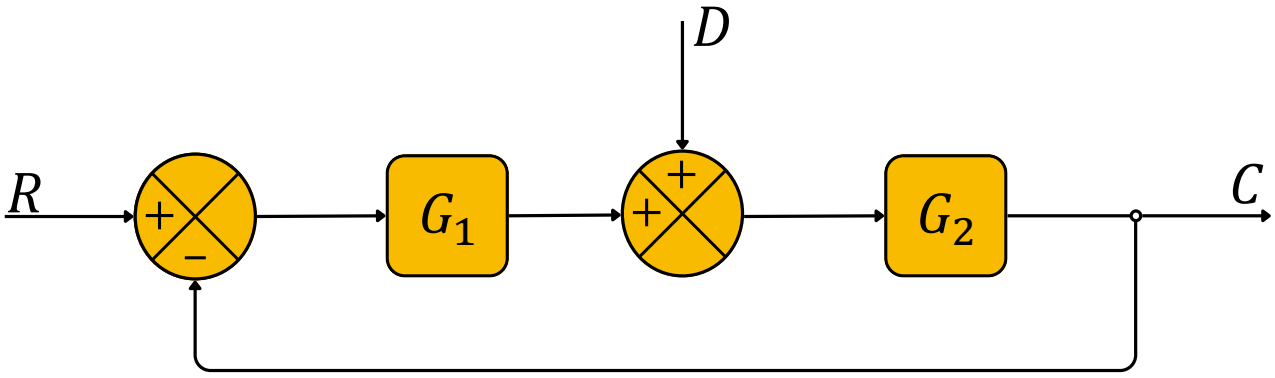
▶ The step-by-step procedure is:

1. Identify the input of interest and the output of interest.
2. Set all other independent inputs to zero.
3. Find transfer fusing block diagram reduction.
4. Repeat for each input-output pair if needed, then sum individual responses for the total output.

- If the summator do a subtraction operation, you can set minus sign out of the summator as a block to change it to sum operation.



▶ **Q. 7.7:** Consider this multi-input block diagram:



Since R is the reference input and D is disturbance input.

Find (as a feedback system form):

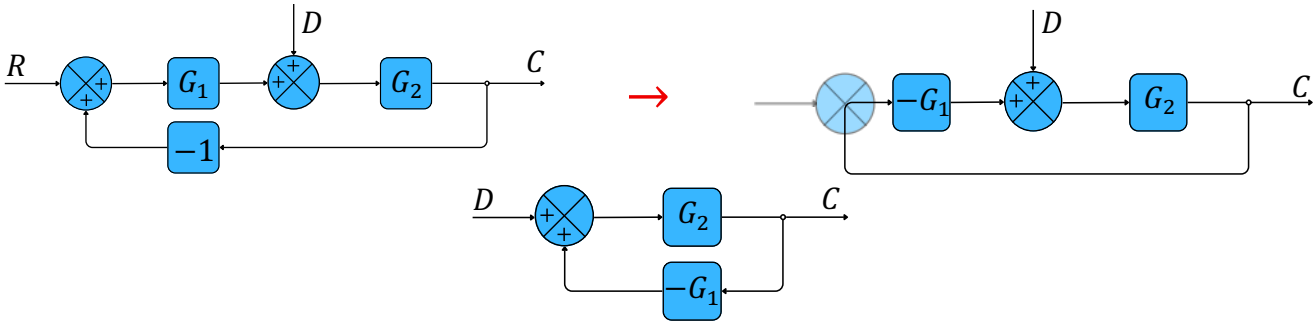
$\therefore$ C due to R "$\frac{C}{R}$".

Firstly, set $D = 0$, then the summator of D acts like an arrow because it adds 0 to G_1 only "$G_1 + 0 = G_1$":

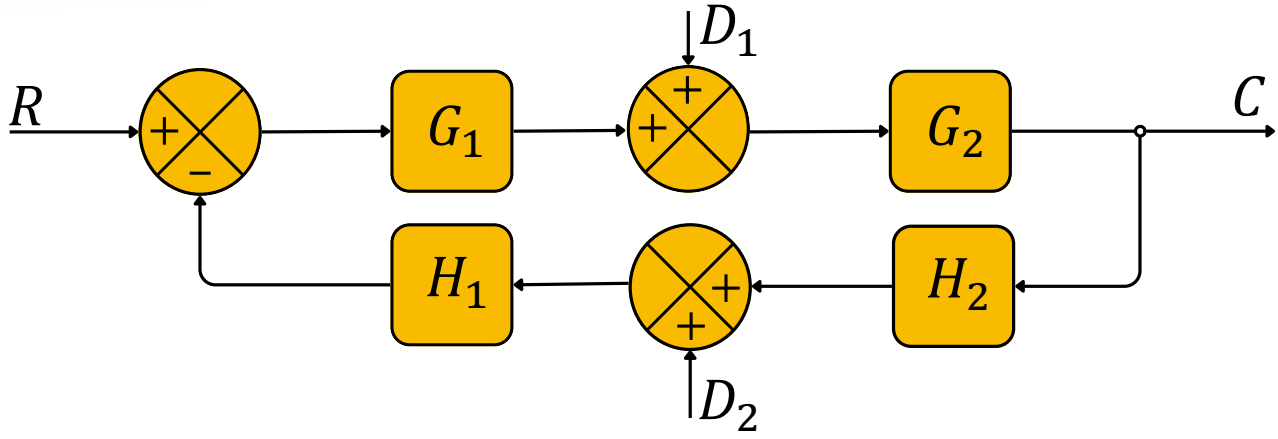


$\therefore$ C due to D "$\frac{D}{R}$".

set $R = 0$, but don't forget the sign of R's summator "$-$", so make it positive and add -1 block at its position:



Q. 7.8: Consider this multi-input block diagram:

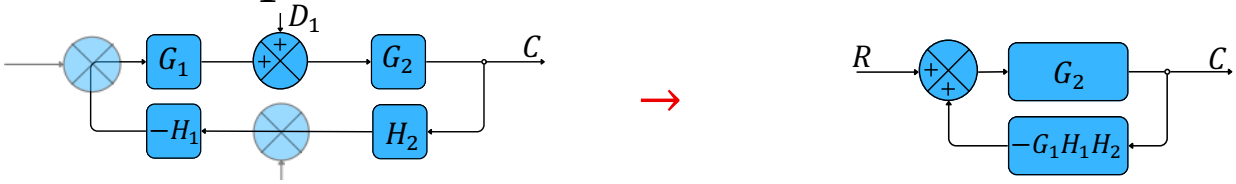


Find (as a feedback system form):

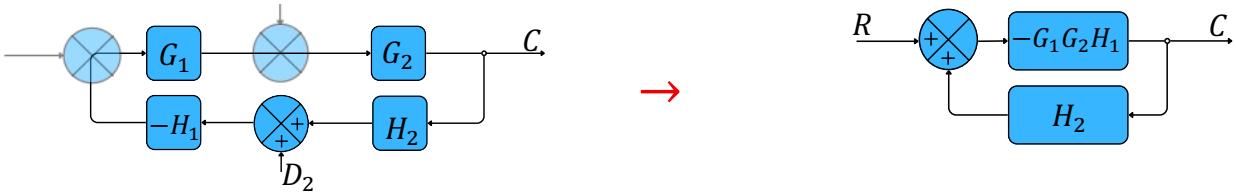
$\therefore$ C due to R.



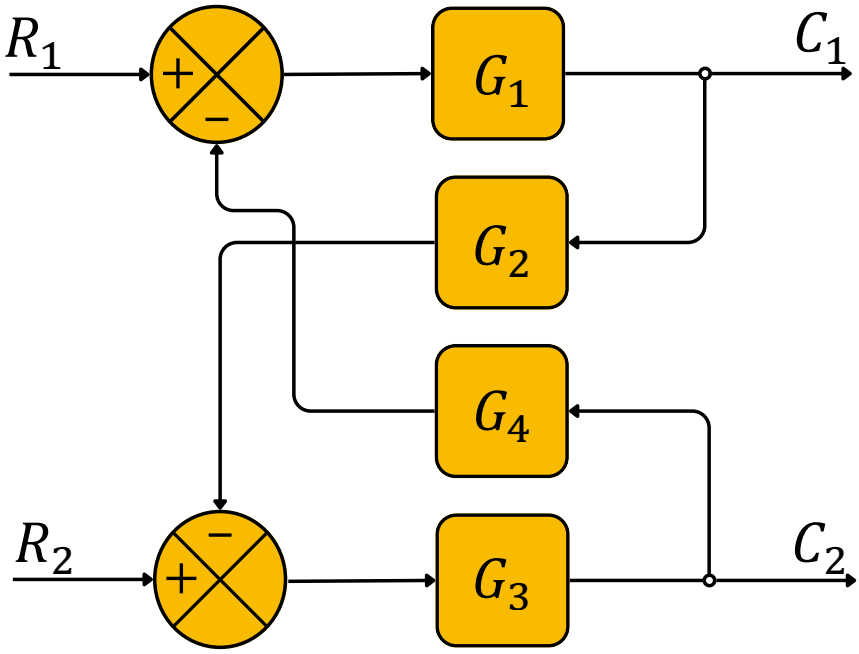
$\therefore$ C due to D_1.



$\therefore$ C due to D_2.

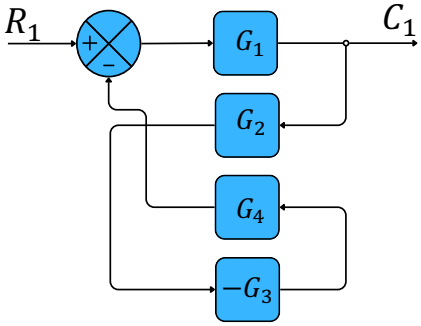


▶ **Q. 7.9:** Consider this multi-input block diagram:

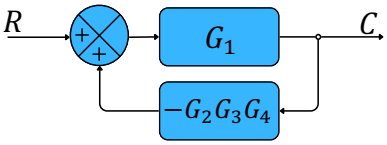


Find (as a feedback system form):

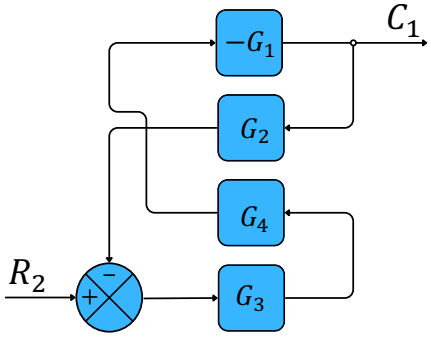
∴ C_1 due to R_1.



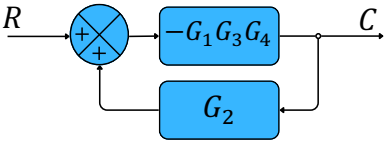
→



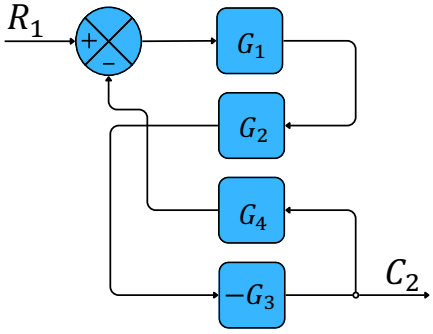
∴ C_1 due to R_2.

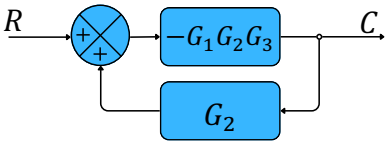


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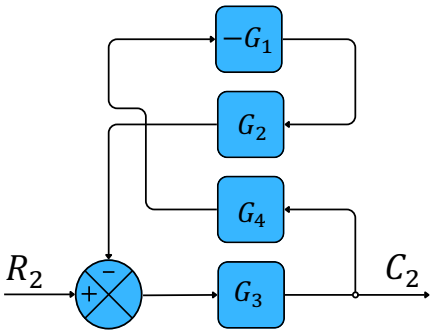


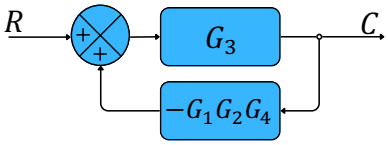
$\therefore$ C_2 due to R_1.





$\therefore$ C_2 due to R_2.



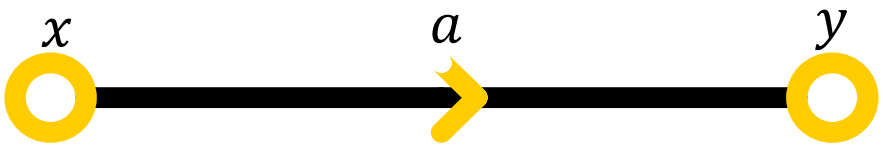


8. Signal Flow Graphs

- **Signal Flow Graph:** A network of nodes connected by directed branches, and shows the flow and relationships between signals in a system.
- You can say that it is an alternative to block diagrams for system representation.
- Consider this equation and draw its signal flow graph:

$$y = ax$$

- The signal flow graph of the equation is shown below:



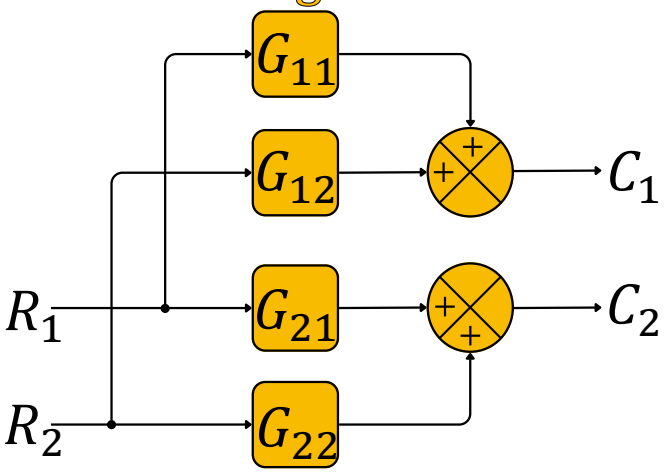
- Every variable in a signal flow graph is designed by a **Node** (like x and y here).
- Every T.F in a signal flow graph is designed by a **Branch** (like a here).

Example:

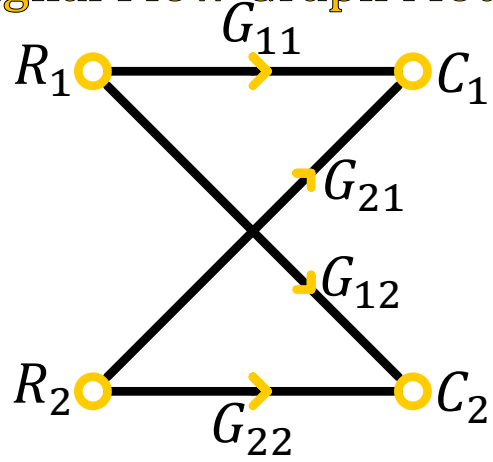
- Take this system to represent it as block diagram and signal flow graph:

$$C_1(s) = G_{11}R_1(s) + G_{12}R_2(s)$$
$$C_2(s) = G_{21}R_1(s) + G_{22}R_2(s)$$

Block Diagram Model

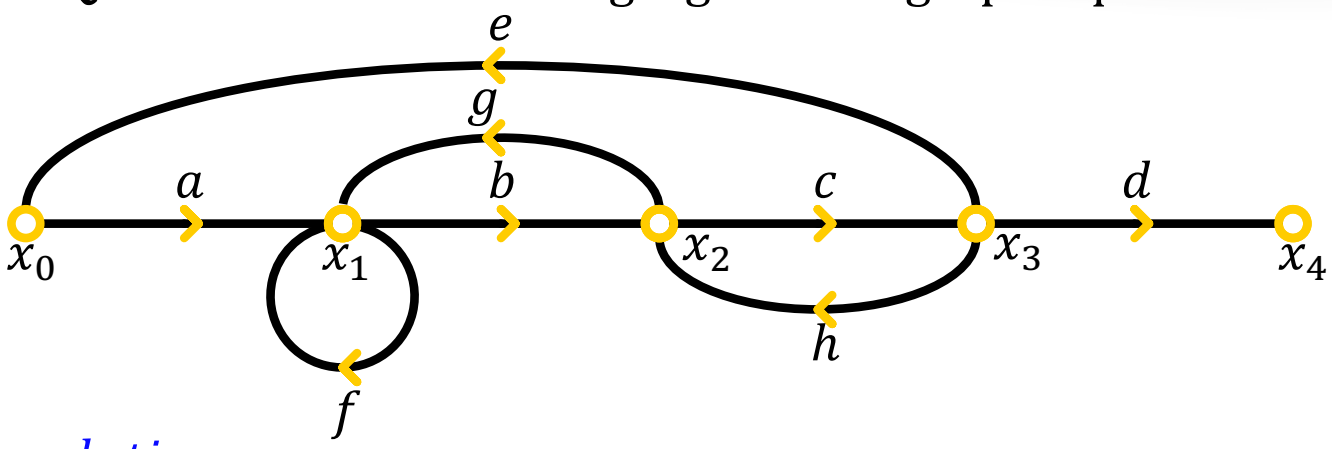


Signal Flow Graph Model



- In these 2 methods to model a system, "in my opinion" I see that the block diagram model of this system is complex, in signal flow graph one it is simpler to understand and to obtain its overall transfer function "will be explained later".

▶ **Q. 8.1:** Find the following signal flow graph equations:



∴ *solution*:

To obtain the equations, you must see the node and the arrows that indicates to that node:

$$x_3 \xrightarrow{e} x_0, \qquad x_0 = ex_3$$

$$x_0 \xrightarrow{a} x_1, x_1 \xrightarrow{f} x_1, x_2 \xrightarrow{g} x_1, \qquad x_1 = ax_0 + fx_1 + gx_2$$

$$x_1 \xrightarrow{b} x_2, x_3 \xrightarrow{h} x_2, \qquad x_2 = bx_1 + hx_3$$

$$x_2 \xrightarrow{c} x_3, \qquad x_3 = cx_2$$

$$x_3 \xrightarrow{d} x_4, \qquad x_4 = dx_3$$

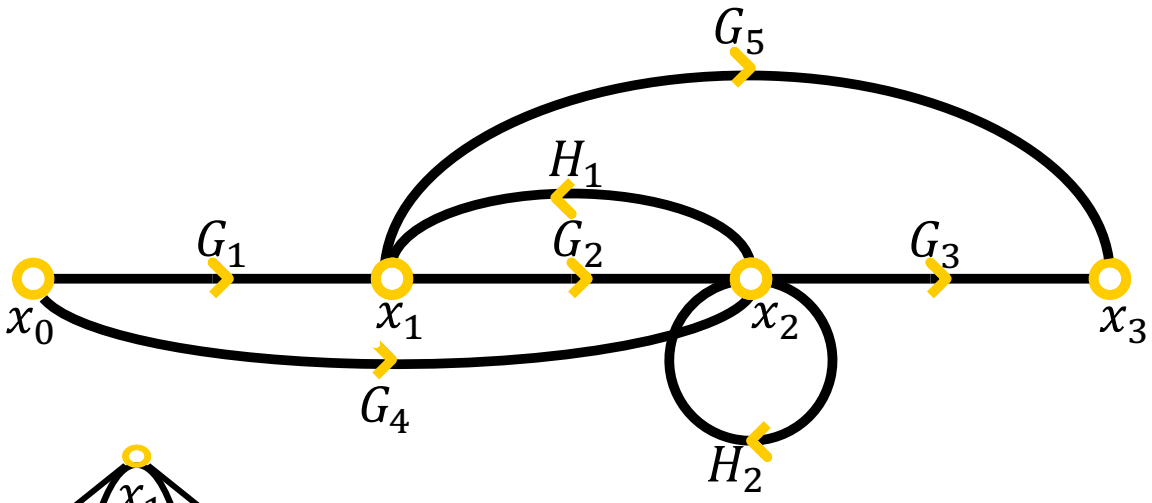
▶ Q. 8.2: Draw signal flow graph using the equations:

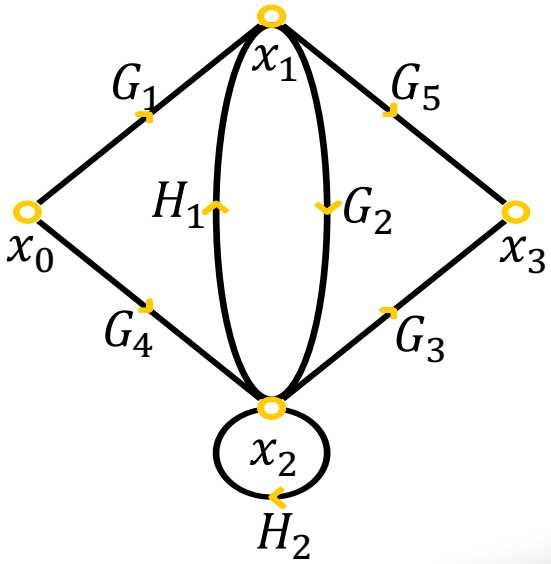
$$x_1 = G_1x_0 + H_1x_2$$
$$x_2 = G_4x_0 + G_2x_1 + H_2x_2$$
$$x_3 = G_5x_1 + G_3x_2$$

∴ *solution*:

To obtain the signal flow graph, you must draw the nodes first, then from the equation draw a line with arrow from input to output with its transfer function:

$$x_1 = G_1x_0 + H_1x_2, \quad x_0 \xrightarrow{G_1} x_1, x_2 \xrightarrow{H_1} x_1,$$
$$x_2 = G_4x_0 + G_2x_1 + H_2x_2, \quad x_0 \xrightarrow{G_4} x_2, x_1 \xrightarrow{G_2} x_2, x_2 \xrightarrow{H_2} x_2$$
$$x_3 = G_5x_1 + G_3x_2, \quad x_1 \xrightarrow{G_5} x_3, x_2 \xrightarrow{G_3} x_3$$



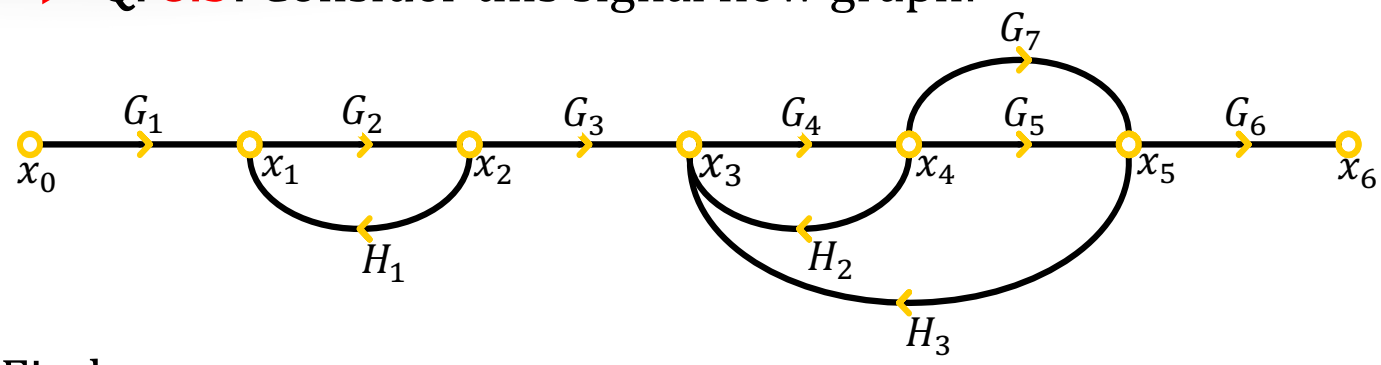


←You can also draw it like this.

Definitions:

- **Input Node:** A node contains only the outgoing branches.
- **Output Node:** A node contains only the incoming branches.
- **Path:** A continuous, unidirectional succession of branches along which no node is passed more than ones.
- **Forward Path:** A path from input node to output node.
- **Loop:** A path which start and return at the same node.
- **Self-Loop:** A feedback loop consisting of a single branch.
- **Gain:** The transfer function of the branch.
- **Path Gain P:** The product of branch gains of the path.
- **Loop Gain L:** The product of the branch gains of the loop.
- **Non-Touching Loops:** Loops have no common nodes.

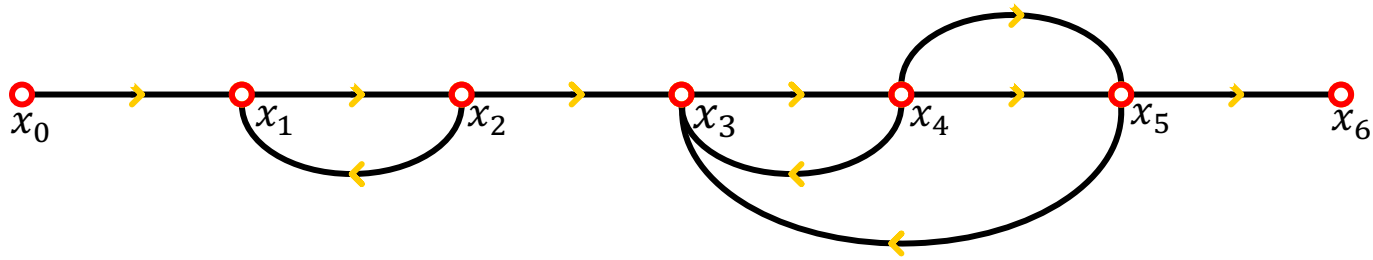
▶ Q. 8.3: Consider this signal flow graph:



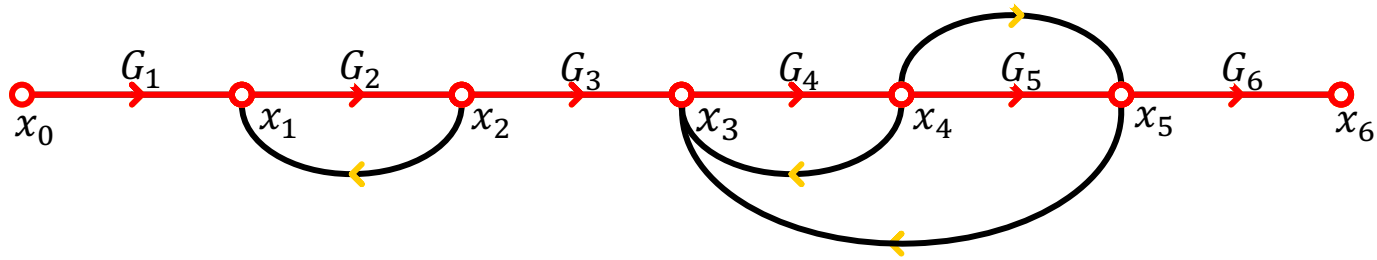
Find:

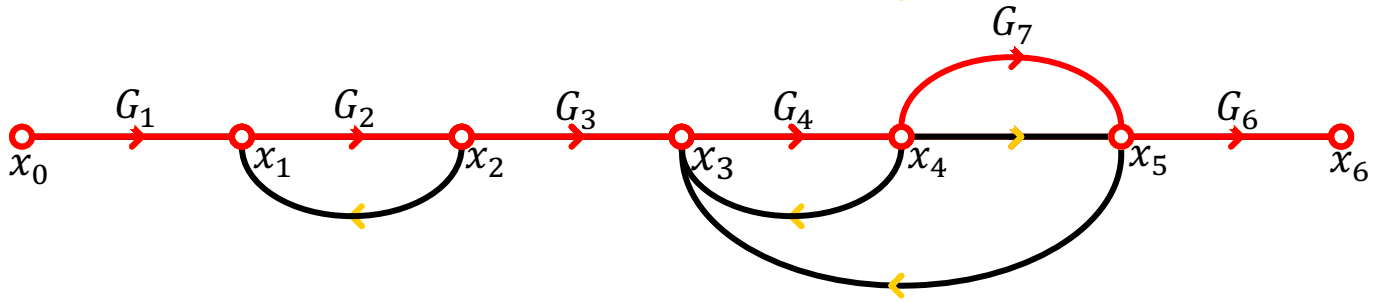
∴ All paths gain:

Only one path: $x_0 \rightarrow x_1 \rightarrow x_2 \rightarrow x_3 \rightarrow x_4 \rightarrow x_5 \rightarrow x_6$



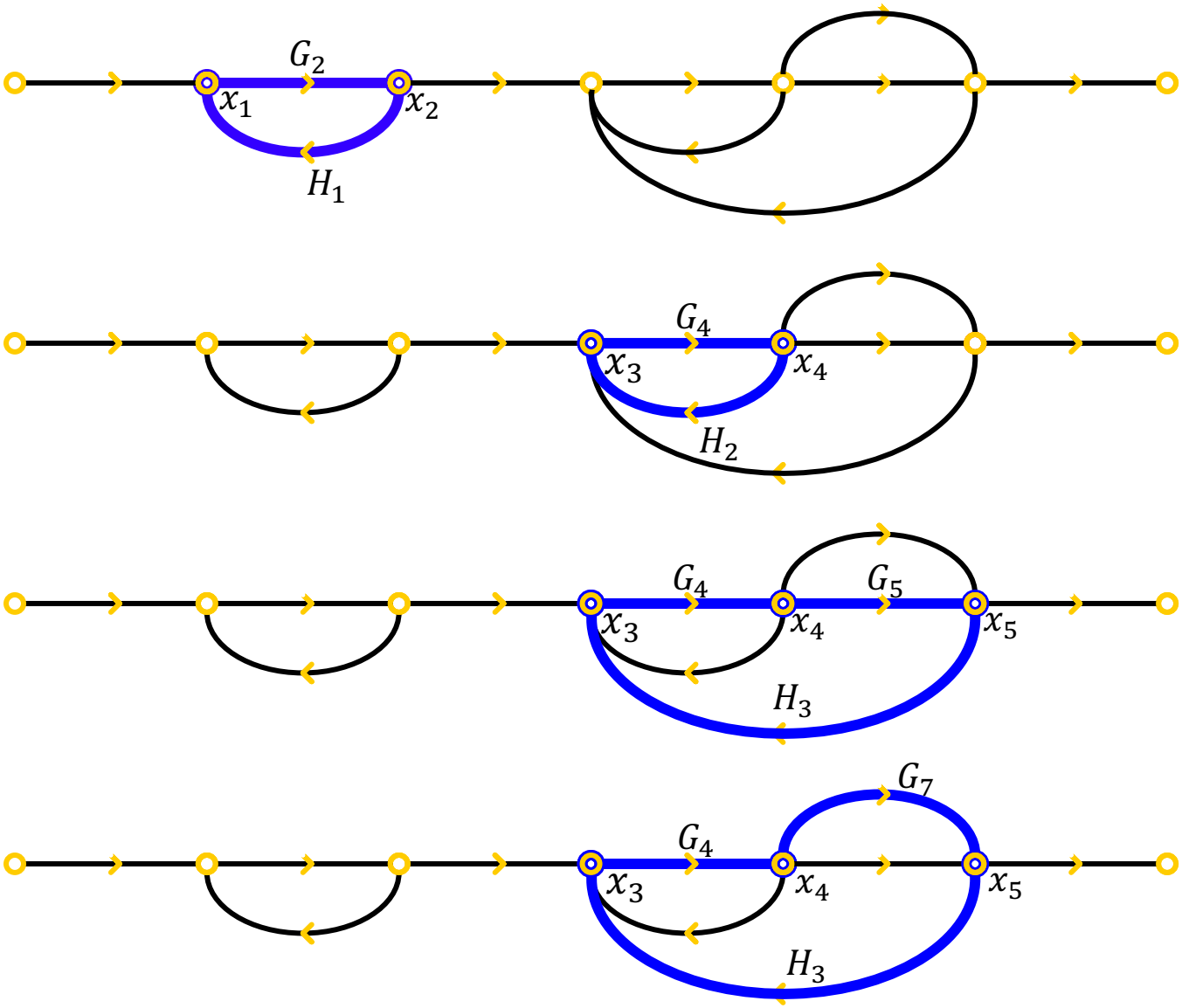
With 2 gain paths "we have $x_4 \xrightarrow{G_5} x_5$ and $x_4 \xrightarrow{G_7} x_5$"





$$P_1 = G_1G_2G_3G_4G_5G_6,\ P_2 = G_1G_2G_3G_4G_7G_6$$

∴ All loops gain.



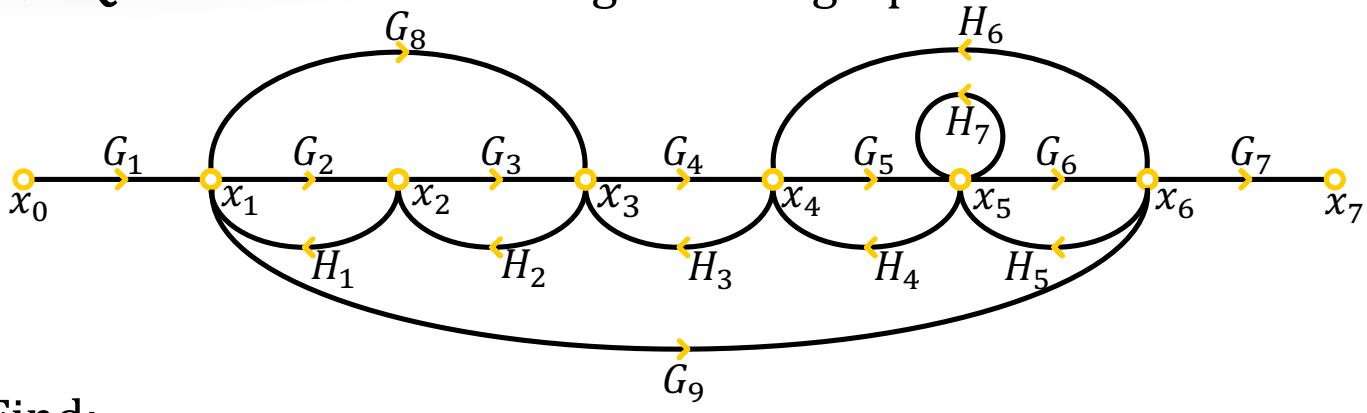
$L_1 = G_2H_1,$ $L_2 = G_4H_2,$ $L_3 = G_4G_5H_3,$ $L_4 = G_4G_7H_3$

∴ All non-touching loops.

As we see, L_2, L_3 and L_4 have a common node x_3, so they are touching each other, but L_1 doesn't touch L_2, L_3 and L_4 at any node "they have no common node", so:

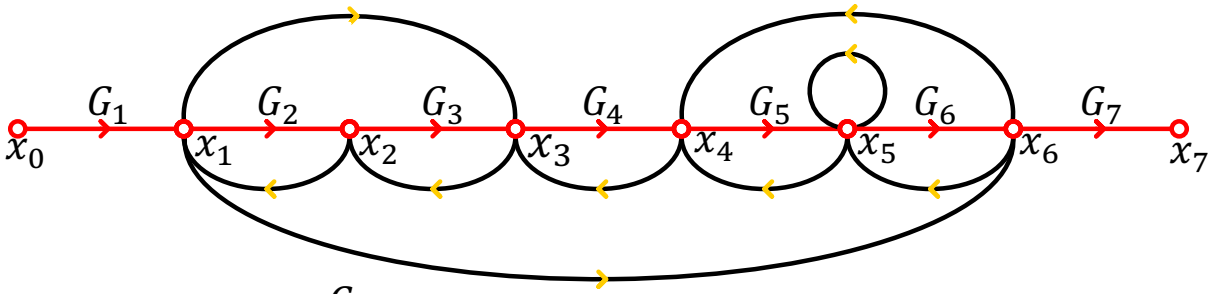
$[L_1L_2], [L_1L_3], [L_1L_4]$ are non-touching each other.

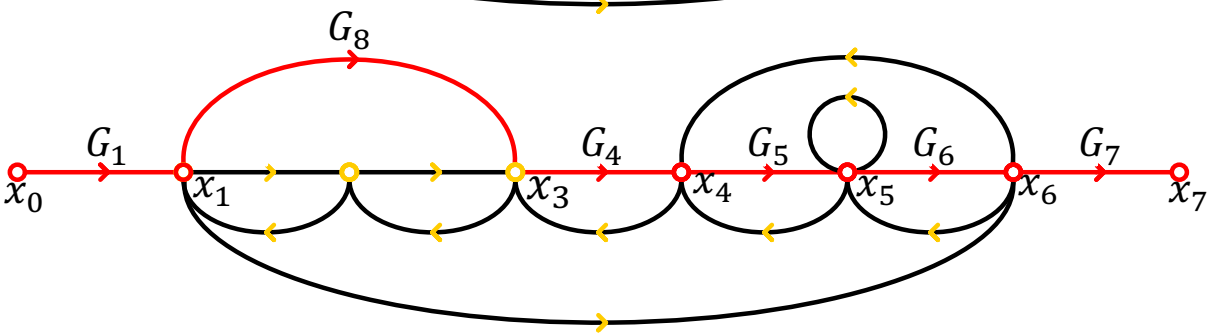
▶ Q. 8.4: Consider this signal flow graph:

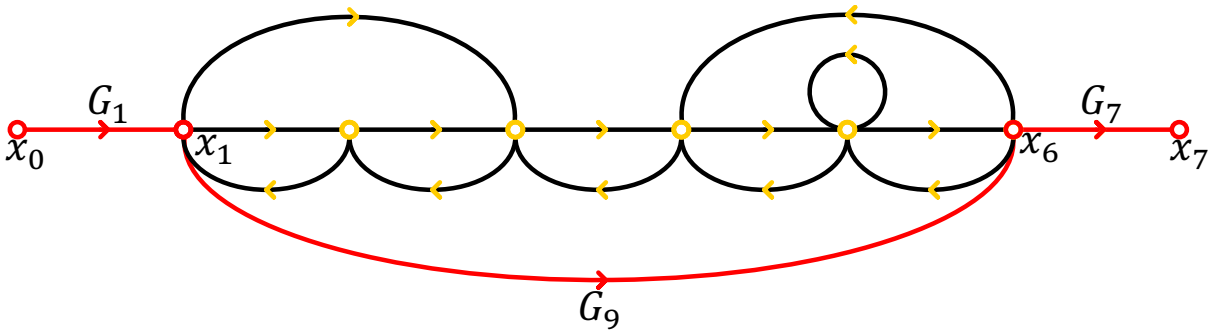


Find:

∴ All paths gain:

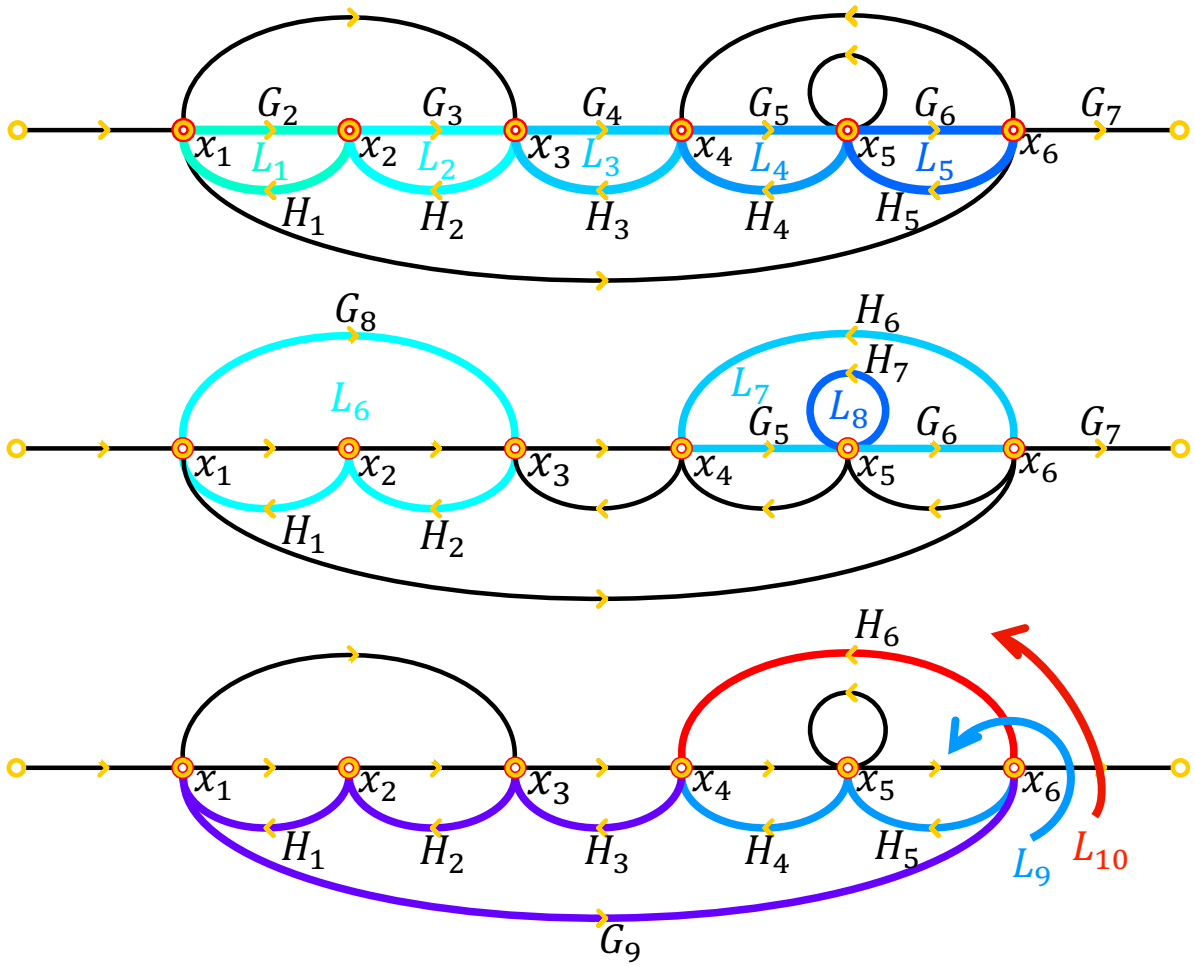






$P_1 = G_1G_2G_3G_4G_5G_6G_7,$ $P_2 = G_1G_8G_4G_5G_6G_7,$ $P_3 = G_1G_9G_7$

∴ All loops gain.



$$L_1 = G_2H_1,\ L_2 = G_3H_2,\ L_3 = G_4H_3,\ L_4 = G_5H_4,\ L_5 = G_6H_5,$$
$$L_6 = G_8H_1H_2,\ L_7 = G_5G_6H_6,\ L_8 = H_7,$$
$$L_9 = G_9H_5H_4H_3H_2H_1,\ L_{10} = G_9H_6H_3H_2H_1$$

∴ All non-touching loops.

When 2 loops are non-touching each other:

$$[L_1L_3], [L_1L_4], [L_1L_5], [L_1L_7], [L_1L_8], [L_2L_4], [L_2L_5], [L_2L_7], [L_2L_8]$$
$$[L_3L_5], [L_3L_7], [L_4L_6], [L_5L_6], [L_6L_7], [L_6L_8], [L_7L_{10}], [L_8L_{10}]$$

When 3 loops are non-touching each other:

$$[L_1L_3L_5], [L_1L_3L_8]$$

Mason's Rule:

- Block diagram reduction involves multiple steps using fundamental rules to find the system's transfer function.
- **Mason's rule** (for signal-flow graphs) uses this single formula to directly compute the overall transfer function:

$$\frac{C(s)}{R(s)} = \sum_{i=1}^{n} \frac{P_i \Delta_i}{\Delta}$$

∴ n is the number of forward paths.

∴ P_i is the ith forward path gain.

∴ Δ is the Determinant of the system.

∴ Δ_i is the Determinant of the ith forward path.

- The Determinant Δ can be calculated using this formula:

$$\Delta = 1 - \sum L_I + \sum L_{n2} - \sum L_{n3} + \cdots$$

∴ $\sum L_I$ is the sum of all individual (single) loop gains.

∴ $\sum L_{n2}$ is sum of all gain products of 2 non-touching loops.

∴ $\sum L_{n3}$ is sum of all gain products of 3 non-touching loops.

• The Determinant of the ith forward path Δ_i can be calculated by removing the path P_i "with its branches and nodes" then defining the remain loops and non-touching loops "with their branches and nodes":

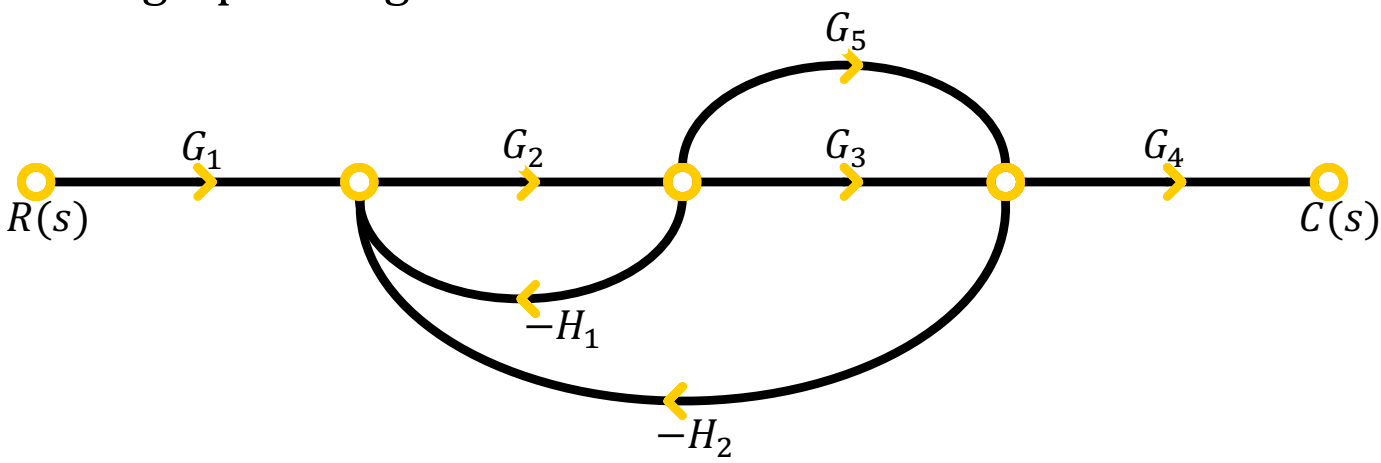
$$\Delta_i = 1 - \sum L_R + \sum L_{n2} - \sum L_{n3} + \cdots$$

$\therefore \sum L_R$ is the sum of all **remained** individual loop gains.
$\therefore \sum L_{n2}$ is sum of all gain products of 2 non-touching loops.
$\therefore \sum L_{n3}$ is sum of all gain products of 3 non-touching loops.

▶ **Rules for Applying Mason's Formula:**

1. Identify all forward path gains P_i.
2. Identify all loop gains L_I.
3. Identify sum of procuct of non-touching loops $L_{2n}, L_{3n}, \ldots$ (Note that Δ_i is equal to 1 when you delete the ith path and there are no remained loops).
4. Calculate Δ using determinant law.
5. Calculate Δ_i for each path (after deleting the path and identifing the remained loop gains and production of non-touching loop gains)
6. Plug values into the formula.

▶ **Q. 8.5:** Find the overall transfer function for this signal flow graph using mason's rule.



$\therefore$ *solution*:

Path gains:

$$P_1 = G_1G_2G_3G_4, \qquad P_2 = G_1G_2G_4G_5$$

Loop gains:

$$L_1 = -G_2H_1, \qquad L_2 = -G_2G_3H_2, \qquad L_3 = -G_2G_5H_2$$

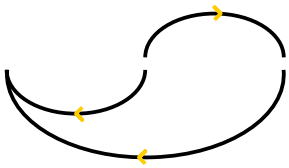
There are no non-touching loop gains:

$$\sum L_{2n} = 0$$

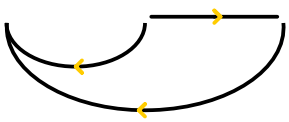
Determinant calculations:

$$\Delta = 1 - \sum L_I = 1 + [G_2H_1 + G_2G_3H_2 + G_2G_5H_2]$$

After deleting P_1 and P_2, there will not remain any loop, so:



$\Delta_1 = 1$

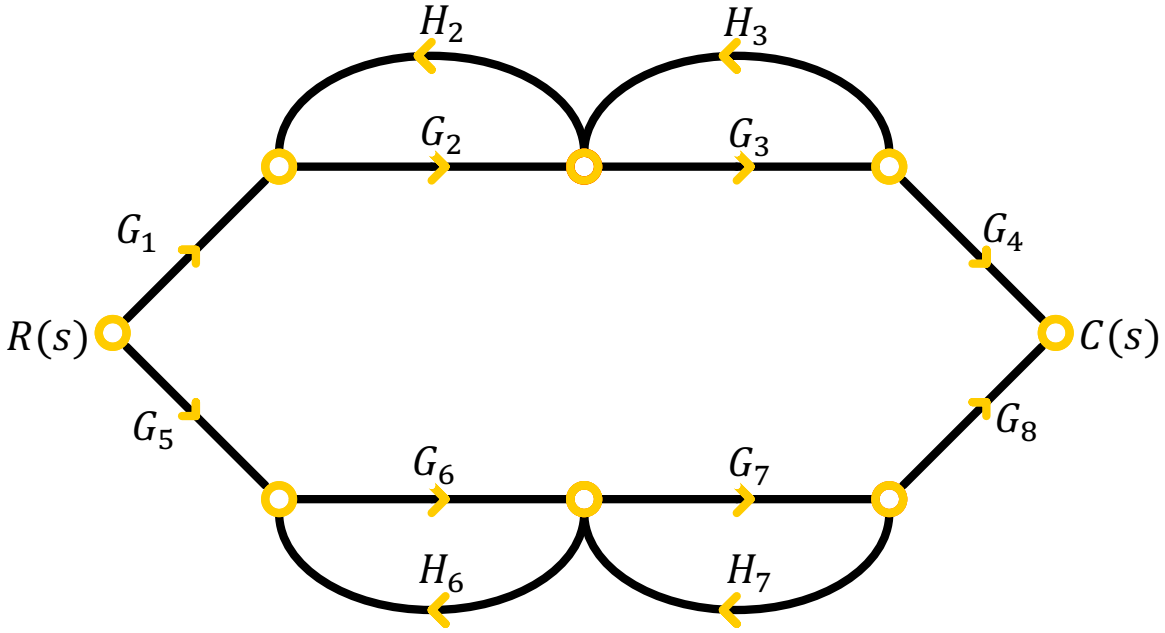


$\Delta_2 = 1$

Overall T.F:

$$\frac{C(s)}{R(s)} = \frac{P_1\Delta_1 + P_2\Delta_2}{\Delta} = \frac{G_1G_2G_3G_4 + G_1G_2G_4G_5}{1 + G_2H_1 + G_2G_3H_2 + G_2G_5H_2}$$

▶ **Q. 8.6:** Find the overall transfer function for this signal flow graph using mason's rule.



$\therefore$ *solution*:

Path gains:

$$P_1 = G_1G_2G_3G_4, \qquad P_2 = G_5G_6G_7G_8$$

Loop gains:

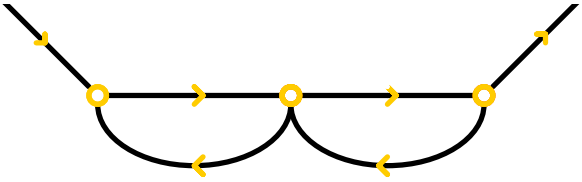
$$L_1 = G_2H_2, \qquad L_2 = G_3H_3, \qquad L_3 = G_6H_6, \qquad L_4 = G_7H_7$$

There are non-touching loop gains "2's only":

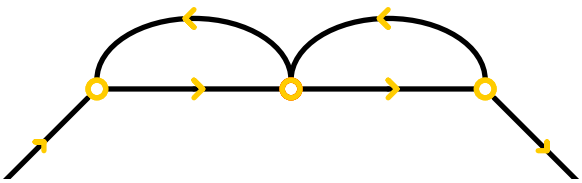
$$\sum L_{2n} = L_1L_3 + L_1L_4 + L_2L_3 + L_2L_4 = (L_1 + L_2)(L_3 + L_4)$$
$$= (G_2H_2 + G_3H_3)(G_6H_6 + G_7H_7)$$

Determinant calculations:

$$\Delta = 1 - \sum L_I + \sum L_{2n}$$
$$= 1 - (G_2H_2 + G_3H_3 + G_6H_6 + G_7H_7) + (G_2H_2 + G_3H_3)(G_6H_6 + G_7H_7)$$



$$\Delta_1 = 1 - [L_3 + L_4] = 1 - (G_6H_6 + G_7H_7)$$



$$\Delta_2 = 1 - [L_1 + L_2] = 1 - (G_2H_2 + G_3H_3)$$

Overall T.F:

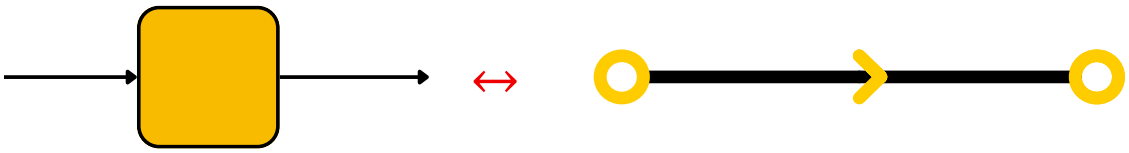
$$\frac{C(s)}{R(s)} = \frac{P_1\Delta_1 + P_2\Delta_2}{\Delta}$$

$$= \frac{G_1G_2G_3G_4\big(1 - (G_6H_6 + G_7H_7)\big) + G_5G_6G_7G_8\big(1 - (G_2H_2 + G_3H_3)\big)}{1 - (G_2H_2 + G_3H_3 + G_6H_6 + G_7H_7) + (G_2H_2 + G_3H_3)(G_6H_6 + G_7H_7)}$$

- Can you find this T.F but using block diagram reduction? :)
- By the way, how could we convert between block diagram model with signal flow graph and vice versa?

Converting between B.D and S.F.G:

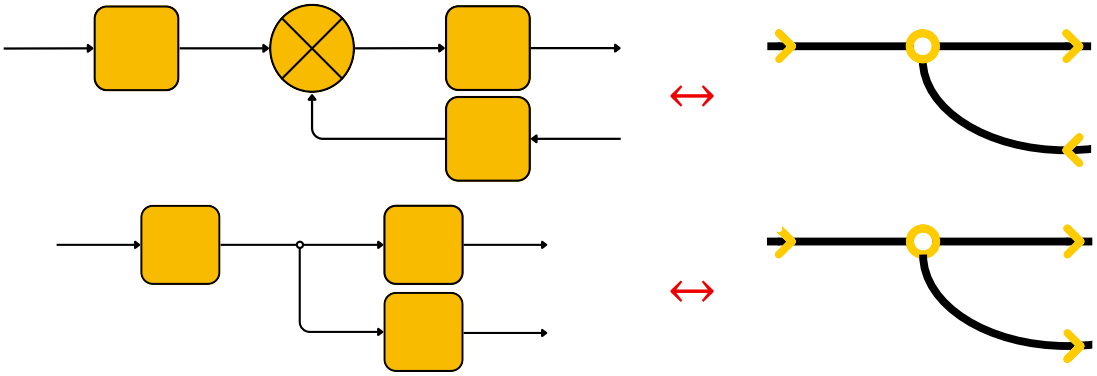
- Blocks are represented as branches.



- Summators and takeoff points are represented as nodes

∴ It is a **summator** if the arrow branch **enters** the node.

∴ It is a **takeoff point** if the arrow branch **exits** the node.

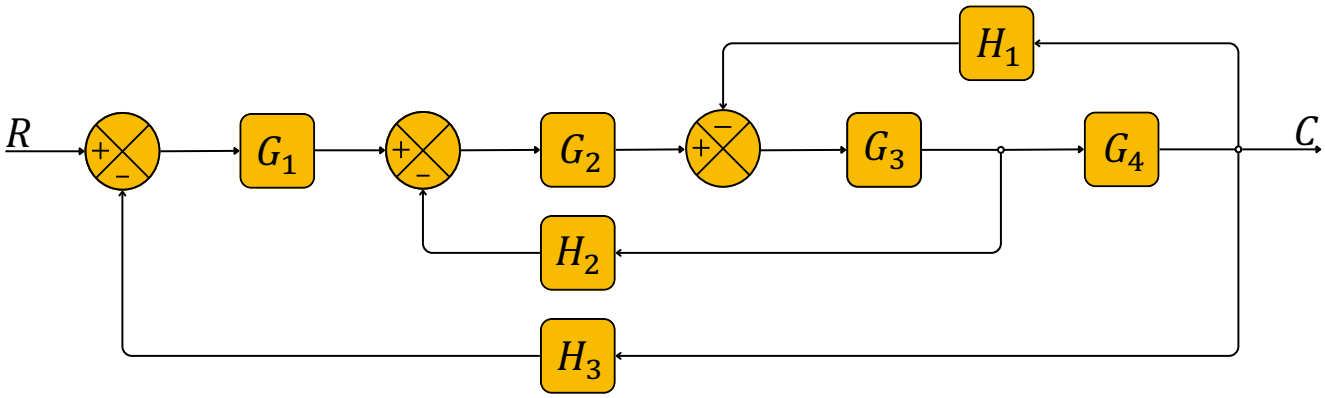


- Connect between nodes depending on arrow direction.

▶ **Key Notes:**

- If there is "1" gain between 2 nodes, it's the same node.
- DO NOT FORGET THE SIGN OF THE SUMMATOR IF IT WAS ADDER OR SUBSTRACTOR!

▶ **Q. 8.7:** Convert this block diagram to a signal flow graph:

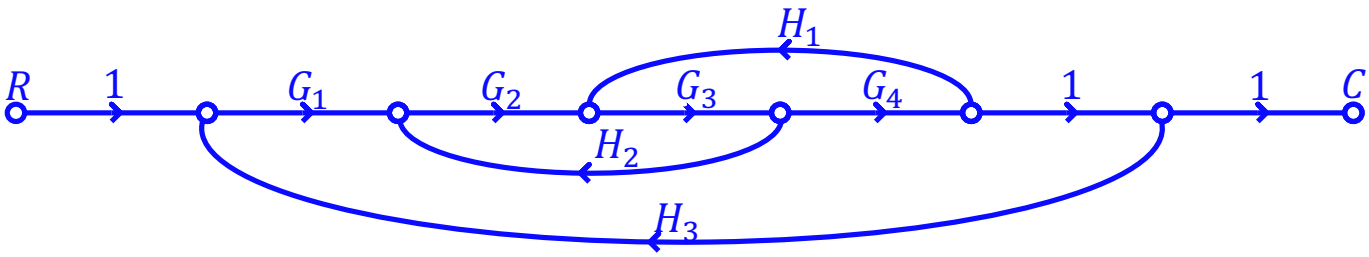


∴ *solution*:

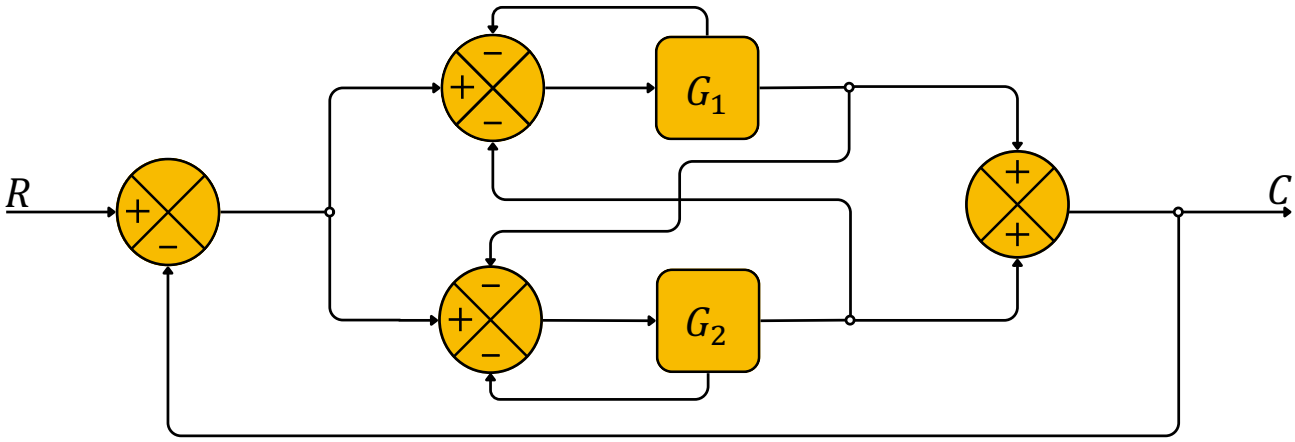
Firstly, replace input, output, summators and takeoff points by nodes then draw the forward path:



then, connect between nodes for the negative feedbacks "H_1, H_2 and H_3":

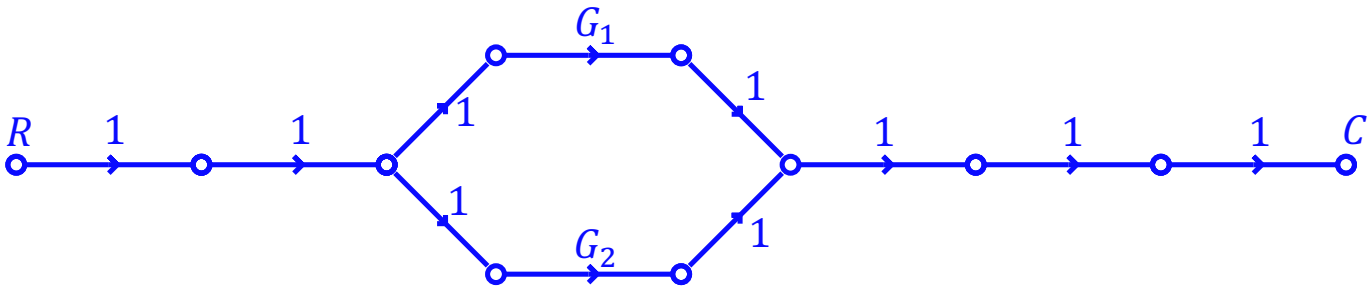


▶ **Q. 8.8:** Convert this block diagram to a signal flow graph:

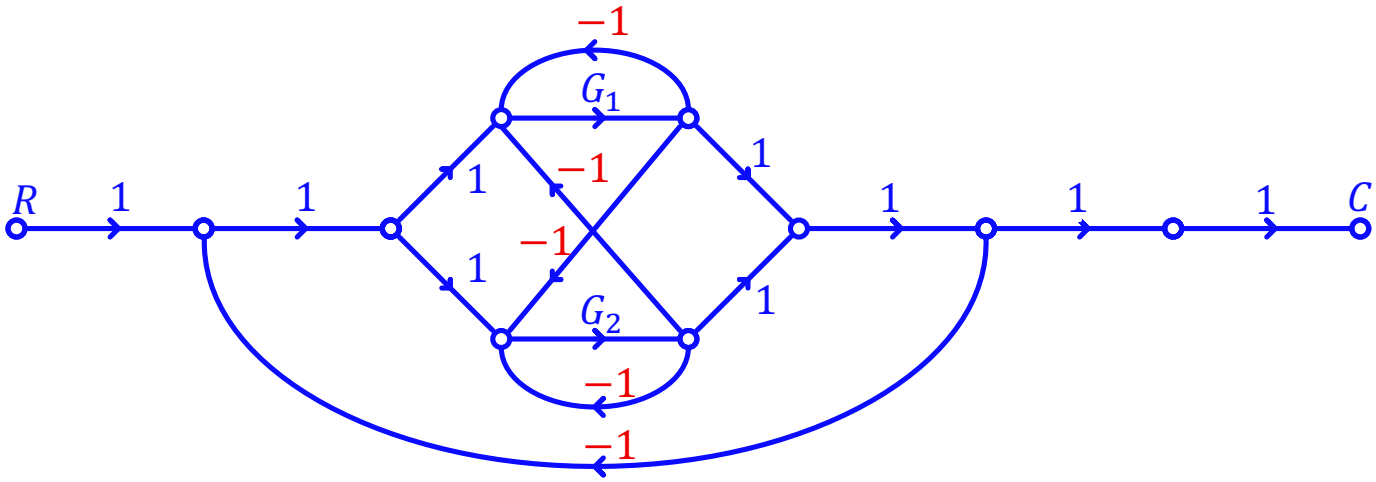


∴ *solution*:

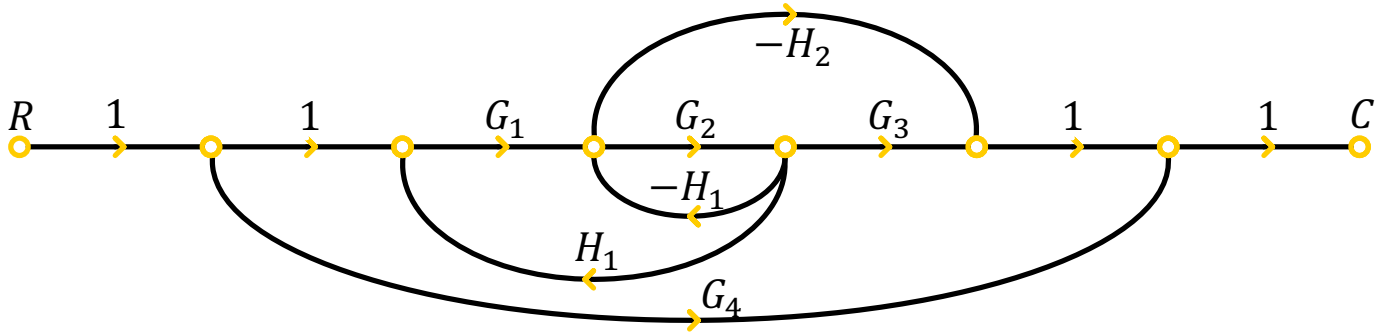
Firstly, replace input, output, summators and takeoff points by nodes then draw the forward paths:



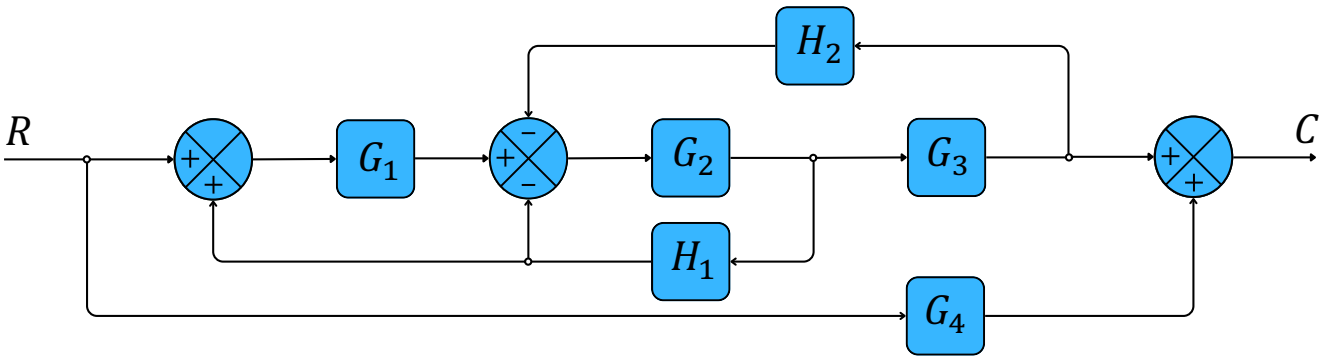
then, connect between nodes for the "−1" feedbacks:



Q. 8.9: Convert this signal flow graph to a block diagram:



$\therefore$ *solution*:



2024-2025's MID Exam (2nd Semester)



9. Time Domain Analysis

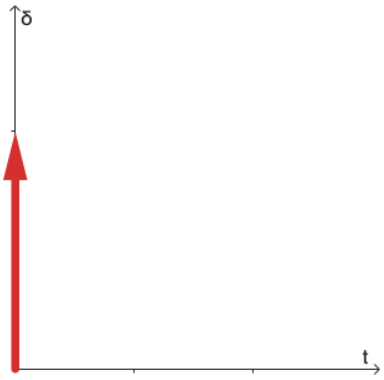
- Time-domain analysis shows how a system responds over time.
- If the system model and input are known, its response can be calculated and inputs are often unpredictable or not fully known.
- For example, in radar tracking, a target's movement may change randomly, so it's hard to describe such inputs with simple equations.
- Real input signals often involve sudden shocks, changes, constant velocity, or constant acceleration.
- To evaluate system behavior, standard test signals are used: impulse, step, constant velocity, and constant acceleration.
- Sinusoidal signals are also important standard test inputs.

◈ Standard Signals:

1. Impulse Signal:

- The impulse signal imitates the sudden shock characteristic of actual input signal.

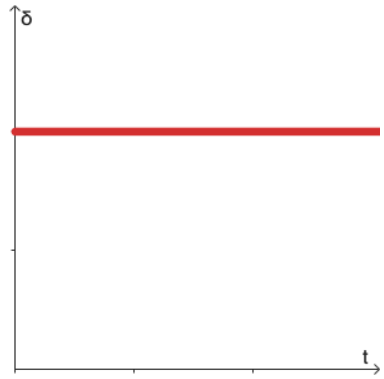
$$\delta(t) = \begin{cases} A, & t = 0 \\ 0, & t \neq 0 \end{cases}$$



2. Step Signal:

- The step signal imitates the sudden change characteristic of actual input signal.

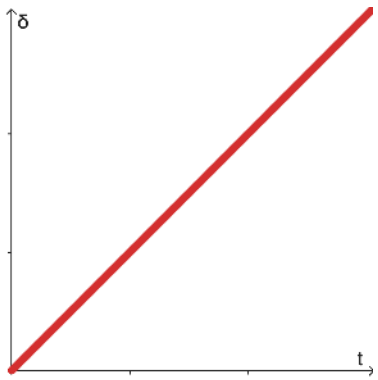
$$u(t) = \begin{cases} A, & t \geq 0 \\ 0, & t < 0 \end{cases}$$



3. Ramp Signal:

- The ramp signal imitates the constant velocity characteristic of actual input signal.

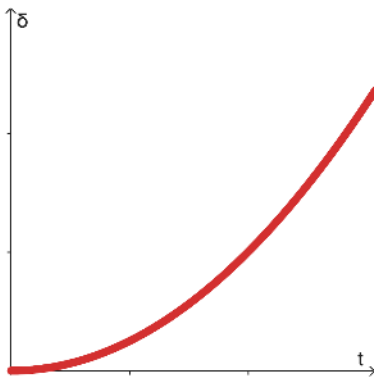
$$r(t) = \begin{cases} At, & t \geq 0 \\ 0, & t < 0 \end{cases}$$



4. Parabolic Signal:

- The parabolic signal imitates the constant acceleration characteristic of actual input signal.

$$p(t) = \begin{cases} \frac{At^2}{2}, & t \geq 0 \\ 0, & t < 0 \end{cases}$$



▶ **Key Notes:**

- In all of these signals, if $A = 1$, we say that is a unit signal; (unit impulse, unit step, unit ramp, unit parabolic).
- The constant A represents the amplitude in impulse and step signals, and the slope in ramp and parabolic signals.
- The relation between these signals is

$$\delta(t) \underset{d/dt}{\overset{\int dt}{\rightleftarrows}} u(t) \underset{d/dt}{\overset{\int dt}{\rightleftarrows}} r(t) \underset{d/dt}{\overset{\int dt}{\rightleftarrows}} p(t)$$

- The Laplace transform for each signal is:

$$\delta(t) \rightarrow \delta(s) = A, \qquad u(t) \rightarrow U(s) = \frac{A}{s}$$

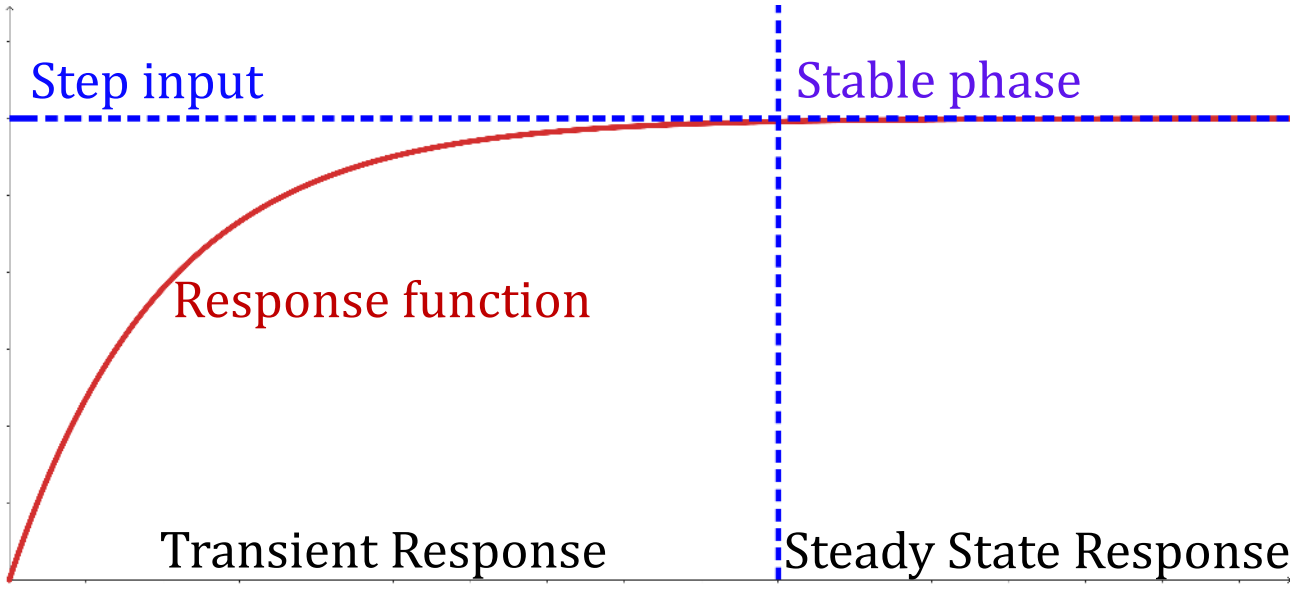
$$r(t) \rightarrow R(s) = \frac{A}{s^2}, \qquad p(t) \rightarrow P(s) = \frac{A}{s^3}$$

- DON'T GET CONFUSED ABOUT $R(s)$ AND $r(t)$ AS RAMP OR REFERENCE SIGNAL!

Time Response of Control Systems:

- The time response shows how a system reacts over time to an input and it has two parts:

1. **Transient Response:** The initial reaction as the system moves from rest to steady state.
2. **Steady State Response:** The Behavior after the system has stabilized.



- When a system changes from rest, it takes time to stabilize this is the transient phase.
- The transient response depends only on the system's poles, not the input type, so a step input is often used to analyze it.
- The steady state response depends on both the system and the input, and it is often evaluated using different test signals and the final value theorem.

First Order Systems:

- The order of a system is the highest degree (power) in the denominator of the system's transfer function.
- The first order systems have this formula in general:

$$\frac{C(s)}{R(s)} = \frac{K}{\tau s + 1}$$

$\therefore$ K is DC gain that is a ratio between the input signal and the steady state value of output.

$\therefore$ τ is Time constant that is a measure of how quickly a first order system responds to a unit step input.

- To find the time response $c(t)$ for each signal, we say that:

$$\frac{C(s)}{R(s)} = \frac{K}{\tau s + 1} \rightarrow C(s) = \frac{K \cdot R(s)}{\tau s + 1}$$

$$\mathcal{L}^{-1}\{C(s)\} = \mathcal{L}^{-1}\left\{\frac{K \cdot R(s)}{\tau s + 1}\right\}$$

$$c(t) = \mathcal{L}^{-1}\left\{\frac{K \cdot R(s)}{\tau s + 1}\right\}$$

- Now we replace $R(s)$ with the signal we want then take the inverse Laplace transform for this function.

Time Response for Standard Unit Signals:

1. Impulse Response:

$$R(s) = \mathcal{L}\{\delta(t)\} = 1$$

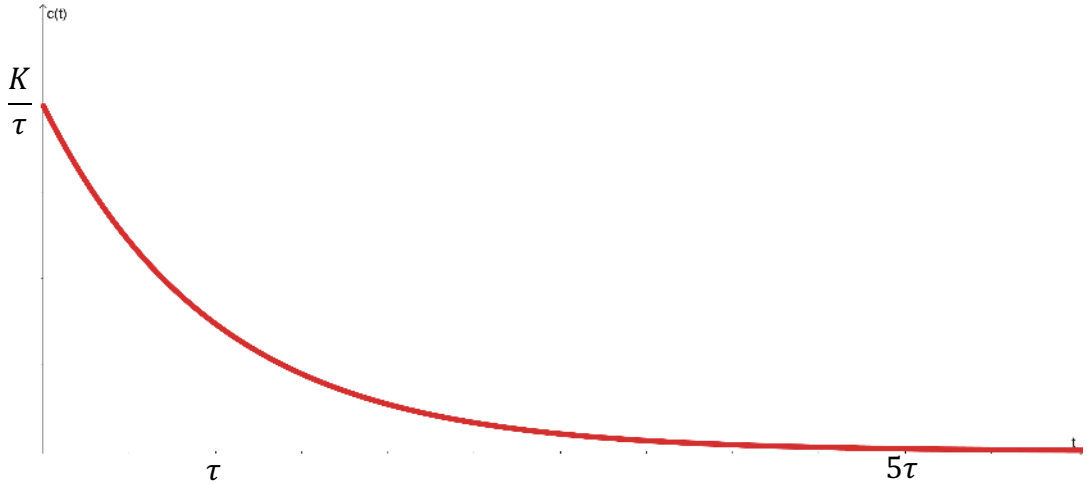
$$C(s) = \frac{K}{\tau s + 1} = \frac{K}{\tau(s + 1/\tau)}$$

$$c(t) = \mathcal{L}^{-1}\left\{\frac{K}{\tau(s + 1/\tau)}\right\} = \frac{K}{\tau}\mathcal{L}^{-1}\left\{\frac{1}{s + 1/\tau}\right\}$$

$$c_\delta(t) = \frac{K}{\tau}e^{-\frac{t}{\tau}} \cdot u(t)$$

Why $\cdot u(t)$? Because all values of time are positive, but you can ignore it.

- Graphically:



2. Step Response:

$$R(s) = \mathcal{L}\{u(t)\} = \frac{1}{s}$$

$$C(s) = \frac{K}{\tau s + 1} \cdot \frac{1}{s} = \frac{K}{s(\tau s + 1)}$$

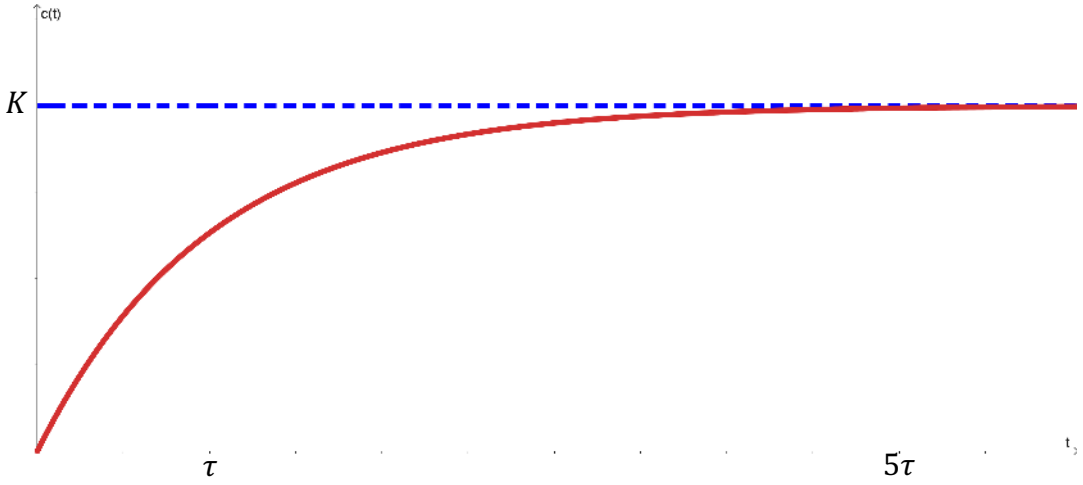
$$c(t) = \mathcal{L}^{-1}\left\{\frac{K}{s(\tau s + 1)}\right\} = K\mathcal{L}^{-1}\left\{\frac{1}{s(\tau s + 1)}\right\}$$

Using partial fractions:

$$= K\mathcal{L}^{-1}\left\{\frac{1}{s} - \frac{\tau}{\tau s + 1}\right\} = K\mathcal{L}^{-1}\left\{\frac{1}{s} - \frac{1}{s + 1/\tau}\right\}$$

$$c_u(t) = K\left(1 - e^{-\frac{t}{\tau}}\right) \cdot u(t)$$

- Graphically:



3. Ramp Response:

$$R(s) = \mathcal{L}\{r(t)\} = \frac{1}{s^2}$$

$$C(s) = \frac{K}{\tau s + 1} \cdot \frac{1}{s^2} = \frac{K}{s^2(\tau s + 1)}$$

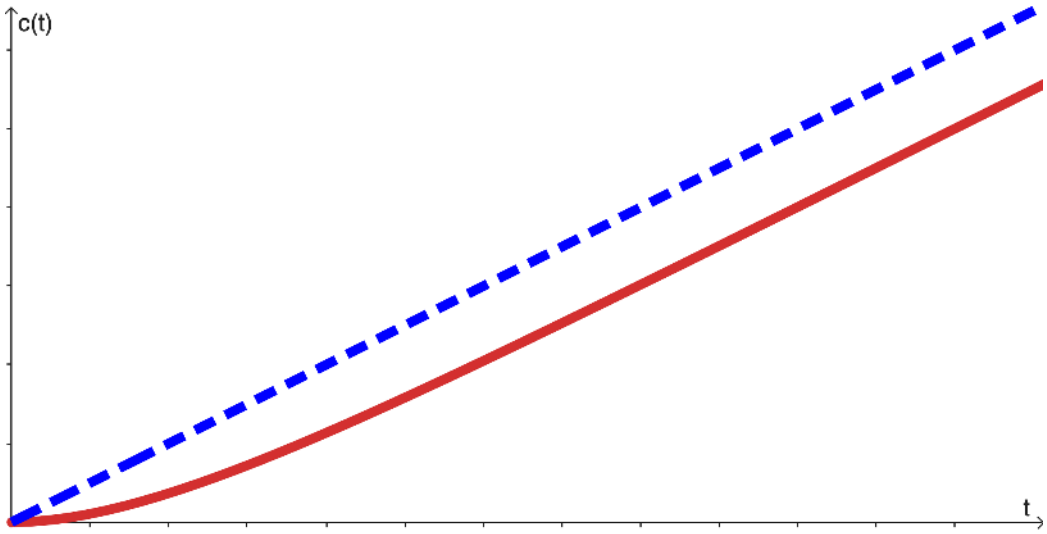
$$c(t) = \mathcal{L}^{-1}\left\{\frac{K}{s^2(\tau s + 1)}\right\} = K\mathcal{L}^{-1}\left\{\frac{1}{s^2(\tau s + 1)}\right\}$$

Using partial fractions:

$$= K\mathcal{L}^{-1}\left\{\frac{1}{s^2} - \frac{\tau}{s} + \frac{\tau^2}{\tau s + 1}\right\} = K\mathcal{L}^{-1}\left\{\frac{1}{s^2} - \frac{\tau}{s} + \frac{\tau}{s + 1/\tau}\right\}$$

$$c_r(t) = K\left(t - \tau\left(1 - e^{-\frac{t}{\tau}}\right)\right) \cdot u(t)$$

- Graphically:



4. Parabolic Response:

$$R(s) = \mathcal{L}\{p(t)\} = \frac{1}{s^3}$$

$$C(s) = \frac{K}{\tau s + 1} \cdot \frac{1}{s^3} = \frac{K}{s^3(\tau s + 1)}$$

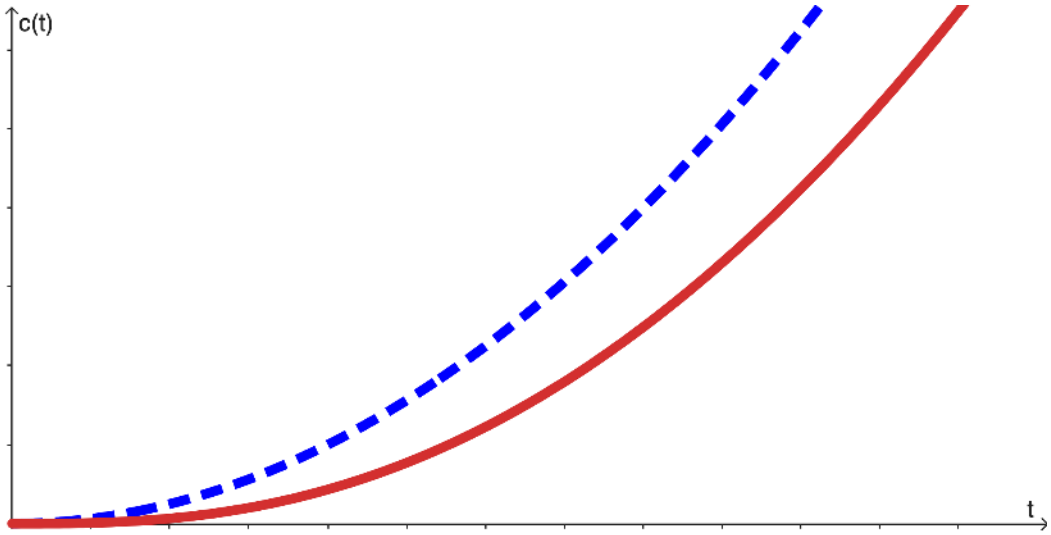
$$c(t) = \mathcal{L}^{-1}\left\{\frac{K}{s^3(\tau s + 1)}\right\} = K\mathcal{L}^{-1}\left\{\frac{1}{s^3(\tau s + 1)}\right\}$$

Using partial fractions:

$$= K\mathcal{L}^{-1}\left\{\frac{\tau^2}{s} - \frac{\tau}{s^2} + \frac{1}{s^3} - \frac{\tau^3}{\tau s + 1}\right\}$$

$$c_p(t) = K\left(\frac{t^2}{2} - \tau\left(t - \tau\left(1 - e^{-\frac{t}{\tau}}\right)\right)\right) \cdot u(t)$$

- Graphically:



- Also, we can say from signals relations:

$$K\left(\frac{1}{\tau}e^{-\frac{t}{\tau}}\right)$$

$$d/dt \uparrow \text{ Input: } \delta(t) \downarrow \int dt$$

$$K\left(1-e^{-\frac{t}{\tau}}\right)$$

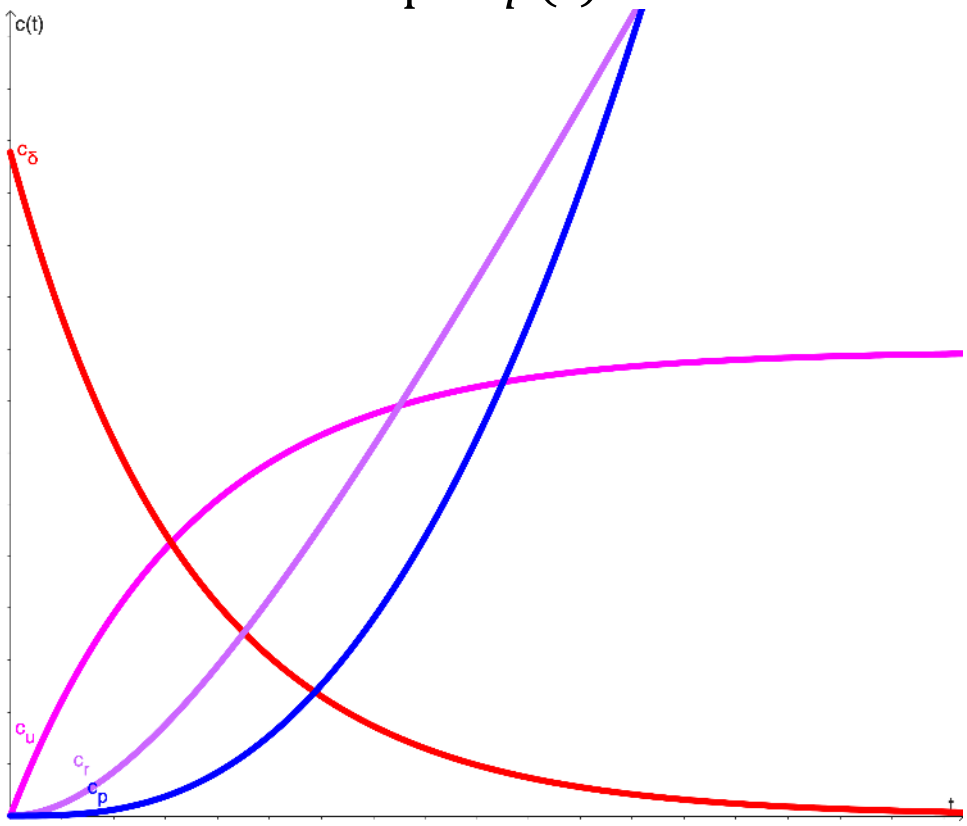
$$d/dt \uparrow \text{ Input: } u(t) \downarrow \int dt$$

$$K\left(t-\tau\left(1-e^{-\frac{t}{\tau}}\right)\right)$$

$$d/dt \uparrow \text{ Input: } r(t) \downarrow \int dt$$

$$K\left(\frac{t^2}{2}-\tau\left(t+\tau\left(1-e^{-\frac{t}{\tau}}\right)\right)\right)$$

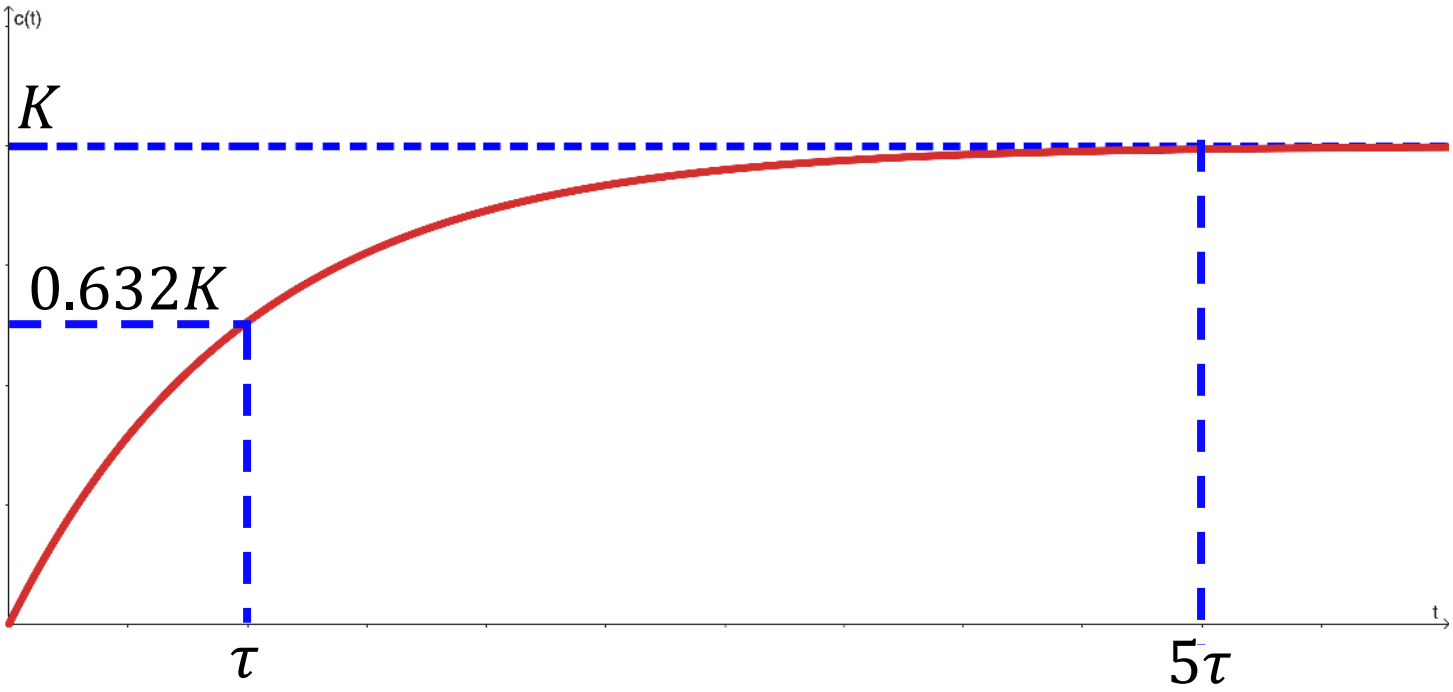
Input: $p(t)$



- When it's difficult or impossible to obtain a system's transfer function analytically, you can still model it using its step response, by applying a step input, we can measure the time constant τ and DC gain K by supposing that the system after 1 time constant it will reach 0.632 of its DC gain value:

$$c_u(\tau) = K\left(1 - e^{-\frac{\tau}{\tau}}\right) = 0.632K$$

- Then you will know the value of τ from graph.



▶ **Q. 9.1:** Find K and τ for the following transfer functions:

$\therefore G(s) = \frac{5}{2s+1}$

Compare directly with the main formula:

$$\frac{C(s)}{R(s)} = \frac{K}{\tau s + 1} = \frac{5}{2s + 1} \rightarrow K = 5, \qquad \tau = 2$$

$\therefore G(s) = \frac{3}{s+5}$

To find τ, we must make it as the same formula above "+1" is essential, so:

$$\frac{3}{s + 5} = \frac{3}{5\left(\frac{s}{5} + 1\right)} = \frac{\frac{3}{5}}{\frac{1}{5}s + 1} \rightarrow K = \frac{3}{5}, \qquad \tau = \frac{1}{5}$$

▶ **Q. 9.2:** For this first order impulse response:

$$c(t) = 3e^{-\frac{t}{2}}$$

Find:

∴ Time constant τ and DC gain K.

From the main formula of impulse response:

$$c_\delta(t) = \frac{K}{\tau} e^{-\frac{t}{\tau}} = 3e^{-\frac{t}{2}}$$

$$e^{-\frac{t}{\tau}} = e^{-\frac{t}{2}} \rightarrow \tau = 2$$

$$\frac{K}{\tau} = 3 \rightarrow 3\tau = K = 6$$

∴ Transfer Function.

From the main formula of first order systems:

$$C(s) = \frac{K}{\tau s + 1} = \frac{6}{2s + 1}$$

∴ Step response.

From signals relations:

$$c_u(t) = \int c_\delta \, dt = \int 3e^{-\frac{t}{2}} \, dt = -6e^{-\frac{t}{2}} + c$$

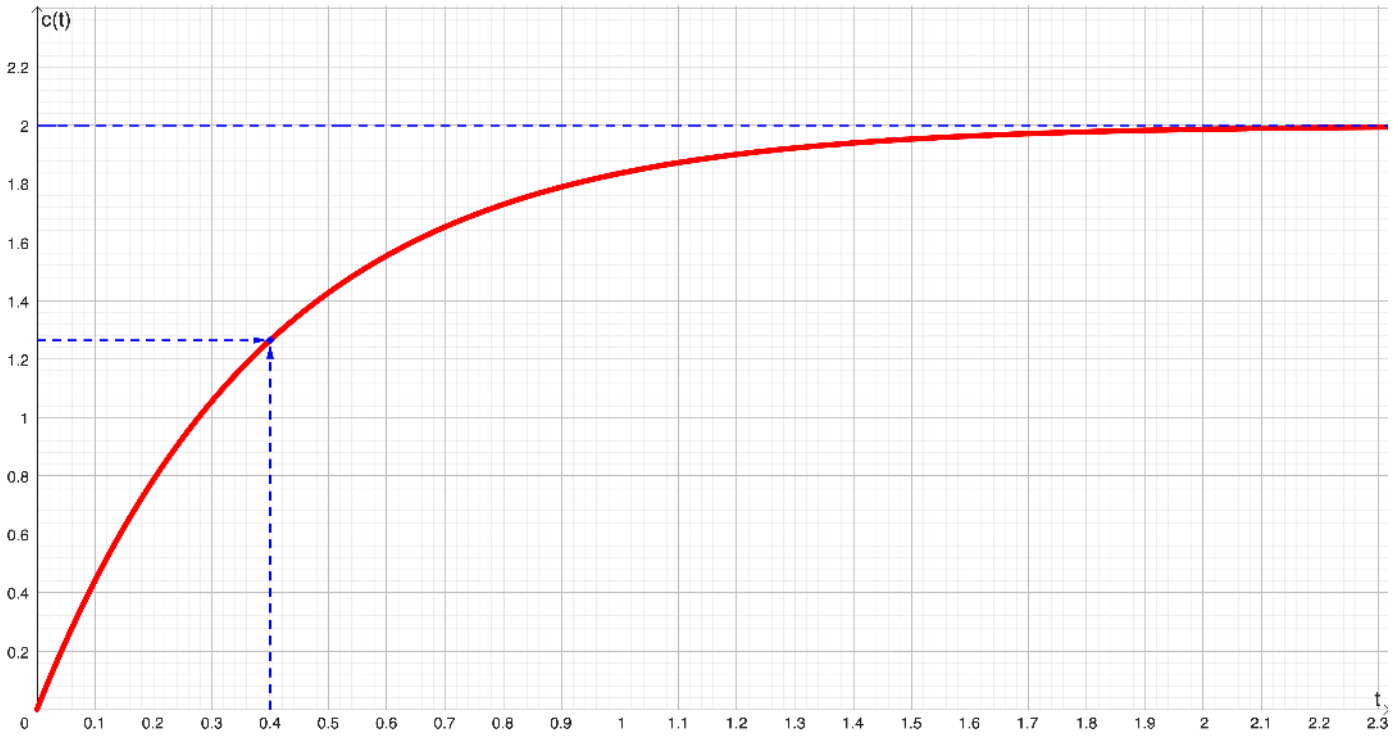
We know that $c_u(0) = 0$ "from graph" so:

$$0 = -6e^0 + c \rightarrow c = 6 \rightarrow c_u(t) = 6 - 6e^{-\frac{t}{2}}$$

OR from the main formula of step response:

$$c_u(t) = K\left(1 - e^{-\frac{t}{\tau}}\right) = 6\left(1 - e^{-\frac{t}{2}}\right)$$

▶ Q. 9.3: For this first order step response graph:



Find:

∴ Time constant τ and DC gain K

From graph, the system settles on 2 after too many times, so:

$$K = 2$$

$$\tau \rightarrow 0.632K = 1.264$$

From graph, the system reaches the value of 1.264 after 0.4s:

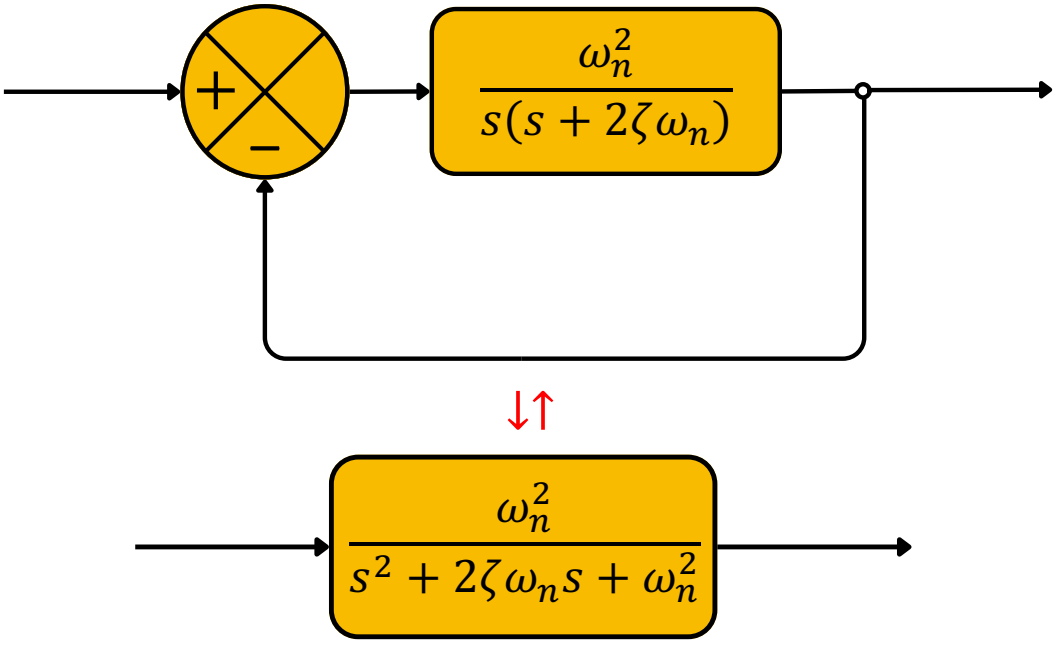
$$\tau = 0.4$$

∴ Transfer function.

$$G(s) = \frac{K}{\tau s + 1} = \frac{2}{0.4s + 1}$$

Second Order Systems:

- Second order systems can display characteristics much like a first-order system or, depending on component values, display damped or pure oscillations for its transient response.



- The second order systems have this formula in general:

$$\frac{C(s)}{R(s)} = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

$\therefore$ ζ is Damping Ratio that is a measure of the degree of resistance to change in the system output "pronounced zeta".

$\therefore$ ω_n is Un-Damped Natural Frequency that is the frequency of oscillation of the system without damping "when $\zeta = 0$".

- The poles of second order system are given by:

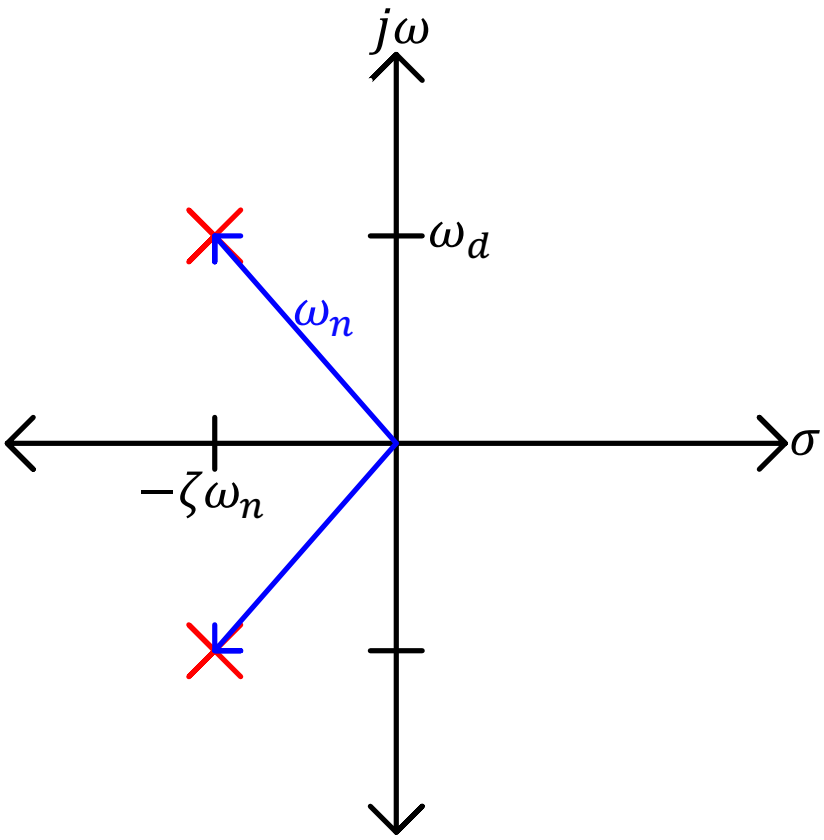
$$s^2 + 2\zeta\omega_n s + \omega_n^2 = 0$$

$$s = \frac{-2\zeta\omega_n \pm \sqrt{(2\zeta\omega_n)^2 - 4\omega_n^2}}{2} \rightarrow s = -\omega_n\zeta \pm \omega_n\sqrt{\zeta^2 - 1}$$

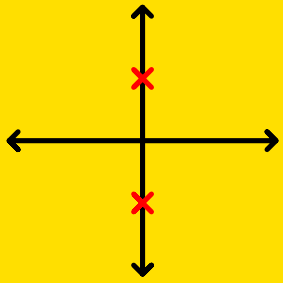
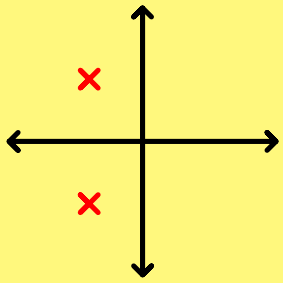
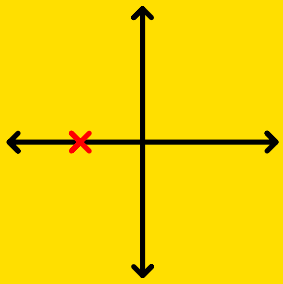
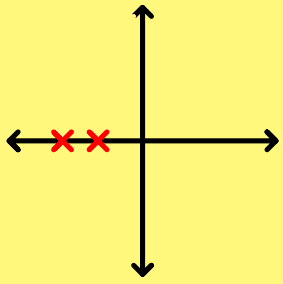
- So, the constant that characterize the nature of the system is zeta ζ due to the value of the pole.
- Know this information:

$$\omega_d = \omega_n\sqrt{1 - \zeta^2}$$

$\therefore$ ω_d is the Damped Natural Frequency that is the frequency at which the system actually oscillates due to damping.



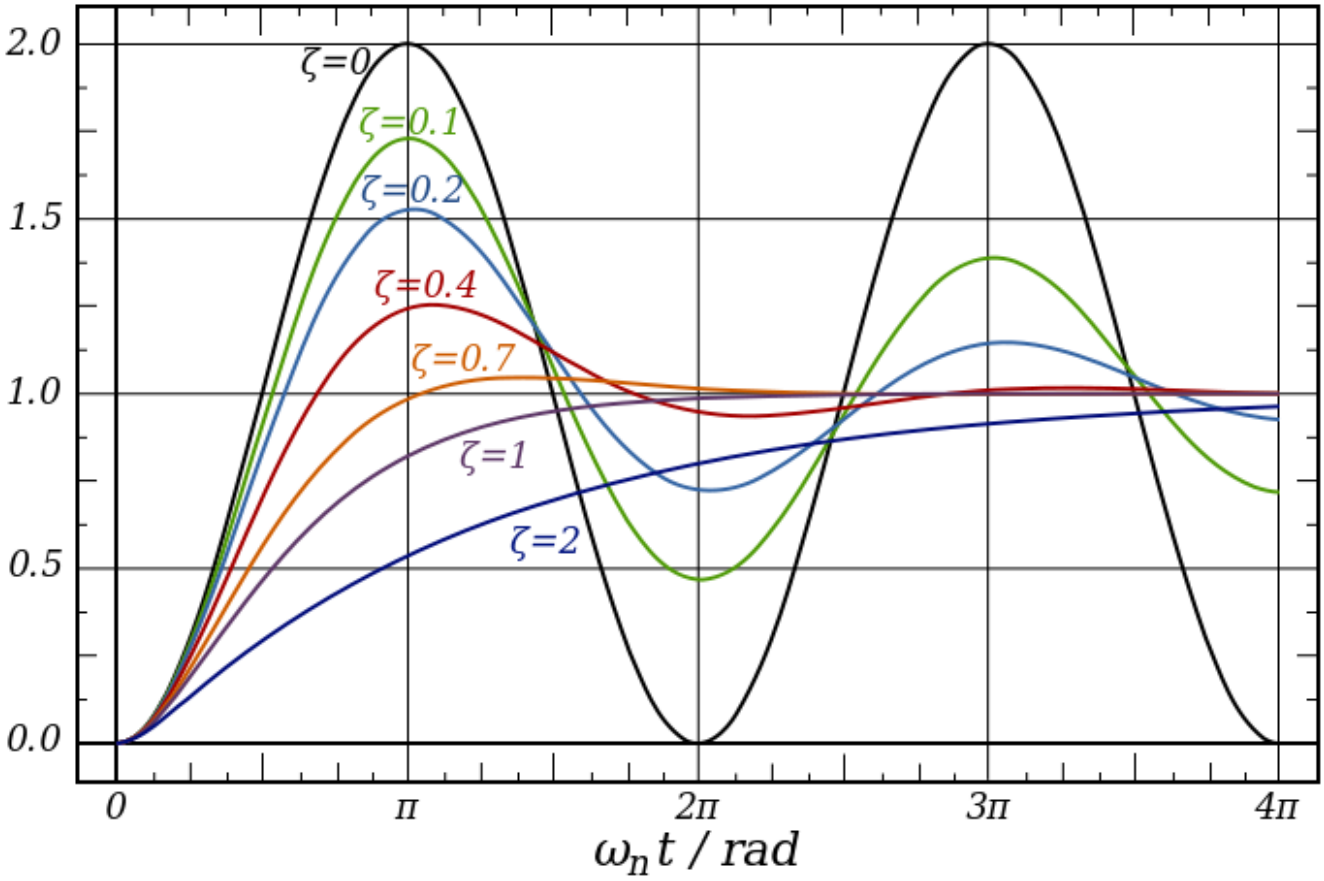
Cases of ζ:

Value of ζ	Case name	Nature of poles	Poles graph
$\zeta = 0$	Undamped	2 Imaginaries $\pm bj$	
$0 < \zeta < 1$	Underdamped	2 Complex conjugates $-a \pm bj$	
$\zeta = 1$	Critically Damped	2 Real repeated $-a, -a$	
$\zeta > 1$	Overdamped	2 Real distinct $-a_1, -a_2$	

Key Note:

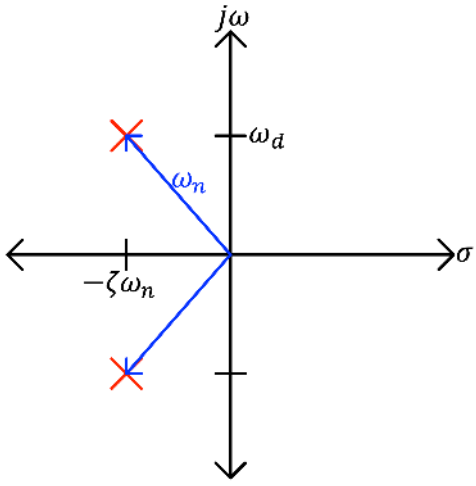
- a and b are positives real numbers.
- $(x, y) \rightarrow (\{Re\}, \{Im\}) \rightarrow (\sigma,\ j\omega)$

- In the following graph we see the differences between zeta's cases:



- The amplitude of the poles is:

$$|s| = \sqrt{(-\omega_n \zeta)^2 + \left(\omega_n \sqrt{\zeta^2 - 1}\right)^2} = \omega_n$$



- Take the underdamped case ($0 < \zeta < 1$):

$$\cos\theta = \frac{\text{adjacent}}{\text{hypotenuse}} = \frac{\zeta\omega_n}{\omega_n} = \zeta$$

$$\theta = \cos^{-1}\zeta$$

▶ Step Response of Underdamped Systems:

- As we do in first order systems:

$$\frac{C(s)}{R(s)} = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} \rightarrow C(s) = \frac{R(s) \cdot \omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

$$R(s) = \mathcal{L}\{u(t)\} = \frac{1}{s} \rightarrow C(s) = \frac{\omega_n^2}{s \cdot (s^2 + 2\zeta\omega_n s + \omega_n^2)}$$

Using partial fractions:

$$C(s) = \frac{1}{s} - \frac{s + 2\zeta\omega_n}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

Using square complement ($\pm b^2/2$):

$$C(s) = \frac{1}{s} - \frac{s + 2\zeta\omega_n}{(s + \zeta\omega_n)^2 + \left(\omega_n\sqrt{1 - \zeta^2}\right)^2}$$

Let $\omega_d = \omega_n\sqrt{1 - \zeta^2}$:

$$C(s) = \frac{1}{s} - \frac{s + 2\zeta\omega_n}{(s + \zeta\omega_n)^2 + (\omega_d)^2} \dots$$

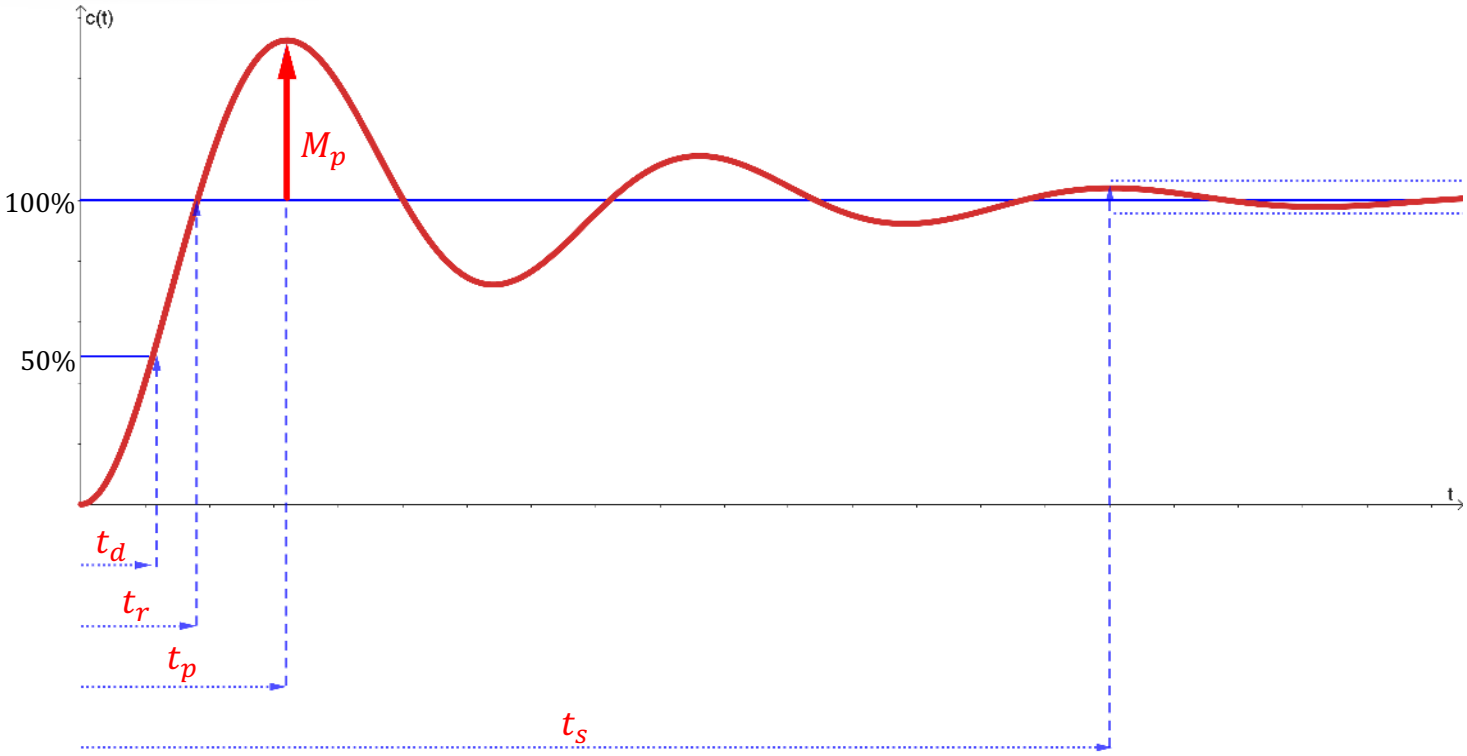
After simplifying:

$$C(s) = \frac{1}{s} - \frac{s + \zeta\omega_n}{(s + \zeta\omega_n)^2 + (\omega_d)^2} - \frac{\zeta}{\sqrt{1 - \zeta^2}} \frac{\omega_d}{(s + \zeta\omega_n)^2 + (\omega_d)^2}$$

Take $\mathcal{L}^{-1}\{C(s)\}$:

$$c(t) = 1 - e^{-\zeta\omega_n t}\left(\cos\omega_d t + \frac{\zeta}{\sqrt{1 - \zeta^2}}\sin\omega_d t\right)$$

- We can conclude from its graph:



∴ **Delay Time t_d**: the time required for the response to reach half the final value.

∴ **Rise Time t_r**: the time required for the response to rise from 10%→90%, 5%→95%, or 0%→100% of its final value.

∴ **Maximum Overshoot M_p**: the maximum peak value of the response curve measured from the final value.

∴ **Peak Time t_p**: the time required for the response to reach the first peak of the overshoot.

∴ **Settling Time t_s**: the time required for the response curve to reach and stay within a range about the final value of size specified by absolute percentage of the final value (usually 2% or 5%).

- We can conclude them mathematically here:

$$t_r = \frac{\pi - \theta}{\omega_d} = \frac{\pi - \cos^{-1}\zeta}{\omega_n\sqrt{1-\zeta^2}}$$

$$t_p = \frac{\pi}{\omega_d} = \frac{\pi}{\omega_n\sqrt{1-\zeta^2}}$$

$$M_p = e^{-\frac{\pi\zeta}{\sqrt{1-\zeta^2}}} \cdot 100\%$$

$$\zeta = \frac{-\ln M_p}{\sqrt{\pi^2 + (\ln M_p)^2}}$$

$$t_s = \begin{cases} 3\tau = \dfrac{3}{\zeta\omega_n}, & \text{settling on 5\% criterion} \\ 4\tau = \dfrac{4}{\zeta\omega_n}, & \text{settling on 2\% criterion} \end{cases}$$

▶ **Q. 9.4:** For this second order system:

$$\frac{C(s)}{R(s)} = \frac{4}{s^2 + 2s + 4}$$

Find:

∴ Undamped natural frequency ω_n.

We compare this second order system with main formula:

$$\frac{C(s)}{R(s)} = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} = \frac{4}{s^2 + 2s + 4}$$

We conclude directly that:

$$\omega_n^2 = 4 \rightarrow \omega_n = 2 \text{ rad/s}$$

∴ Damping ratio ζ.

From s factor:

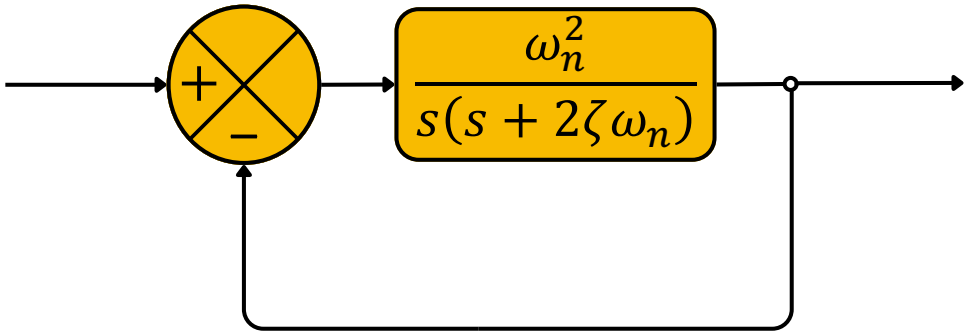
$$2\zeta\omega_n = 2 \rightarrow \zeta\omega_n = 1 \rightarrow \zeta(2) = 1 \rightarrow \zeta = \frac{1}{2}$$

∴ Damped natural frequency ω_d.

$$\omega_d = \omega_n\sqrt{1 - \zeta^2} = 2\sqrt{1 - \left(\frac{1}{2}\right)^2} = 2\sqrt{\frac{3}{4}}$$

$$\omega_d = \sqrt{3} \text{ rad/s}$$

▶ **Q. 9.5:** Consider this system:



Where damping ratio ζ =0.6, natural undamped frequency ω_n =5 rad/sec and the input is unit step. Find:

∴ Rise Time t_r.

$$t_r = \frac{\pi - \theta}{\omega_d} = \frac{\pi - \cos^{-1}(\zeta)}{\omega_n\sqrt{1-\zeta^2}} = \frac{\pi - \cos^{-1}(0.6)}{5\sqrt{1-(0.6)^2}} = 0.553\ \text{s}$$

∴ Peak Time t_p.

$$t_p = \frac{\pi}{\omega_d} = \frac{\pi}{\omega_n\sqrt{1-\zeta^2}} = \frac{\pi}{5\sqrt{1-0.6^2}} = 0.785\ \text{s}$$

∴ Maximum Overshoot M_p.

$$M_p = e^{-\frac{\pi\zeta}{\sqrt{1-\zeta^2}}} \cdot 100\% = e^{-\frac{\pi(0.6)}{\sqrt{1-(0.6)^2}}} \cdot 100\% = 9.47\%$$

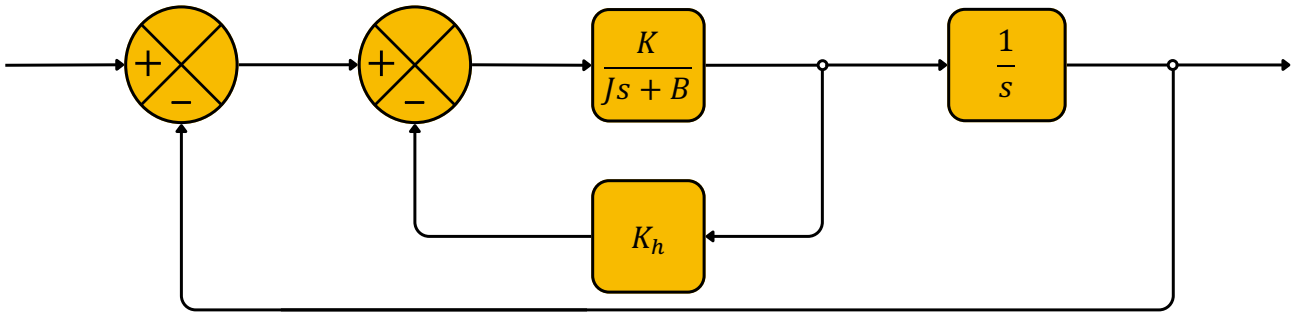
∴ Settling Time 5% criterion.

$$t_{s_{(5\%)}} = \frac{3}{\zeta\omega_n} = \frac{3}{5 \cdot 0.6} = 1\ \text{s}$$

∴ Settling Time 2% criterion.

$$t_{s_{(2\%)}} = \frac{4}{\zeta\omega_n} = \frac{4}{5 \cdot 0.6} = 1.33\ \text{s}$$

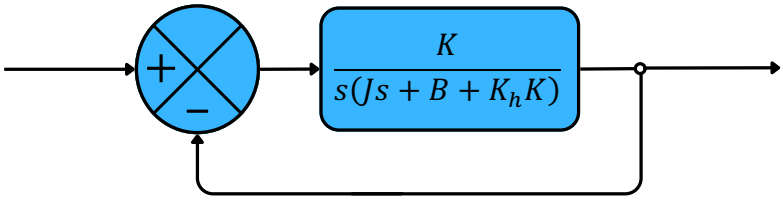
Q. 9.6: Consider this system:



Where maximum overshoot M_p =20%, peak time t_p =1 sec and the constants $J = 1\ \text{Kg}\cdot\text{m}^2$ and $B = 1\ \frac{\text{N}\cdot\text{m}}{\text{rad/s}}$. Find:

$\therefore$ Damping ratio ζ.

First, simplify the block diagram (take the internal feedback loop then multiply it with $1/s$):



Then the overall transfer function is:

$$\frac{K}{s(Js+B+K_hK)+K} = \frac{K}{s^2+(1+K_hK)s+K}$$

Compare it with the main formula or 2nd order systems:

$$\frac{\omega_n^2}{s^2+2\zeta\omega_n s+\omega_n^2} = \frac{K}{s^2+(1+K_hK)s+K}$$

$$K=\omega_n^2, \qquad 2\zeta\omega_n = (1+K_hK) \rightarrow K_h = \frac{2\zeta\omega_n - 1}{K}$$

From the information of maximum overshoot = 20%

$$M_p = e^{-\frac{\pi\zeta}{\sqrt{1-\zeta^2}}} \rightarrow e^{-\frac{\pi\zeta}{\sqrt{1-\zeta^2}}} = 0.2 \rightarrow -\frac{\pi\zeta}{\sqrt{1-\zeta^2}} = \ln 0.2$$

$$\rightarrow -\pi\zeta = \ln 0.2\left(\sqrt{1-\zeta^2}\right) \rightarrow \pi^2\zeta^2 = \ln^2 0.2\,(1-\zeta^2)$$

$$\rightarrow (\pi^2 + \ln^2 0.2)\zeta^2 = \ln^2 0.2 \rightarrow \zeta^2 = \frac{\ln^2 0.2}{\pi^2 + \ln^2 0.2}$$

$$\rightarrow \zeta = \sqrt{\frac{\ln^2 0.2}{\pi^2 + \ln^2 0.2}} \rightarrow \zeta = 0.456$$

$\therefore$ Undamped natural frequency ω_n.

From the information of peak time = 5 sec:

$$t_p = \frac{\pi}{\omega_n\sqrt{1-\zeta^2}} \rightarrow \omega_n = \frac{\pi}{t_p\sqrt{1-\zeta^2}} = \frac{\pi}{\sqrt{1-0.456^2}}$$

$$\omega_n = 3.53 \text{ rad/s}$$

$\therefore$ Gain constant K and Velocity-feedback constant K_h.

$$K = \omega_n^2 = 3.53^2 = 12.46$$

$$K_h = \frac{2\zeta\omega_n - 1}{K} = \frac{2(0.456)(3.53) - 1}{12.46} = 0.178$$

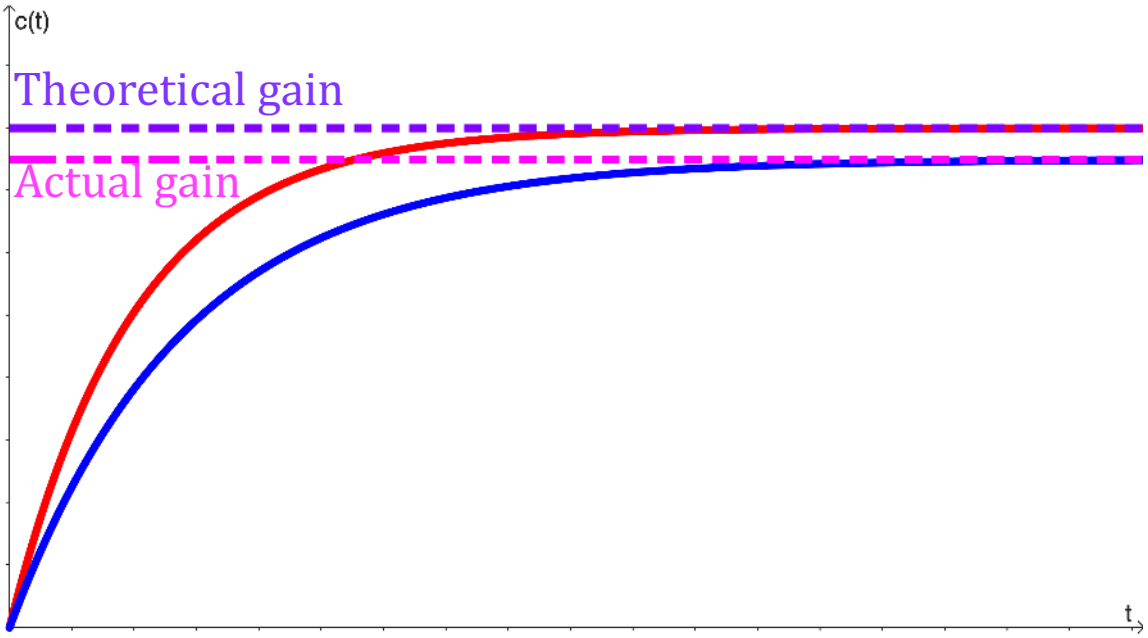
$\therefore$ Settle time on 5% and 2%.

$$t_{s_{5\%}} = \frac{3}{\zeta\omega_n} = \frac{3}{(0.456)(3.53)} = 1.86\text{s}$$

$$t_{s_{2\%}} = \frac{4}{\zeta\omega_n} = \frac{4}{(0.456)(3.53)} = 2.48\text{s}$$

10. Steady State Error

• We have learned before in time domain analysis that at step response the system will almost be stable on the value of DC gain K after 5 time constants τ theoretically, actually, the system will not reach K exactly, there will be a difference between the actual value and the calculated value.
[this difference is known as steady-state error]



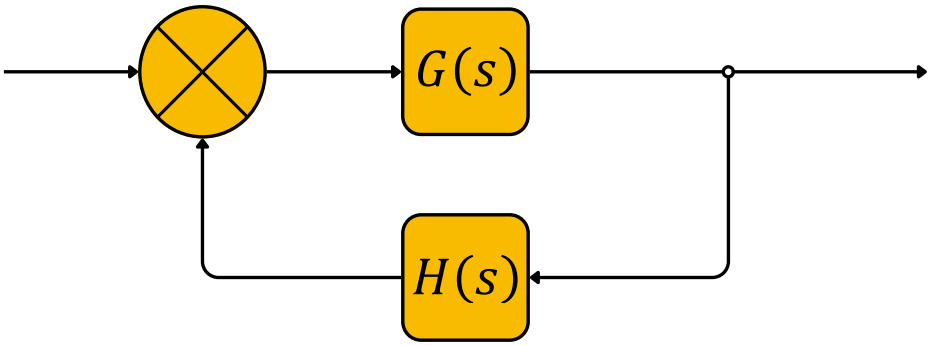
• All physical control systems exhibit some degree of steady-state error in response to certain input types.

• The presence and magnitude of steady-state error depend on the open-loop transfer function of the system.

• The magnitude of the steady-state error for these standard inputs reflects the performance quality of the control system.

▶ Steady State Error of Unity Feedback Systems:

- Consider the unity-feedback control system with the following open-loop transfer function:



$$GH(s) = \frac{K(\tau_a s + 1)(\tau_b s + 1)\cdots(\tau_m s + 1)}{s^N(\tau_1 s + 1)(\tau_2 s + 1)\cdots(\tau_n s + 1)}$$

- N is Type Number that classifies the type of the system:
- As the type number is increased, accuracy is improved, and the stability is aggravated, so compromising between steady-state accuracy and relative stability is always necessary.
- From error ratio:

$$\frac{E(s)}{R(s)} = \frac{1}{1 + GH(s)} \rightarrow E(s) = \frac{R(s)}{1 + GH(s)}$$

- As we see, to find error in time domain, we can use final value theorem from Laplace transform's properties:

$$e_{ss}(t) = \lim_{t\to\infty} e(t) = \lim_{s\to 0} s \cdot E(s)$$
$$= \lim_{s\to 0} s \frac{R(s)}{1 + GH(s)}$$

Static Error Constants:

- Static Error Constants: figures of merit of control systems. The higher the constants, the smaller the steady-state error.
- In different systems, the output might be something like position, speed, pressure, or temperature. To keep things simple, we'll use "position" as a general term to refer to the system's output. Likewise, we'll use "velocity" to mean the rate at which the output is changing over time.

For Example:

- in a temperature control system:

∴ The term "Position" will represent the temperature itself.

∴ The term "Velocity" will represent the rate of change of the temperature.

- This convention helps standardize the terminology across various types of control systems.

▶ We Will Study About 3 Static Error Constants:

1. Static Position Error Constant K_p:

- This Constant is the steady state error for the input $u(t)$:

$$GH(s) = \frac{K(\tau_a s + 1)(\tau_b s + 1)\cdots}{s^N(\tau_1 s + 1)(\tau_2 s + 1)\cdots}$$

$$R(s) = \frac{1}{s} \rightarrow E(s) = \frac{R(s)}{1 + GH(s)} = \frac{1}{s\big(1 + GH(s)\big)}$$

$$e_{ss}(t) = \lim_{s\to 0} s \cdot E(s) = \lim_{s\to 0} s \cdot \frac{1}{s(1 + GH(s))} = \frac{1}{1 + \lim_{s\to 0} GH(s)}$$

$$K_p = \lim_{s\to 0} GH(s)$$

$\therefore$ For type 0 systems:

$$\text{Type } 0 \rightarrow N = 0:$$

$$GH(s) = \frac{K(\tau_a s + 1)(\tau_b s + 1)\cdots}{s^0(\tau_1 s + 1)(\tau_2 s + 1)\cdots} = K\frac{(\tau_a s + 1)(\tau_b s + 1)\cdots}{(\tau_1 s + 1)(\tau_2 s + 1)\cdots}$$

$$K_p = \lim_{s\to 0} GH(s) = K\frac{(\tau_a(0) + 1)(\tau_b(0) + 1)\cdots}{(\tau_1(0) + 1)(\tau_2(0) + 1)\cdots} = K$$

$$e_{ss}(t) = \frac{1}{1 + K_p} = \frac{1}{1 + K}$$

∴ For type 1 systems:

$$\text{Type } 1 \rightarrow N = 1:$$

$$GH(s) = \frac{K(\tau_a s + 1)(\tau_b s + 1)\cdots}{s^1(\tau_1 s + 1)(\tau_2 s + 1)\cdots} = K\frac{(\tau_a s + 1)(\tau_b s + 1)\cdots}{s(\tau_1 s + 1)(\tau_2 s + 1)\cdots}$$

$$K_p = \lim_{s\to 0} GH(s) = K\frac{(\tau_a(0) + 1)(\tau_b(0) + 1)\cdots}{(0)(\tau_1(0) + 1)(\tau_2(0) + 1)\cdots} = \infty$$

$$e_{ss}(t) = \frac{1}{1 + K_p} = \frac{1}{1 + \infty} = 0$$

∴ For type 2 systems:

$$\text{Type } 2 \rightarrow N = 2:$$

$$GH(s) = \frac{K(\tau_a s + 1)(\tau_b s + 1)\cdots}{s^2(\tau_1 s + 1)(\tau_2 s + 1)\cdots} = K\frac{(\tau_a s + 1)(\tau_b s + 1)\cdots}{s^2(\tau_1 s + 1)(\tau_2 s + 1)\cdots}$$

$$K_p = \lim_{s\to 0} GH(s) = K\frac{(\tau_a(0) + 1)(\tau_b(0) + 1)\cdots}{(0)^2(\tau_1(0) + 1)(\tau_2(0) + 1)\cdots} = \infty$$

$$e_{ss}(t) = \frac{1}{1 + K_p} = \frac{1}{1 + \infty} = 0$$

2. Static Velocity Error Constant K_v:

- This Constant is the steady state error for the input $r(t)$:

$$GH(s) = \frac{K(\tau_a s + 1)(\tau_b s + 1)\cdots}{s^N(\tau_1 s + 1)(\tau_2 s + 1)\cdots}$$

$$R(s) = \frac{1}{s^2} \rightarrow E(s) = \frac{R(s)}{1 + GH(s)} = \frac{1}{s^2\big(1 + GH(s)\big)}$$

$$e_{ss}(t) = \lim_{s\to 0} s \cdot E(s) = \lim_{s\to 0} s \cdot \frac{1}{s^2\big(1 + GH(s)\big)}$$

$$= \lim_{s\to 0} \frac{1}{s\big(1 + GH(s)\big)} = \lim_{s\to 0} \frac{1}{s + s \cdot GH(s)}$$

$$= \frac{1}{0 + \lim_{s\to 0} s \cdot GH(s)} = \frac{1}{\lim_{s\to 0} s \cdot GH(s)}$$

$$K_v = \lim_{s\to 0} s \cdot GH(s)$$

$\therefore$ For type 0 systems:

$$\text{Type } 0 \rightarrow N = 0:$$

$$GH(s) = K\frac{(\tau_a s + 1)(\tau_b s + 1)\cdots}{s^0(\tau_1 s + 1)(\tau_2 s + 1)\cdots} = K\frac{(\tau_a s + 1)(\tau_b s + 1)\cdots}{(\tau_1 s + 1)(\tau_2 s + 1)\cdots}$$

$$K_v = \lim_{s\to 0} s \cdot GH(s) = \lim_{s\to 0} Ks\frac{(\tau_a s + 1)(\tau_b s + 1)\cdots}{(\tau_1 s + 1)(\tau_2 s + 1)\cdots}$$

$$= K(0)\frac{(\tau_a(0) + 1)(\tau_b(0) + 1)\cdots}{(\tau_1(0) + 1)(\tau_2(0) + 1)\cdots} = 0$$

$$e_{ss}(t) = \frac{1}{K_v} = \frac{1}{0} = \infty$$

∴ For type 1 systems:

$$\text{Type } 1 \rightarrow N = 1:$$

$$GH(s) = \frac{K(\tau_a s + 1)(\tau_b s + 1)\cdots}{s^1(\tau_1 s + 1)(\tau_2 s + 1)\cdots} = K\frac{(\tau_a s + 1)(\tau_b s + 1)\cdots}{s(\tau_1 s + 1)(\tau_2 s + 1)\cdots}$$

$$K_v = \lim_{s\to 0} s \cdot GH(s) = \lim_{s\to 0} Ks\frac{(\tau_a s + 1)(\tau_b s + 1)\cdots}{s(\tau_1 s + 1)(\tau_2 s + 1)\cdots}$$

$$= \lim_{s\to 0} K\frac{(\tau_a s + 1)(\tau_b s + 1)\cdots}{(\tau_1 s + 1)(\tau_2 s + 1)\cdots}$$

$$= K\frac{(\tau_a(0) + 1)(\tau_b(0) + 1)\cdots}{(\tau_1(0) + 1)(\tau_2(0) + 1)\cdots} = K$$

$$e_{ss}(t) = \frac{1}{K_v} = \frac{1}{K}$$

∴ For type 2 systems:

$$\text{Type } 2 \rightarrow N = 2:$$

$$GH(s) = \frac{K(\tau_a s + 1)(\tau_b s + 1)\cdots}{s^2(\tau_1 s + 1)(\tau_2 s + 1)\cdots} = K\frac{(\tau_a s + 1)(\tau_b s + 1)\cdots}{s^2(\tau_1 s + 1)(\tau_2 s + 1)\cdots}$$

$$K_v = \lim_{s\to 0} s \cdot GH(s) = \lim_{s\to 0} Ks\frac{(\tau_a s + 1)(\tau_b s + 1)\cdots}{s^2(\tau_1 s + 1)(\tau_2 s + 1)\cdots}$$

$$= \lim_{s\to 0} K\frac{(\tau_a s + 1)(\tau_b s + 1)\cdots}{s(\tau_1 s + 1)(\tau_2 s + 1)\cdots} = \infty$$

$$e_{ss}(t) = \frac{1}{K_v} = 0$$

3. Static Acceleration Error Constant K_a:

- This Constant is the steady state error for the input $p(t)$:

$$GH(s) = \frac{K(\tau_a s + 1)(\tau_b s + 1)\cdots}{s^N(\tau_1 s + 1)(\tau_2 s + 1)\cdots}$$

$$R(s) = \frac{1}{s^3} \rightarrow E(s) = \frac{R(s)}{1 + GH(s)} = \frac{1}{s^3\big(1 + GH(s)\big)}$$

$$e_{ss}(t) = \lim_{s\to 0} s \cdot E(s) = \lim_{s\to 0} s \cdot \frac{1}{s^3\big(1 + GH(s)\big)}$$

$$= \lim_{s\to 0} \frac{1}{s^2\big(1 + GH(s)\big)} = \lim_{s\to 0} \frac{1}{s^2 + s^2 \cdot GH(s)}$$

$$= \frac{1}{0 + \lim_{s\to 0} s^2 \cdot GH(s)} = \frac{1}{\lim_{s\to 0} s^2 \cdot G(s)}$$

$$K_a = \lim_{s\to 0} s^2 \cdot GH(s)$$

$\therefore$ For type 0 systems:

$$\text{Type } 0 \rightarrow N = 0:$$

$$GH(s) = \frac{K(\tau_a s + 1)(\tau_b s + 1)\cdots}{s^0(\tau_1 s + 1)(\tau_2 s + 1)\cdots} = K\frac{(\tau_a s + 1)(\tau_b s + 1)\cdots}{(\tau_1 s + 1)(\tau_2 s + 1)\cdots}$$

$$K_a = \lim_{s\to 0} s^2 \cdot GH(s) = \lim_{s\to 0} Ks^2 \frac{(\tau_a s + 1)(\tau_b s + 1)\cdots}{(\tau_1 s + 1)(\tau_2 s + 1)\cdots}$$

$$= K(0)^2 \frac{(\tau_a(0) + 1)(\tau_b(0) + 1)\cdots}{(\tau_1(0) + 1)(\tau_2(0) + 1)\cdots} = 0$$

$$e_{ss}(t) = \frac{1}{K_a} = \frac{1}{0} = \infty$$

∴ For type 1 systems:

$$\text{Type } 1 \rightarrow N = 1:$$

$$GH(s) = \frac{K(\tau_a s + 1)(\tau_b s + 1)\cdots}{s^1(\tau_1 s + 1)(\tau_2 s + 1)\cdots} = K\frac{(\tau_a s + 1)(\tau_b s + 1)\cdots}{s(\tau_1 s + 1)(\tau_2 s + 1)\cdots}$$

$$K_a = \lim_{s\to 0} s^2 \cdot GH(s) = \lim_{s\to 0} Ks^2 \frac{(\tau_a s + 1)(\tau_b s + 1)\cdots}{s(\tau_1 s + 1)(\tau_2 s + 1)\cdots}$$

$$= \lim_{s\to 0} Ks \frac{(\tau_a s + 1)(\tau_b s + 1)\cdots}{(\tau_1 s + 1)(\tau_2 s + 1)\cdots}$$

$$= K(0)\frac{(\tau_a(0) + 1)(\tau_b(0) + 1)\cdots}{(\tau_1(0) + 1)(\tau_2(0) + 1)\cdots} = 0$$

$$e_{ss}(t) = \frac{1}{K_a} = \frac{1}{0} = \infty$$

∴ For type 2 systems:

$$\text{Type } 2 \rightarrow N = 2:$$

$$GH(s) = \frac{K(\tau_a s + 1)(\tau_b s + 1)\cdots}{s^2(\tau_1 s + 1)(\tau_2 s + 1)\cdots} = K\frac{(\tau_a s + 1)(\tau_b s + 1)\cdots}{s^2(\tau_1 s + 1)(\tau_2 s + 1)\cdots}$$

$$K_a = \lim_{s\to 0} s^2 \cdot GH(s) = \lim_{s\to 0} Ks^2 \frac{(\tau_a s + 1)(\tau_b s + 1)\cdots}{s^2(\tau_1 s + 1)(\tau_2 s + 1)\cdots}$$

$$= \lim_{s\to 0} K \frac{(\tau_a s + 1)(\tau_b s + 1)\cdots}{(\tau_1 s + 1)(\tau_2 s + 1)\cdots}$$

$$= K\frac{(\tau_a(0) + 1)(\tau_b(0) + 1)\cdots}{(\tau_1(0) + 1)(\tau_2(0) + 1)\cdots} = K$$

$$e_{ss}(t) = \frac{1}{K_a} = \frac{1}{K}$$

- This table summarize all cases:

e_{ss} for:	Step Input $r(t) = 1$	Ramp Input $r(t) = t$	Parabolic Input $r(t) = t^2/2$
Type 0 System	$\frac{1}{1+K}$	∞	∞
Type 1 System	0	$\frac{1}{K}$	∞
Type 2 System	0	0	$\frac{1}{K}$

▶ Key Note:

- If there are a system of type N and input in time domain with degree n:

$\therefore$ If $N > n$:

$$e_{ss} = 0$$

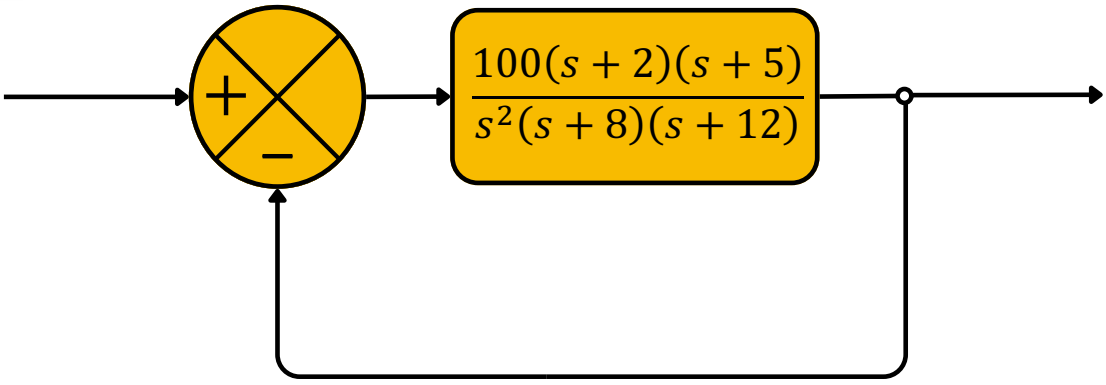
$\therefore$ If $N = n$:

$$e_{ss} = \begin{cases} \frac{1}{1+K}, & N = 0 \\ \frac{1}{K}, & N > 0 \end{cases}$$

$\therefore$ If $N < n$:

$$e_{ss} = \infty$$

▶ **Q. 10.1:** Consider this unity feedback system:



Determine the static error constants and the expected steady state errors for the standard step, ramp and parabolic inputs.

∴ *solution*:

$$K_p = \lim_{s\to 0} GH(s) = \frac{100(0+2)(0+5)}{0^2(0+8)(0+12)} = \infty$$

$$K_v = \lim_{s\to 0} sGH(s) = \frac{100(0+2)(0+5)}{0(0+8)(0+12)} = \infty$$

$$K_a = \lim_{s\to 0} s^2GH(s) = \frac{100(0+2)(0+5)}{(0+8)(0+12)} = \frac{1000}{96}$$

$$e_{ss}|_{u(t)} = \frac{1}{1+\infty} = 0$$

$$e_{ss}|_{r(t)} = \frac{1}{\infty} = 0$$

$$e_{ss}|_{p(t)} = \frac{1}{\frac{1000}{96}} = 0.096$$

11. Routh-Hurwitz Criterion

- **Routh-Hurwitz Criterion** is a method for determining continuous system stability.

it states that: "the number of roots of the characteristic equation with positive real parts is equal to the number of changes in sign of the first column of the Routh array".

- This method yields stability information without the need to solve for the closed-loop system poles.
- Using this method, we can tell how many closed-loop system poles are in the left half-plane, in the right half-plane, and on the $j\omega$-axis.
- WE SAY HOW MANY, NOT WHERE.
- The method requires two steps:
  1. Generate a data table called a Routh table.
  2. interpret the Routh table to tell how many closed-loop system poles are in the left-hand poles "LHP", the right-hand poles "RHP", and on the $j\omega$-axis poles.

- Let this formula represents any characteristic equation:

$$q(s) = a_n s^n + a_{n-1} s^{n-1} + a_{n-2} s^{n-2} + \cdots + a_1 s + a_0$$

- Routh table is:

s^n	a_n	a_{n-2}	a_{n-4}	$\cdots$	a_0
s^{n-1}	a_{n-1}	a_{n-3}	a_{n-5}	$\cdots$	a_1
s^{n-2}	b_1	b_2	b_3	$\cdots$	
s^{n-2}	c_1	c_2	c_3	$\cdots$	
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\ddots$	
s^0					

Key Notes:

- $a_n, a_{n-2}, \cdots, a_0$ are the coefficients for the highest power's
- The first two rows based directly on coefficients of $q(s)$.
- The constants b's and c's is determined like this:

$$b_1 = \frac{\begin{vmatrix} a_{n-1} & a_{n-3} \\ a_n & a_{n-2} \end{vmatrix}}{a_{n-1}} = \frac{a_{n-1} \cdot a_{n-2} - a_n \cdot a_{n-3}}{a_{n-1}}$$

$$b_2 = \frac{\begin{vmatrix} a_{n-1} & a_{n-5} \\ a_n & a_{n-4} \end{vmatrix}}{a_{n-1}} = \frac{a_{n-1} \cdot a_{n-4} - a_n \cdot a_{n-5}}{a_{n-1}}$$

$$c_1 = \frac{\begin{vmatrix} b_1 & a_{n-1} \\ b_2 & a_{n-3} \end{vmatrix}}{b_1} = \frac{b_1 \cdot a_{n-3} - a_{n-1} \cdot b_2}{b_1}$$

$$c_2 = \frac{\begin{vmatrix} b_1 & a_{n-1} \\ b_3 & a_{n-5} \end{vmatrix}}{b_1} = \frac{b_1 \cdot a_{n-5} - a_{n-1} \cdot b_3}{b_1}$$

▶ Example:

- To check the stability for this characteristic equation:

$$q(s) = s^3 + 6s^2 + 2s + 8$$

Routh table is:

+	s^3	1	12	0
+	s^2	6	8	0
+	s^1	$\frac{32}{3}$	0	0
+	s^0	8	0	0

$$\frac{6 \cdot 12 - 1 \cdot 8}{6} = \frac{32}{3}$$

$$\frac{6 \cdot 0 - 8 \cdot 0}{6} = 0$$

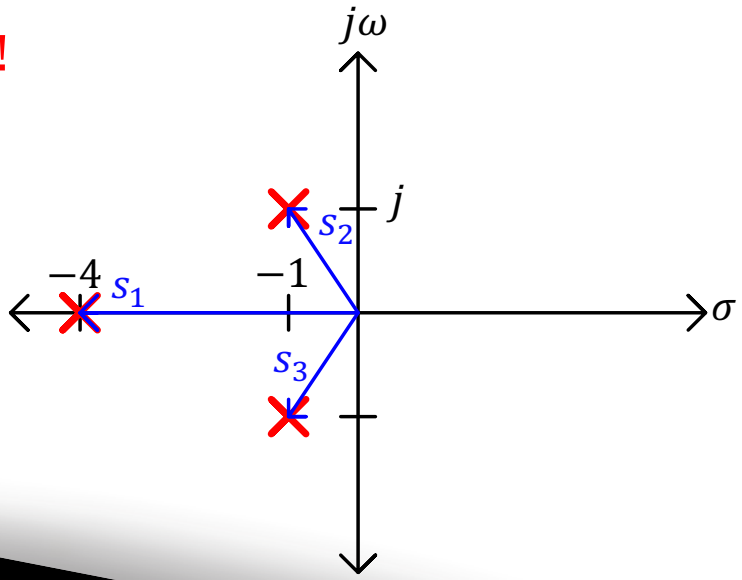
$$\frac{32/3 \cdot 8 - 6 \cdot 0}{32/3} = 8$$

- Since all elements of the first column has the same sign "+" and there is no change in sign elements, the system is stable.
- In R.H.C results, we say that this system has 3 poles "because the system is 3rd order (s^3)" in the left-hand side in the complex plane ($\sigma, j\omega$). Let's check if this right:

$$s^3 + 6s^2 + 2s + 8 = 0 \cdots \rightarrow (s+4)\big(s+(1-j)\big)\big(s+(1+j)\big)$$

$$s_1 = -4, \qquad s_2 = -1 + j, \qquad s_3 = -1 - j$$

So, this is right!



▶ Configurations of Routh's table:

1. No zeros in the first column:

- When you fill Routh's table and there are no zeros in the first column, continue the solution normally to check the stability of the system.

▶ **Q. 11.1:** Check the stability for these system's characteristic equations:

$\therefore q(s) = s^3 + 3s^2 + 2s + 4$

+	s^3	1	2	0
+	s^2	3	4	0
+	s^1	2	0	0
+	s^0	4	0	0

3 poles
no changes in sign:
no poles in positive real axis:
stable system.

$\therefore q(s) = 2s^4 + s^3 + 5s^2 + 2s + 8$

+	s^4	2	5	8
+	s^3	1	2	0
−	s^2	−1	8	0
+	s^1	10	0	0
+	s^0	8	0	0

4 poles
2 changes in sign (+ → − & − → +):
2 poles in positive real axis:
unstable system.

2. A zero in the first column:

- This is a problem when we try to check the stability for a system (because you would to divide over zero and this is impossible); so, we have 2 methods to fix this:

▶ Epsilon Method:

- Let epsilon ϵ a very small positive number, replace that zero in first column by ϵ, then continue the solution by approaching the number that is multiplied by ϵ to zero

▶ **Q. 11.2:** Check the stability for this system:

$$q(s) = s^3 + 2s^2 + 2s + 4$$

+	s^3	1	2	0
+	s^2	2	4	0
+	s^1	ϵ	0	0
?	s^0	a	0	0

$$a = \frac{4\epsilon - 2}{\epsilon} \approx -\frac{2}{\epsilon}$$

Since ϵ is very close to zero, that means 4ϵ is also ≈ 0, so the sign of a is $-$, so the system is unstable:

▶ **Q. 11.3:** Find all values of K that makes the system stable:

$\therefore q(s) = s^3 + 2s^2 + s + K$

+	s^3	1	2	0
+	s^2	1	K	0
?	s^1	a	0	0
?	s^0	K	0	0

$$a = \frac{2 - K}{1} = 2 - K$$

The system is sable when all values of the first column are positive "greater than zero", so:

$$a > 0 \rightarrow 2 - K > 0 \rightarrow K < 2$$

From last element in the first raw: $K > 0$, so the values of K:

$$K < 2 \cup K > 0 \rightarrow 0 < K < 2$$

So, the system is stable when K is between 0 and 2.

$\therefore q(s) = s^4 + s^3 + s^2 + s + K$

+	s^4	1	1	K
+	s^3	1	1	0
+	s^2	ϵ	K	0
?	s^1	a	0	0
?	s^0	K	0	0

$$a = \frac{\epsilon - K}{\epsilon} \approx -\frac{K}{\epsilon}$$

The system is sable when all values of the first column are positive "greater than zero", so:

$$a > 0 \rightarrow -\frac{K}{\epsilon} > 0 \rightarrow \frac{K}{\epsilon} < 0 \rightarrow K < \epsilon \rightarrow K < 0$$

From last element in the first raw: $K > 0$, so the values of K:

$$K < 0 \cup K > 0 \rightarrow 0 < K < 0 \rightarrow no\ value$$

So, the system is unstable for all values of K.

▶ Reverse Coefficients Method:

- This method represented as exchange the coefficient in the first 2 rows (up→down and left→right) like this:

s^n	a	b	c
s^{n-1}	d	e	f
⋮	⋮	⋮	⋮
⋮	⋮	⋮	⋮
⋮	⋮	⋮	⋮

→

s^n	f	e	d
s^{n-1}	c	b	a
⋮	⋮	⋮	⋮
⋮	⋮	⋮	⋮
⋮	⋮	⋮	⋮

Then continue normally.

▶ Q. 11.4: Check the stability for this system:

$$q(s) = s^5 + 2s^4 + 3s^3 + 6s^2 + 5s + 3$$

s^5	1	3	5
s^4	2	6	3
s^3	0	3.5	0
s^2			
s^1			
s^0			

→

+	s^5	3	6	2
+	s^4	5	3	1
+	s^3	4.2	1.4	0
+	s^2	1.33	1	0
−	s^1	−1.75	0	0
+	s^0	1	0	0

5 poles 2 poles in positive real axis unstable system

3. Entire row is zero:

- Sometimes while making a Routh table, we find that an entire row consists of zero, this happen because there are 2 poles in $j\omega$ axis without any real part.

▶ **To deal with this case:**

1. Find the auxiliary equation $A(s)$ whose coefficients are the first even row preceding the zero row in the table.
2. Find $A'(s)$, then replace the zero row with coefficients of $A'(s)$, then continue the solution normally

▶ **Q. 11.5:** Check the stability for this system:

$$q(s) = s^5 + s^4 + 4s^3 + 24s^2 + 3s + 63$$

s^5	1	4	3
s^4	1	24	63
s^3	-20	-60	0
s^2	21	63	0
s^1	0	0	0
s^0			

→

+	s^5	1	4	3
+	s^4	1	24	63
−	s^3	-20	-60	0
+	s^2	21	63	0
+	s^1	42	0	0
+	s^0	63	0	0

$A(s) = 21s^2 + 63$, $A'(s) = 42s + 0$

5 poles

2 poles in $j\omega$ axis
2 poles in positive real axis

unstable system

12. Root Locus

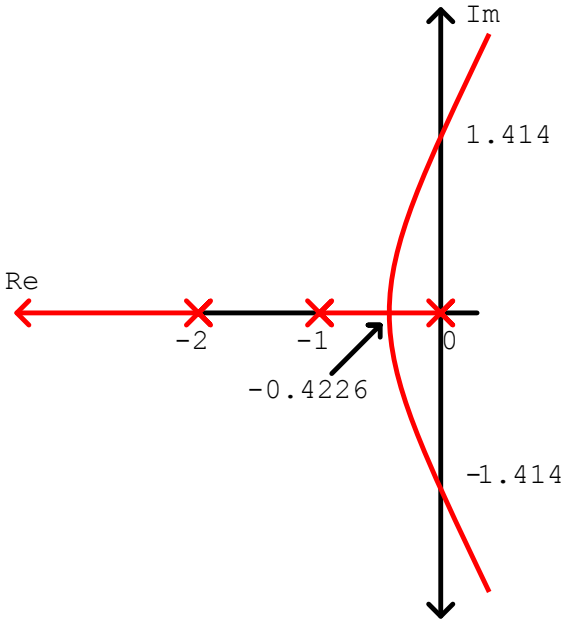
• **Root Locus** is a graphical technique used in to see how the closed-loop poles move in the complex s-plane when a system parameter "$K, \zeta, \omega_n, \ldots$" changes from 0 to ∞.

∴ Helps you to check system's **stability** as the parameter "usually K" changes.

∴ Starts from **open-loop poles** and ends at **open-loop zeros** (including zeros at infinity if there are fewer finite zeros).

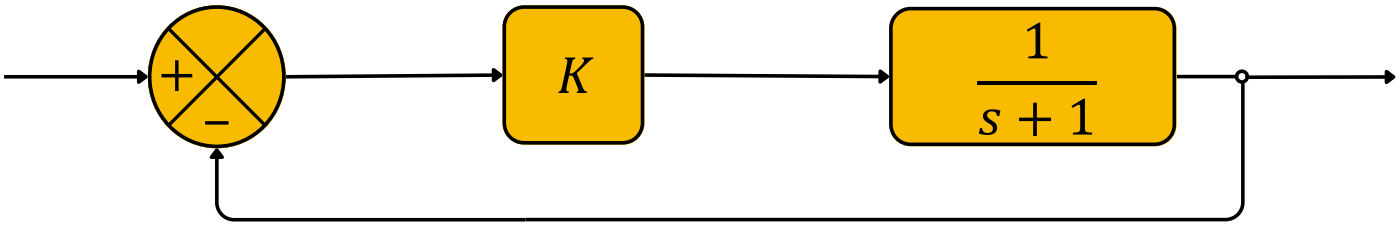
• In other words, the root locus is the path of the roots of the characteristic equation traced out in the s-plane as a system parameter varies from 0 to ∞.

• Usually, it graphed by MATLAB "we will learn it in the lab", but in this course, we will learn how to graph it manually.



Example:

- Consider a unity feedback control system shown below:

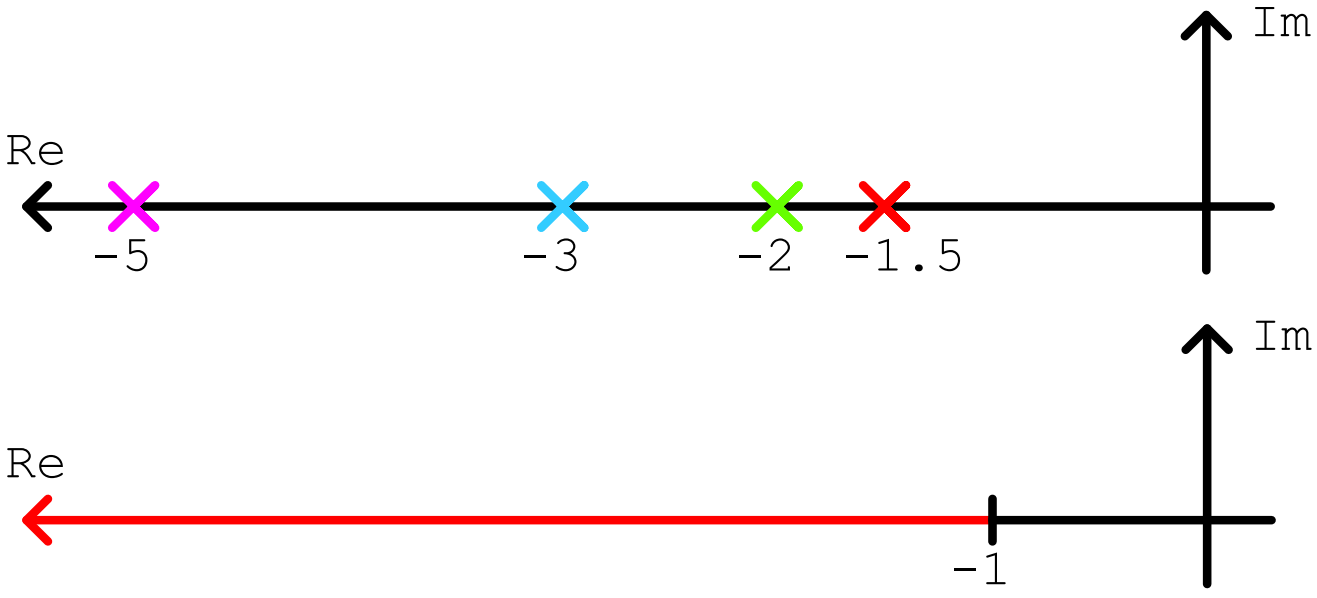


- Overall transfer function and characteristic equation are:

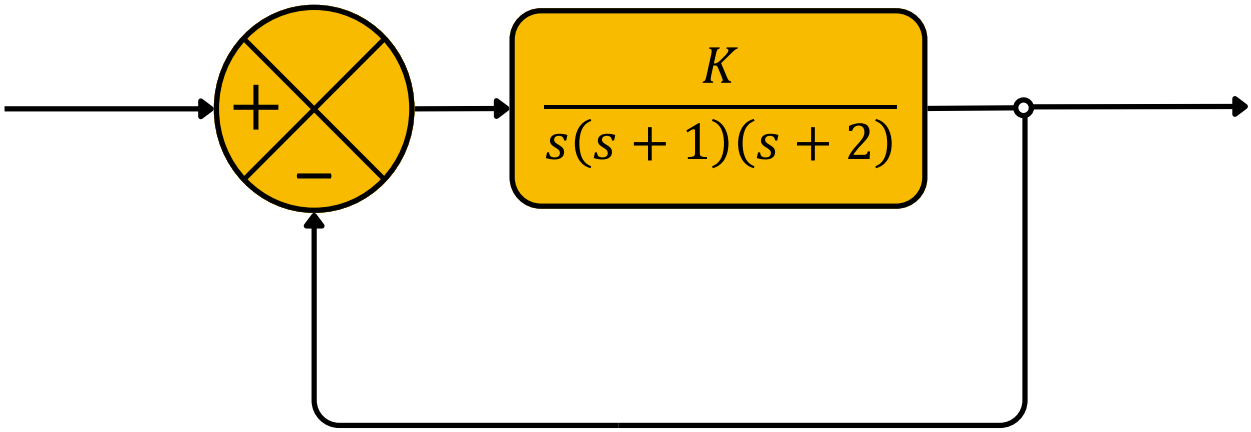
$$\frac{C(s)}{R(s)} = \frac{K}{s+1+K} \rightarrow q(s) = s + 1 + K = 0 \rightarrow s = -(K+1)$$

- So, when we change K value the poles would change also, now let's choose some values for K and see how it will be plotted in $\{Re, Im\}$ and compared with root locus graph "will be explained later".

K	0.5	1	2	4
Pole	-1.5	-2	-3	-5



• It looks easy to draw all K values, but it might be difficult for higher order systems like this:



$$\frac{C(s)}{R(s)} = \frac{K}{s(s+1)(s+2)+K}$$

K	0.5	1	2	4
Pole	?	?	?	?

∴ What are the possible values of K that makes this system stable, unstable, critically stable… *etc*?

∴ What are the possible values of s when we set K value?

∴ What is K value that makes system damping as I want?

• The root locus graph answers all of these questions.

Root Locus Parameters:

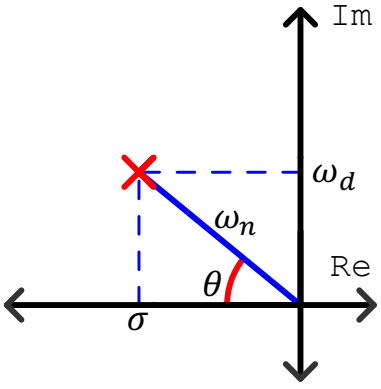
- The parameters that you can plot them into the s-plane:
  1. Gain K.
  2. Un-Damped frequency ω_n.
  3. Damping ratio ζ.
- Since s-plane is complex, you can represent poles as:
  - A. **Rectangular Form:** Real and imaginary components:

$$s = \sigma + j\omega_d$$

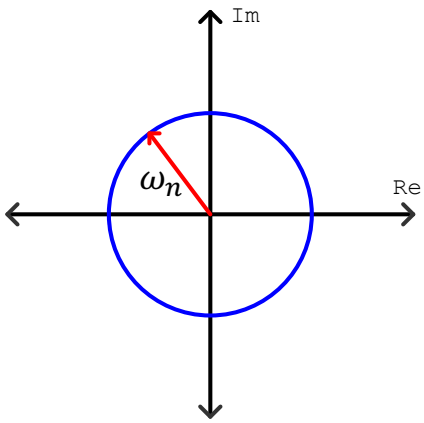
  - B. **Polar Form:** Magnitude and angle:

$$s = \omega_n \angle \theta$$

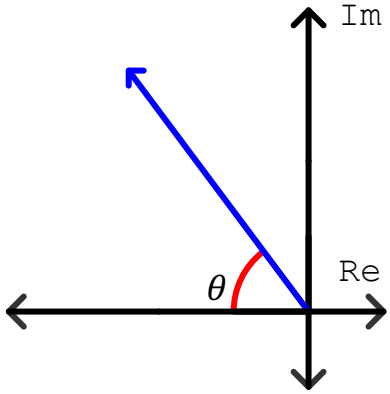
- When you want to make a system that has ω_n value regardless θ value, you represent it in root locus as a circle has a radius with value of ω_n.
- When you want to make a system that has θ value regardless ω_n value, you represent it in root locus as a line has an angle with value of θ.



Pole on s-Plane



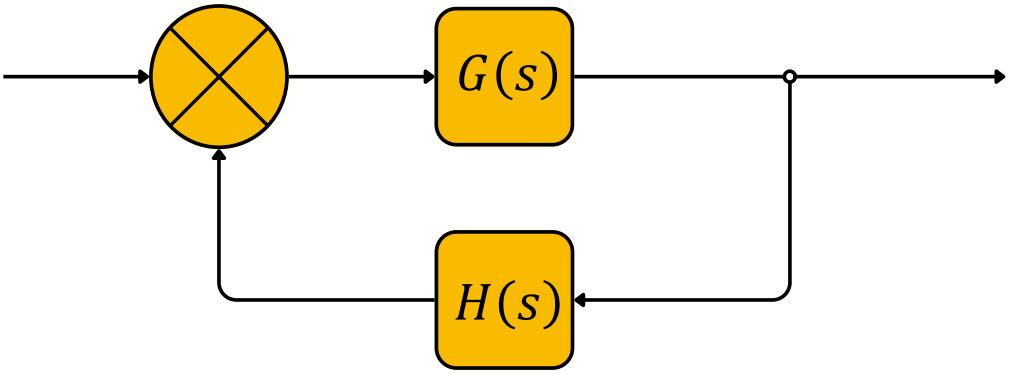
Root Locus for ω



Root Locus for θ

Magnitude and Angle Conditions:

- Consider this system below:



- From characteristic equation:

$$G(s)H(s) + 1 = 0 \rightarrow G(s)H(s) = -1$$

- Since $G(s)H(s)$ is a complex quantity, it can be split into angle and magnitude part:

$$G(s)H(s) = -1 = |1| \angle 180°$$

- So, we can conclude 2 conditions for the roots of characteristic equation that they are:

1. **Angle Condition:**

$$\angle G(s)H(s) = \pm 180°(2n + 1)$$

2. **Magnitude Condition:**

$$|G(s)H(s)| = 1$$

Key Notes:

- Remember that $\pm 180°(2n+1)$ for any value of integer n is always equal to 180°.
- **Angle condition** is used to check any value of s in the system if it in root locus or not.
- **Magnitude condition** is used to determine the value of K at any pole at root locus area.
- Because $G(s)H(s)$ is a complex quantity, you can find the angle and magnitude like this:

$$G(s)H(s) = K\frac{(s+z_1)(s+z_2)\cdots}{(s+p_1)(s+p_2)\cdots}$$

$$\angle G(s)H(s) = \angle(s+z_1) + \angle(s+z_2) + \cdots - \angle(s+p_1) - \angle(s+p_2) - \cdots$$

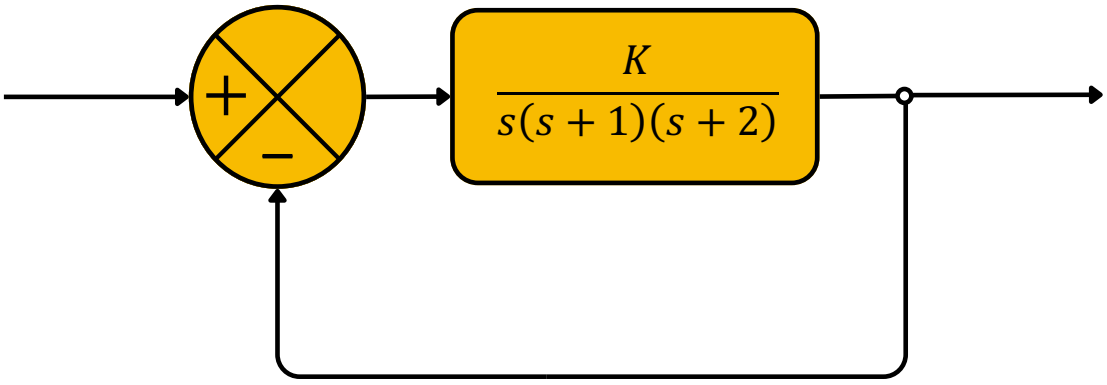
$$|G(s)H(s)| = K\frac{|(s+z_1)||(s+z_2)|\cdots}{|(s+p_1)||(s+p_2)|\cdots}$$

- **Real-Axis Rule for the Root Locus:** "A point on the real s-axis lies on the root locus if the total number of real poles and zeros to its right is odd":

$$\angle G(s)H(s) = \pm 180°(2n+1)$$

$\pm 180° \rightarrow$ right pole or zero, $(2n+1) \rightarrow$ odd numbers totally

▶ **Q. 12.1:** Consider this unity feedback system:



Check whether $s = -0.25$ and $s = -1 + j$ is on the root locus or not, and if it was there, find its gain.

$\therefore$ *solution*:

Apply angle condition:

$$\angle G(s)H(s) = \angle K - \angle s - \angle(s+1) - \angle(s+2)$$

$s = -0.25$:

$$\angle G(s)H(s) = \angle K - \angle(-0.25) - \angle(0.75) - \angle(1.75)$$
$$= 0 - (180°) - (180°) - (180°) = -3(180°)$$

so, $s = -0.25$ is in the root locus, now from Magnitude condition:

$$|G(s)H(s)| = 1 \rightarrow \left|\frac{K}{-0.25(0.75)(1.75)}\right| = 1 \rightarrow K = 0.328$$

$s = -1 + j$:

$$\angle G(s)H(s) = \angle K - \angle(-1+j) - \angle(j) - \angle(1+j)$$
$$= 0 - (135°) - (90°) - (45°) = -270° = 90°$$

so, $s = -1 + j$ is not in the root locus.

Root Locus Sketching:

- To sketch root locus graph, apply these steps:

1. Obtain all finite poles and zeros:

- There are the finite poles and zeros, you can find them by:

$$\frac{C(s)}{R(s)} = K\frac{(s+z_1)(s+z_2)\cdots}{(s+p_1)(s+p_2)\cdots}$$

$p = \{-p_1, -p_2 \dots\}$, $\#p \rightarrow$ number of poles

$z = \{-z_1, -z_2 \dots\}$, $\#z \rightarrow$ number of zeros

2. Set real axis line segments:

- Draw complex plane then plot all poles by X and zeros by O then connect between "2 poles" or "2 zeros" or "1 pole and 1 zero" at the area that sum of all poles and zeros in its right area is odd "to approve angle condition".

3. Count the infinite poles and zeros:

- The poles and zeros must be equal in general, if $\#p \neq \#z$ there will be an infinite poles or zeros that calculated by:

$$i = |\#p - \#z|$$

- if $\#p > \#z$, i is the number of infinite zeros.
- if $\#z > \#p$, i is the number of infinite poles.

4. Find the asymptotes:

• They are the lines that lead to the infinite poles or zeros, to find them you must find its centroid σ and angles ψ:

$$\sigma = \frac{\sum p - \sum z}{i}, \qquad \psi = \frac{180°}{i}(2n+1)$$

• **NOTE:** Substitute n from 0 to $i-1$.

5. Find breakaway point:

$$KG(s)H(s) = |1| \rightarrow K = \frac{1}{G(s)H(s)} \rightarrow \frac{dK}{ds} = 0$$

• After you finding $\frac{dK}{ds}$ and equal it to zero, find s values and we say that the value of s is a breakaway point if s on the root locus and between 2 poles only.

• Then draw curved lines from breakaway point to the asymptotes.

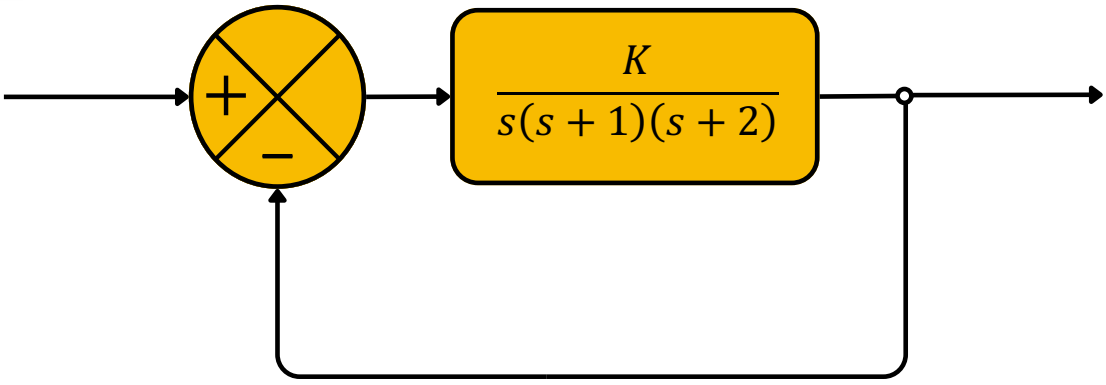
6. Find imaginary axis crossing "if exist":

• If there are crossing between that curved line and $j\omega$ axis, you must find K and ω at that cross, you can find:

$\therefore$ K from Routh-Hurwitz criterion "zero row case".

$\therefore$ ω from axillary equation $A(s)$.

▶ Q. 12.2: Consider this unity feedback system:

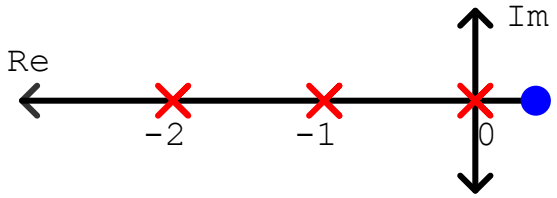


∴ Sketch root locus:

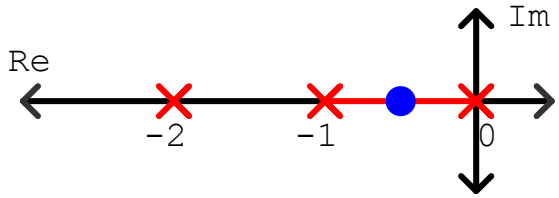
1. Obtain all finite poles and zeros:

$$p = \{-2, -1, 0\} \rightarrow \#p = 3, \qquad z = \{\phi\} \rightarrow \#z = 0$$

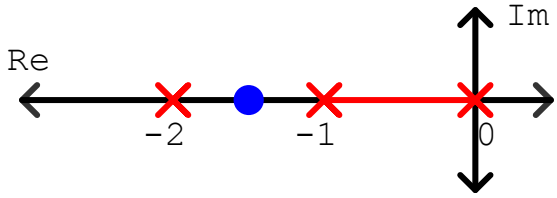
2. Set real axis line segments:



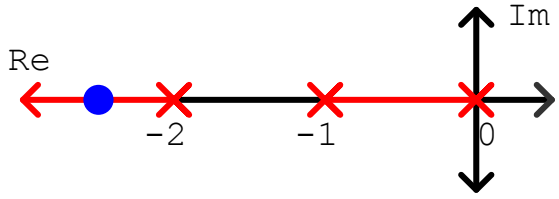
$\sum p_{Right} = 0 \rightarrow$ isn't on the root locus



$\sum p_{Right} = 1 \rightarrow$ on the root locus

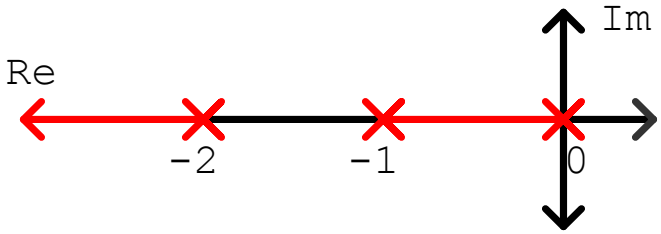


$\sum p_{Right} = 2 \rightarrow$ isn't on the root locus



$\sum p_{Right} = 3 \rightarrow$ on the root locus

So, the final result of real axis line segments is:



3. Count the infinite poles and zeros:

$$\#p = 3, \#z = 0 \rightarrow i = \#p - \#z = 3$$

There are 3 infinite zeros in the root locus.

4. Find the asymptotes:

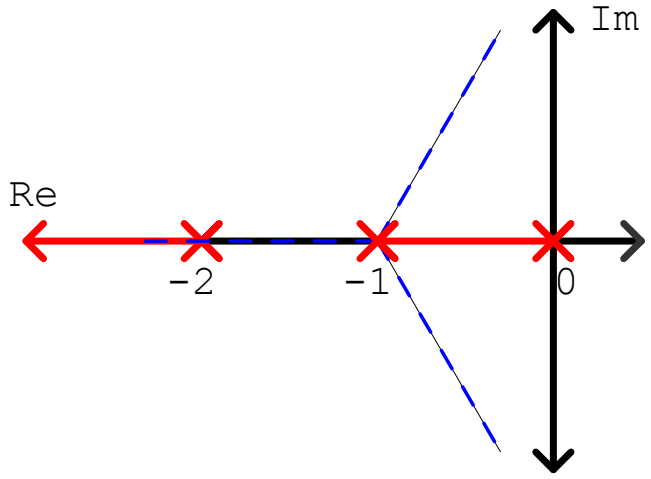
$$\sigma = \frac{\sum p - \sum z}{i} = \frac{(-2 - 1 + 0) - 0}{3} = -1$$

$$\psi = \frac{180°(2n+1)}{i} = \frac{180°}{3}(2n+1) = 60°(2n+1)$$

$$n = 0 \rightarrow \psi_1 = 60°$$

$$n = 1 \rightarrow \psi_2 = 180°$$

$$n = 2 \rightarrow \psi_3 = 300°$$

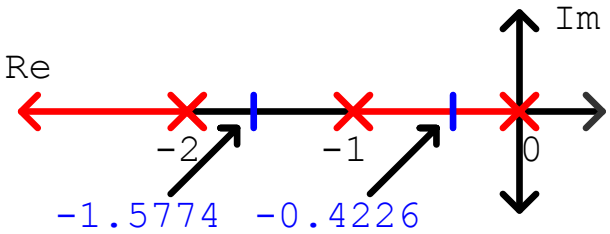


5. Find breakaway point:

$$\frac{K}{s(s+1)(s+2)} = 1 \rightarrow K = s(s+1)(s+2) = s^3 + 3s^2 + 2s$$

$$\frac{dK}{ds} = 3s^2 + 6s + 2 = 0$$

$$s = -0.4226 \text{ or } s = -1.5774$$



The breakaway point must access 2 conditions, so that it is between 2 poles and on the root locus, so we find that when $s = -0.4226$ it achieves these conditions while $s = -1.5774$ is not on root locus. So breakaway point is $s = -0.4226$

6. Find imaginary axis crossing:

$$G(s)H(s) + 1 = 0 \rightarrow \frac{K}{s(s+1)(s+2)} + 1 = 0$$

$$\rightarrow s^3 + 3s^2 + 2s + K = 0$$

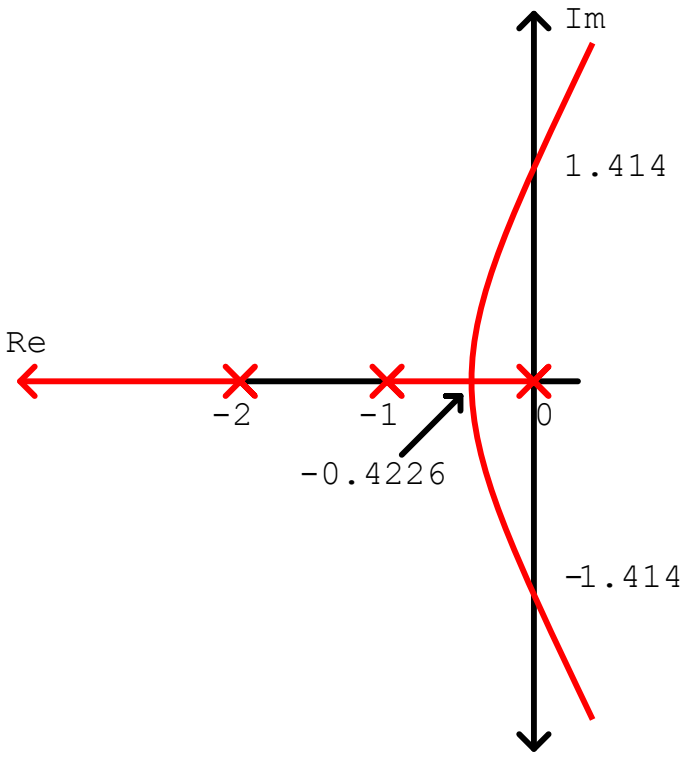
s^3	1	2
s^2	3	K
s^1	$\frac{6-K}{3}$	0
s^0	K	

Using R-H, zero row case achieves when $K = 6$

Now to find ω:

$$A(s) = 3s^2 + K = 0 \rightarrow 3s^2 + 6 = 0 \rightarrow s = \pm\sqrt{2}j$$

The root locus graph:



$\therefore$ Find the value of K that makes the system damping by 0.5:

$$\zeta = 0.5 \rightarrow \theta = \cos^{-1}(\zeta) = 60° \text{ from } \{-Re\} \text{ axis}$$

Now from root locus find the pole that makes $60°$ with $\{-Re\}$ so we found that $s = -0.333 \pm 0.577j$ achieve this.
Now from magnitude condition:

$$\left|\frac{K}{s(s+1)(s+2)}\right| = 1$$

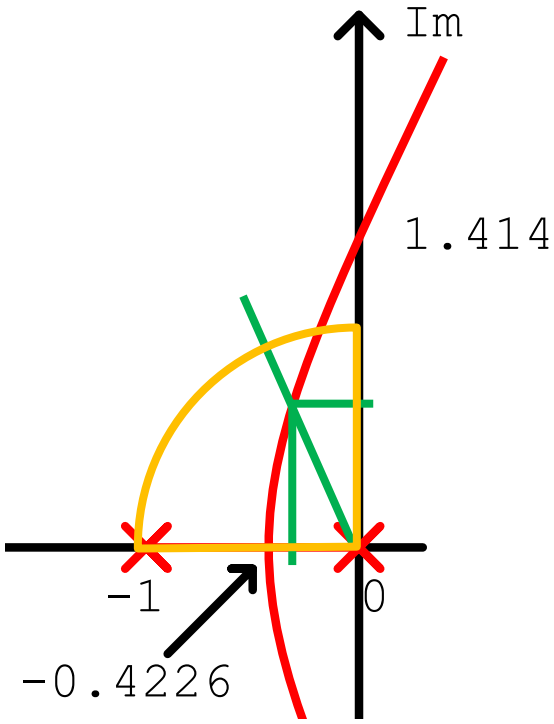
$$\rightarrow K = |s| \cdot |(s+1)| \cdot |(s+2)| \rightarrow K = 1.0383$$

$\therefore$ Find the value of K that makes the system has $\omega_n = 1$:

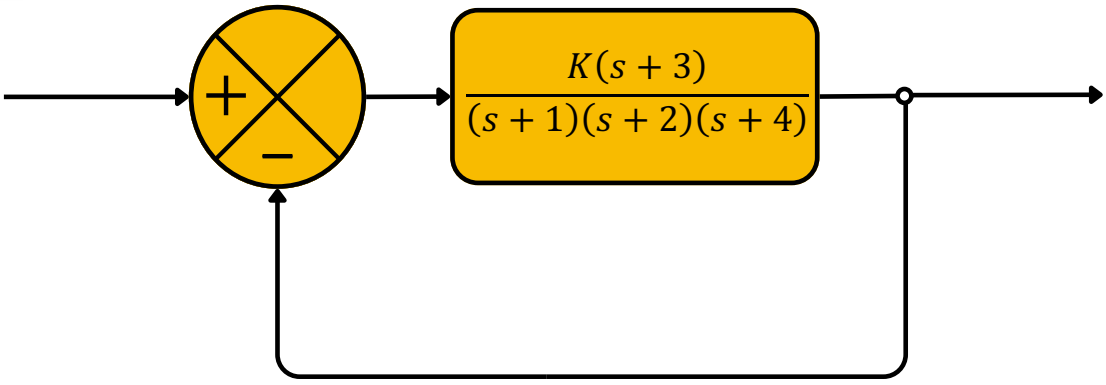
From crossing between the root locus of ω_n with K one, we see the pole is $s = -0.191 + 0.964j$, Now from magnitude condition:

$$\left|\frac{K}{s(s+1)(s+2)}\right| = 1$$

$$\rightarrow K = |s| \cdot |(s+1)| \cdot |(s+2)| \rightarrow K = 2.618$$



▶ Q. 12.3: Sketch root locus for this unity feedback system:

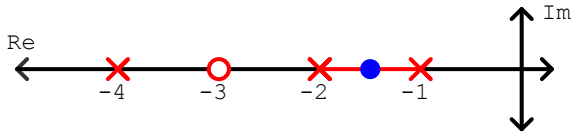


∴ *solution*:

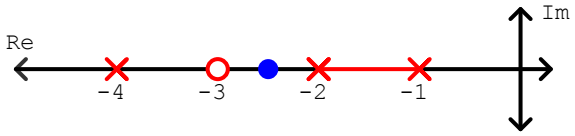
1. Obtain all finite poles and zeros:

$p = \{-4, -2, -1\} \rightarrow \#p = 3, \qquad z = \{-3\} \rightarrow \#z = 1$

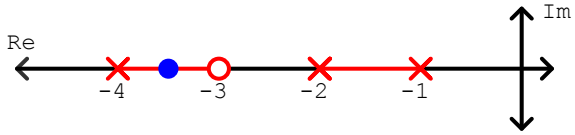
2. Set real axis line segments:



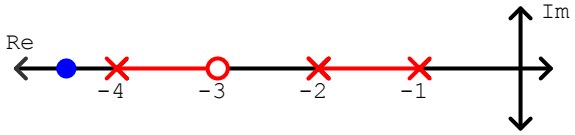
$\sum p_{Right} = 1 \rightarrow$ on the root locus



$\sum p_{Right} = 2 \rightarrow$ isn't on the root locus

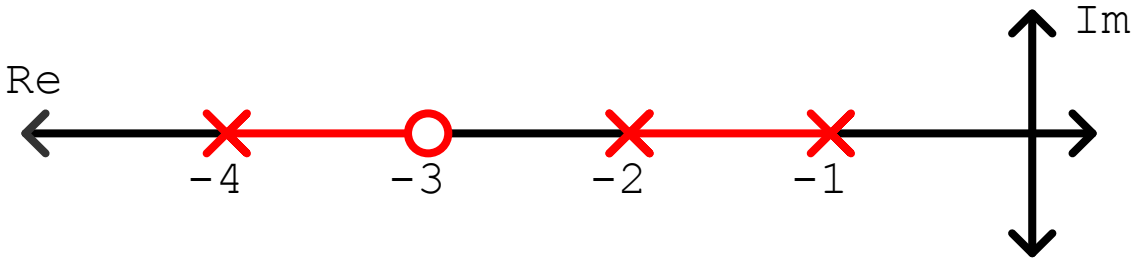


$\sum p_{Right} = 3 \rightarrow$ on the root locus



$\sum p_{Right} = 4 \rightarrow$ isn't on the root locus

So, the final result of real axis line segments is:



3. Count the infinite poles and zeros:

$$\#p = 3, \#z = 1 \rightarrow i = \#p - \#z = 2$$

There are 2 infinite zeros in the root locus.

4. Find the asymptotes:

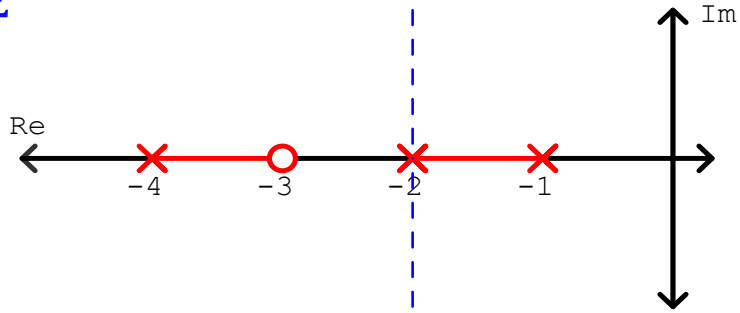
$$\sigma = \frac{\sum p - \sum z}{i} = \frac{(-4-2-1)-(-3)}{2} = -2$$

$$\psi = \frac{180°(2n+1)}{i} = \frac{180°}{2}(2n+1) = 90°(2n+1)$$

$$n = 0 \rightarrow \psi_1 = 90°$$

$$n = 1 \rightarrow \psi_2 = 270°$$

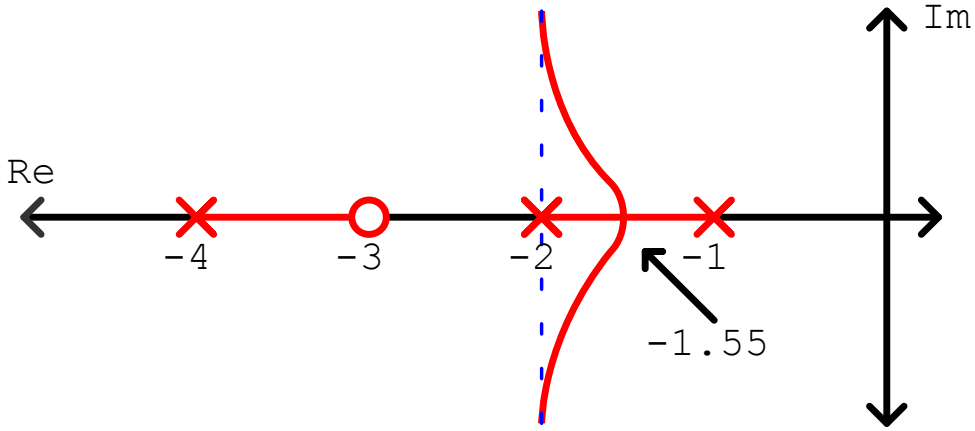
$$z_{inf} = -2 \pm j\infty$$



5. Find breakaway point:

$$\frac{K(s+3)}{(s+1)(s+2)(s+4)} = 1 \rightarrow K = \frac{(s+1)(s+2)(s+4)}{(s+3)}$$

$$= s^2 + 4s + 2 + \frac{2}{s+3} \rightarrow \frac{dK}{ds} = \cdots = 0 \rightarrow s \approx -1.55$$



13. PID Controller

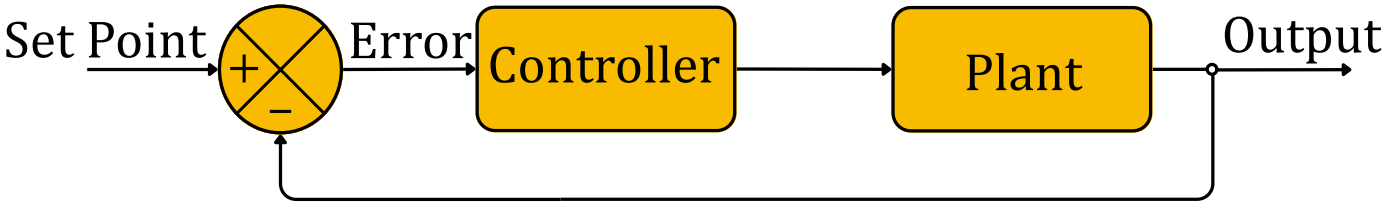
- PID controllers are a widely used feedback control system in engineering and automation. It stands for:

∴ P – Proportional

∴ I – Integral

∴ D – Derivative

- Any feedback system "like PID" is like this:



- We first set the reference value (set point). The controller responds to this reference signal and processes it using its transfer function. The output of the controller is then used to control the plant (like motor).
- Then the output of the plant is measured and compared with the set point to generate the error signal. This error signal is then fed back into the controller, where it is processed again. This loop continues, allowing the system to minimize the error and follow the set point accurately.
- This is the dynamics of any feedback control system.
- We will focus on the controller and see what is the PID.

▶ P Controller:

- Proportional controller is a number takes the error signal and multiplies it by that number "to increase or decrease the error signal's magnitude as you want".

∴ **Idea:**

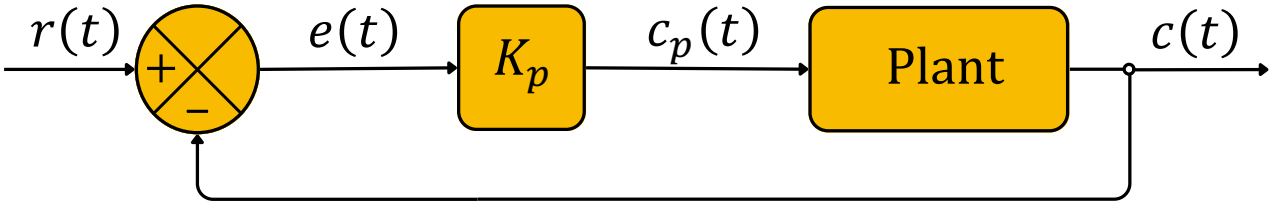
- Output is directly proportional to the current error.
- Large error → large correction
- Small error → small correction

∴ **Effect:**

- Fast, straightforward response
- Reduces but does not fully remove steady-state error

∴ **Block Diagram & Transfer Function:**

- It denoted by K_p "proportional gain" and follows this block diagram (in time domain):



$$e(t) \rightarrow E(s), \qquad c_p(t) = K_p e(t) \rightarrow C_p(s) = K_p E(s)$$

- So, the P controller's transfer function:

$$\frac{C_p(s)}{E(s)} = K_p$$

I Controller:

- Integral controller s a number that takes the error signal and integrates it over time "to keep adding or subtracting until the accumulated error is removed as you need".

∴ **Idea:**

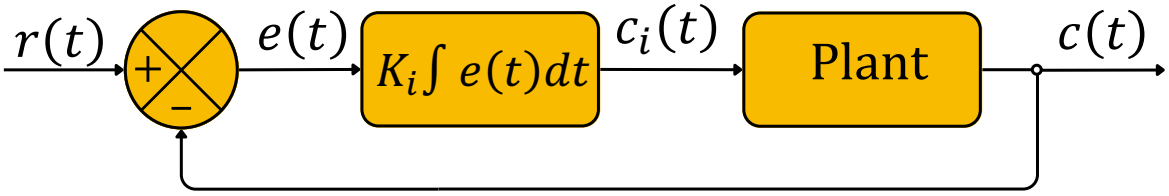
- Output depends on the accumulated past error.
- Keeps integrating until the error is zero.
- Eliminates steady-state offset.

∴ **Effect:**

- Removes steady state error
- Smooths out small persistent deviations.

∴ **Block Diagram & Transfer Function:**

- It denoted by K_i "integral gain" and follows this block diagram (in time domain):



$$e(t) \rightarrow E(s), \qquad c_i(t) = K_i \int e(t)dt \rightarrow C_i(s) = \frac{K_i E(s)}{s}$$

- So, the controller's transfer function:

$$\frac{C_i(s)}{E(s)} = \frac{K_i}{s}$$

▶ D Controller:

• Derivative controller is a number that takes the error signal and differentiates it with respect to time "to react to how fast the error is changing and counteract future".

∴ **Idea:**

• Output reacts to the rate of change of error.

• Predicts where the error is heading.

• Adds damping and stabilizes.

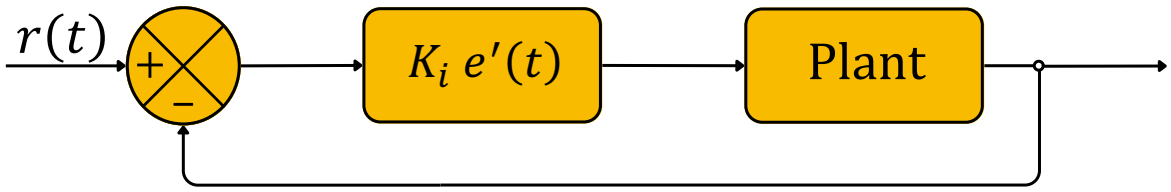
∴ **Effect:**

• Reduces overshoot.

• improves stability.

∴ **Block Diagram & Transfer Function:**

• It denoted by K_d "derivative gain" and follows these block diagrams (in time domain):



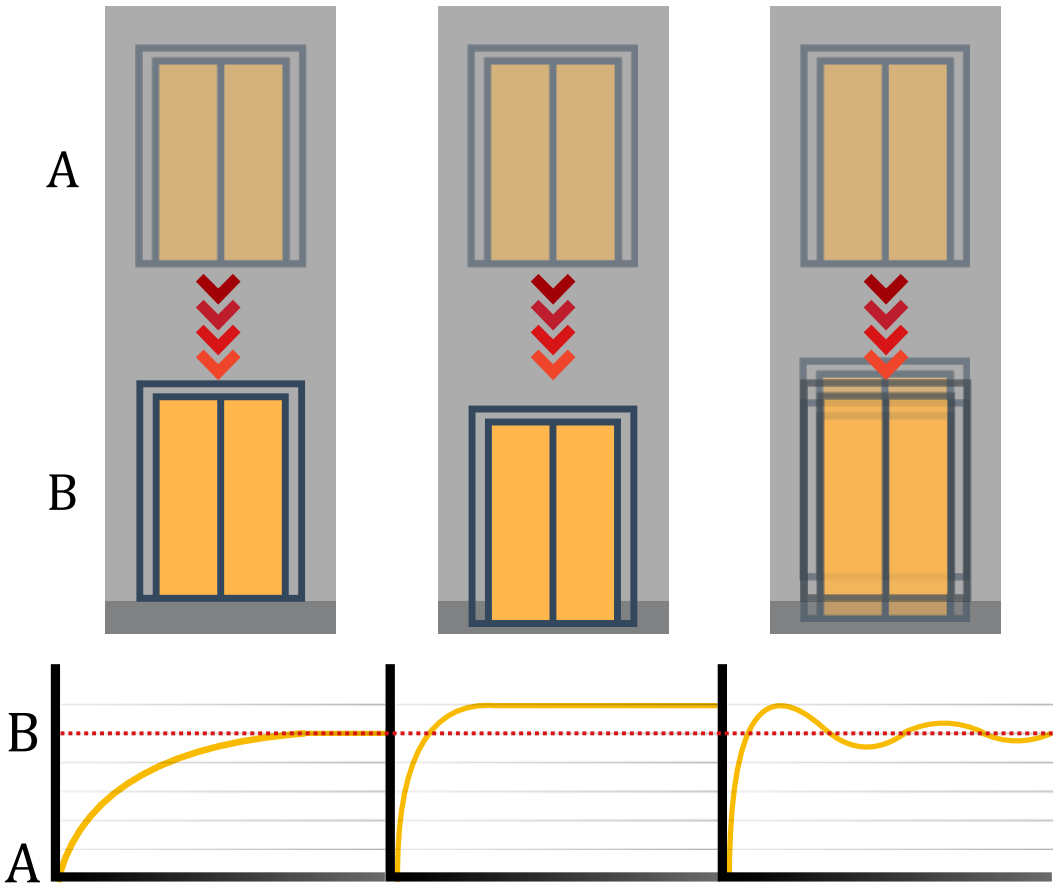
$$e(t) \rightarrow E(s), \qquad c_d(t) = K_d \frac{d}{dt} e(t) \rightarrow C_d(s) = K_d s E(s)$$

• So, the controller's transfer function:

$$\frac{C_i(s)}{E(s)} = \frac{K_i}{s} s$$

Example:

- Let we have an elevator and we want to control its level as we see in these figures with their step responses:



- We see the transition between the level of the elevator from A to B, what's the problems that we may face?

1. You may find the elevator reaches B level correctly, but it takes a lot of time to do this.
2. You may find the elevator reaches B level quickly, but its position is not correctly reached.
3. You may find the elevator reaches B level correctly and quickly, but it does little oscillations around level B.

• All of these models of elevators are open-loop systems, because there is no feedback loop "a value from output compared with reference set point", assume that we make it a feedback system, how could we fix these problems using PID controller to make the elevator reaches B quickly, correctly and with less oscillations as much as possible?

∴ **To Fix Problem 1: Use P Control:**

• Increasing the proportional gain K_p strengthens the immediate reaction to the current error, so the motor drives harder and the elevator travels faster toward the setpoint.

∴ **To Fix Problem 2: Use I Control:**

• The integral action keeps accumulating the error over time and continues adjusting until the offset is driven to zero, guaranteeing the elevator ends exactly at level B.

∴ **To Fix Problem 3: Use D Control:**

• Derivative action reacts to the rate of change of the error, adding damping that counteracts rapid motion and suppresses oscillations for a smooth stop at level B.

Gain	Rise Time	Overshoot	Settling Time	S-S Error
K_p ↑	↓ Faster	↑ More	~ Small change	↓ Lower
K_i ↑	↓ Faster	↑ More	↑ Slower	- Eliminated
K_d ↑	~ Small change	↓ Less	↓ Faster	~ Small change

▶ To Design PID Controller:

- Obtain what is the response that needs to be improved.
- Add P control to improve the rise time.
- Add I control to eliminate the steady-state error.
- Add D control to improve the overshoot.
- Adjust each of K_p, K_i and K_d until you obtain a desired response "Using PID Tuning".

▶ Key Notes:

- keep in mind that you do not need to implement all three controllers (P, I and D) into a single system, if not necessary:

∴ If you have PID controller and you want to reduce it to PI controller, make $K_d = 0$.

∴ If you have PID controller and you want to reduce it to PD controller, make $K_i = 0$.

- Keep the controller as simple as possible.
- For example, if a PI controller gives a good enough response, then you don't need to implement derivative controller to the system.

PID Tuning:

- Tuning is the process of adjusting the three gains of a PID controller (K_p, K_i and K_d) so that a system responds quickly and stably to changes or set-point commands.
- However, the first settings often produce a step response with excessive overshoot, which is usually unacceptable. Therefore, engineers use the Ziegler–Nichols results as an educated starting point and then perform fine adjustments until the system meets the required specifications.
- These rules provide initial parameter values that ensure basic stability.
- The transfer function of PID controller is given as:

$$\frac{C(s)}{R(s)} = K_p + \frac{K_i}{s} + K_d s \rightarrow K_p\left(1 + \frac{K_i}{K_p}\cdot\frac{1}{s} + \frac{K_d}{K_p}\cdot s\right)$$

- Let $T_i = \frac{K_p}{K_i}$ and $T_d = \frac{K_d}{K_p}$:

$$\frac{C(s)}{R(s)} = K_p\left(1 + \frac{1}{T_i s} + T_d s\right)$$

▶ **Ziegler–Nichols tuning rules are:**

∴ First Method “Open-loop”.

∴ Second Method “Closed-loop”.

▶ Ziegler–Nichols First Method:

• This method uses the open-loop step response of the system to characterize the process depending on these procedures:

1. Apply a unit step input to the open-loop system.
2. Observe the response and extract two parameters:

∴ **Dead Time L:** The delay from when the step is applied to when the system first responds.

∴ **Time Constant T**: The time it takes for the system to reach 63.2% of its final value after the dead time.

3. Compute PID parameters using this table:

Controller	K_p	T_i	T_d
P	$\frac{T}{L}$	∞	0
PI	$0.9\frac{T}{L}$	$\frac{L}{0.3}$	0
PID	$1.2\frac{T}{L}$	$2L$	$0.5L$

▶ Key Notes:

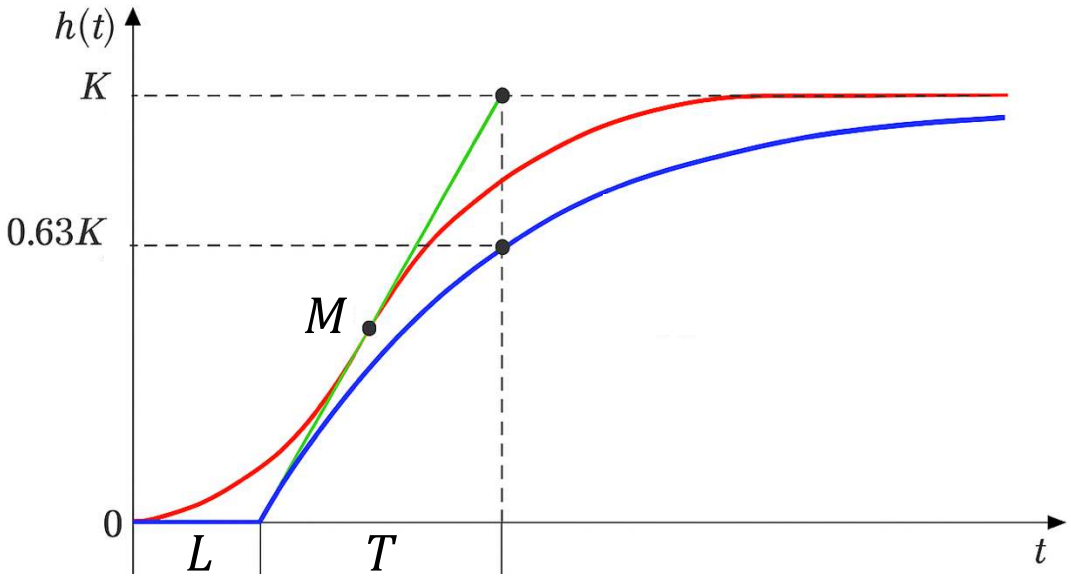
- If the plant involves neither integrator(s) nor dominant complex conjugate poles, then such a unit-step response curve may look S-shaped.
- This method applies if the system you study is:

∴ Theoretically: as delayed step input T.F:

$$\frac{C(s)}{R(s)} = \frac{K}{Ts+1} e^{-Ls}$$

∴ Experimentally: as step input exhibits an S-shaped curve, so you take the inflection point M then draw M's tangent line, then:

- L is the time when that line crosses with 0.
- T is the time when that line crosses with K.



▶ Ziegler–Nichols Second Method:

- This method depending on these procedures:

1. Use P controller to the plant (Set $T_i = \infty$ and $T_d = 0$, keep K only).

2. Observe these two parameters:

∴ **Critical Gain K_{cr}:** The gain value that makes the system's output exhibits sustained oscillations.

∴ **Critical Period P_{cr}:** The period of that oscillation.

3. Compute PID parameters using this table:

Controller	K_p	T_i	T_d
P	$0.5K_{cr}$	∞	0
PI	$0.45K_{cr}$	$\frac{1}{1.2}P_{cr}$	0
PID	$0.6K_{cr}$	$\frac{1}{2}P_{cr}$	$\frac{1}{8}P_{cr}$

▶ Key Notes:

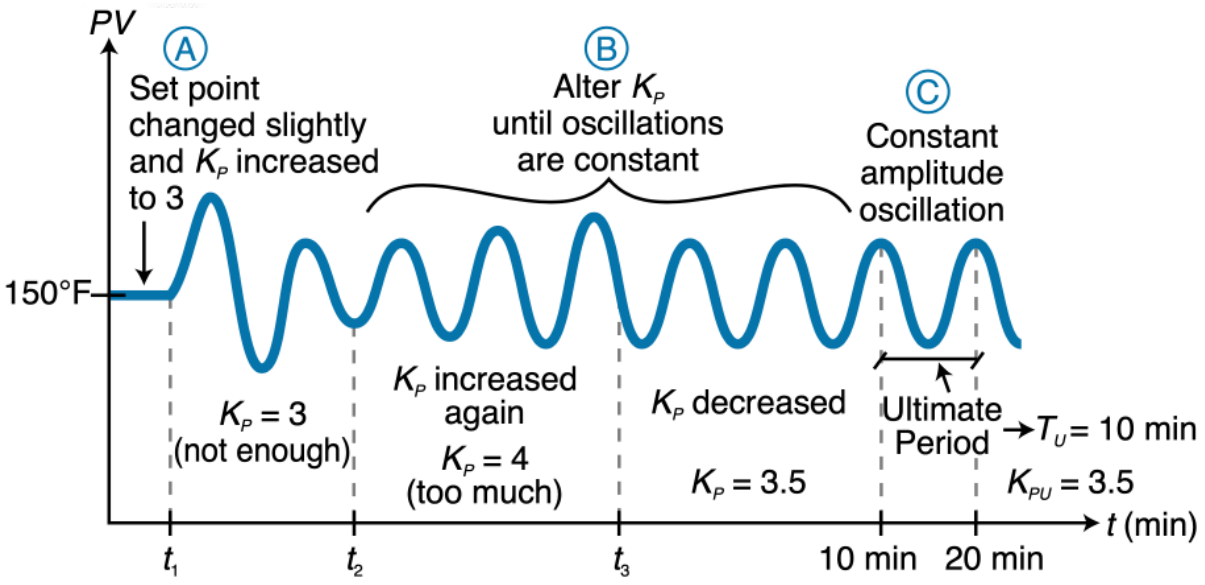
- If the output does not exhibit sustained oscillations for whatever value K may take, then this method does not apply.
- This method applies if the system you study is:

∴ Theoretically:

- You can find K_{cr} using Routh-Hurwitz criterion in entire zero row case, or by graphing root locus and see the value of K at $j\omega$ axis.
- You can find P_{cr} by substituting s with $j\omega$ at characteristic equation and equal it to zero to find ω firstly, then $P_{cr} = \frac{2\pi}{\omega}$.

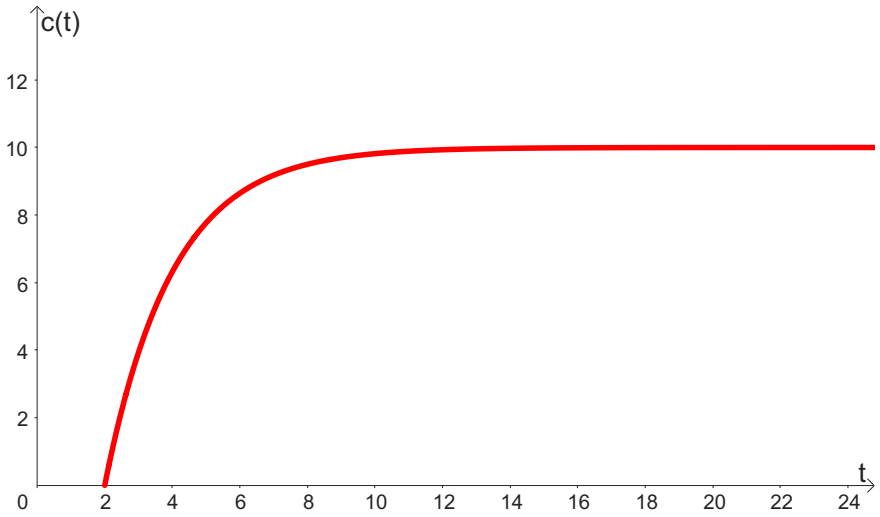
∴ Experimentally:

- Increase the value of K gradually from 0 to reach the value of K_{cr} when the system sustained only oscillations.
- The period of these oscillations is P_{cr}.



▶ **Q. 13.1**: Apply Z–N first method to tune K_p, T_i and T_d:

$$\frac{C(s)}{R(s)} = \frac{10}{2s+1} e^{-2s}$$



$\therefore$ *solution*:

If you want to find L and T even from equation or graph you will find that:

$$L = 2, \qquad T = 2$$

Now from Z-N first method tuning table:

$$K_p = 1.2\frac{T}{L} = 1.2$$

$$T_i = 2L = 4$$

$$T_d = 0.5L = 1$$

▶ **Q. 13.2:** Consider this unity feedback system that has:

$$G(s) = \frac{1}{s(s+1)(s+5)}$$

Apply Z–N second method to tune K_p and T_i:

$\therefore$ *solution*:

Firstly, set P controller has a value of K:

$$G(s) = \frac{K}{s(s+1)(s+5)}$$

$$G(s)H(s) + 1 = 0 \rightarrow \frac{K}{s(s+1)(s+5)} + 1 = 0$$

$$\rightarrow s^3 + 6s^2 + 5s + K = 0$$

s^3	1	5
s^2	6	K
s^1	$\frac{30-K}{6}$	0
s^0	K	

Using R-H, zero row case achieves when $K = 30$, so:

$$K_{cr} = 30$$

Now to find P_{cr} we must find ω_{cr}:

$$A(s) = 6s^2 + K = 0 \rightarrow 6s^2 + 30 = 0 \rightarrow s = \pm\sqrt{5}j$$

$$\omega_{cr} = \frac{2\pi}{P_{cr}} \rightarrow P_{cr} = \frac{2\pi}{\omega_{cr}} = \frac{2\pi}{\sqrt{5}} = 2.81$$

Now from Z-N second method tuning table:

$$K_p = 0.45K_{cr} = 13.5$$

$$T_i = \frac{1}{1.2}P_{cr} = 2.341$$

MAE

2024-2025's Final Exam (2nd Semester)



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